

Stock Market Analysis

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Background

Goal

We are interested in the probability of each stock that returns a positive return in a given period within the market. This information is very crucial for evaluating whether investing in a certain company is a good choice or not.

Description of data

In our dataset, we have three columns: sector, stock, and flip. Each sector represents a different field of business, taking values of 1 through 5 to represent what sector each company belongs to. Within those 5 sectors, we have stock to represent a company. This variable can take on the values 1 through 10 to represent the index for the company number within each sector. Lastly, we have 30 observations for if the stock returned a net positive or negative. This is also called a flip, because our outcome results in either a 1 for a net positive or 0 for a net negative return.

Exploratory Data Analysis

For a given stock, the return periods from one period to the next are iid (independent and identically distributed). we have 1500 observations, comprising of 5 sectors, each with 10 company and their corresponding stock, as well as 30 “flips” for each stock.

Some limitations and potential sources of bias can come from the rates of the stock market. The stock market is volatile and can be unpredictable, especially depending on the state of the economy. The period of which this data was obtained is undefined as well, meaning this dataset could be outdated and may not provide useful insights compared to markets today.

Defining the model

Bayesian Inference

Using Bayesian theory, we can find the probability of each stock that returns a positive return in a given period within the market. First, we need to choose which model we’d like to work with. Since each company in a sector is related to each other (in a sense that one company actions could influence another’s stock), we will use a random effect hierarchical model.

For our likelihood, we need to look at stock returns for companies nested in sectors. This can be modeled by a Bernoulli likelihood of $y_{scf}|\theta_{sc} \sim \text{Bern}(\theta_{sc})$ where

y_{scf} is a binary variable that represents the observed
net gain/loss of stock f at company c in sector s .
= 1 if positive return; 0 otherwise.

θ_{sc} represents the probability of a positive net return of company c in sector s

The full likelihood can be written as

$$L(\theta_{sc}|y_{scf}) \propto \prod_{s=1}^5 \prod_{c=1}^{10} \prod_{f=1}^{30} \theta_{sc}^{y_{scf}} (1 - \theta_{sc})^{1-y_{scf}}$$

For θ_{sc} , we can apply the log-odds because the model involves inter dependencies between stocks within the same company, where a change in one stock's performance could influence others. We can also assume that each sector and their companies are iid as well, leaving us with

$$\begin{aligned} \theta_{sc} &\stackrel{iid}{\sim} p(\theta|\mu_s, \tau_s^{-1}) \\ \text{logit}(\theta_{sc}) &= \log\left(\frac{\theta_{sc}}{1 - \theta_{sc}}\right) = \eta_{sc} \end{aligned}$$

Since we don't know much information about our initial value, we can start with a weak informative prior of $\text{logit}(\theta_{sc}) = \eta_{sc}$, where $\eta_{sc}|\mu_s\tau_s \sim N(\mu_s, \tau_s^{-1})$ where

$$\begin{aligned} \mu_s &\sim N(\mu_0, \sigma_0^2) \\ \tau_s^{-1} &\sim IG(\alpha_0, \beta_0) \end{aligned}$$

Hyperpriors

We will assume a $N(\mu_0 = 0, \sigma_0^2 = 20)$ weak mean prior because we aren't yet sure about this sector's success, and we leave the variance small in order to update our model in small steps.

Lastly, we will assume an $IG(\alpha_0 = 1, \beta_0 = 0.1)$ weak precision prior because it provides flexibility in our precision. In this context, it allows for high or low variability between companies in the same sector.

Overview/Model details

$$y_{scf} \stackrel{ind}{\sim} p(y|\theta_{sc})$$

$$\theta_{sc} \stackrel{iid}{\sim} p(\theta|\mu_s, \tau_s^{-1})$$

$$\text{logit}(\theta_{sc}) = \log\left(\frac{\theta_{sc}}{1 - \theta_{sc}}\right) = \eta_{sc}$$

$$\eta_{sc}|\mu_s\tau_s \sim N(\mu_s, \tau_s^{-1})$$

$$\mu_s \sim N(\mu_0 = 0, \sigma_0^2 = 20)$$

$$\tau_s \sim \text{Gamma}(\alpha_0 = 1, \beta_0 = 0.1)$$

$$s = 1, 2, \dots, 5$$

$$c = 1, 2, \dots, 10$$

$$f = 1, 2, \dots, 30$$

Interpretations of each parameter:

- y_{scf} : The observable response variable where $y_{scf} = 1$ if company c in sector s had a positive return at flip f , with 0 otherwise.
- θ_{sc} : The probability of observing a positive net return of company c in sector s .
- η_{sc} : The log-odds of a positive return at company c in sector s .
- μ_s : The sector mean log-odds of a positive return.
- τ_s^{-1} : The precision log-odds of a positive return at company c in sector s .
- s : Number of sectors (1-5)
- c : Number of companies and their stocks (1-10)
- f : Number of observations (flips) in a stock, returning either 1 or 0 for a positive/negative net return (1-30).

Fitting the model

Now that we have our likelihood and our prior, we can begin building a model using JAGS. Reference R code in appendix.

Model Diagnostics

Numer of iterations: 60,000

Number of burn-in iterations: 20,000

Number of adapt iterations: 20,000

Starting values:

$\eta_{[s,c]}$ = Log odds of flips for company stock c in sector s

$\mu_{[s]}$ = Average log odds of companies' stocks across sector s

$\tau_{[s]}$ = Log odds precision of companies' stocks across sector s

for $s = 1, \dots, 5$; $c = 1, \dots, 10$

Initial Interpretations

Traceplots (1.3) and boxplots (1.4) show that the algorithm has converged, with η_{sc} stabilizing around 0 within a ± 0.5 logit range.

A look at the initial values (1.5) lets us know that the burn-in and adaption period were effective, as our data has converged and become stationary.

ESS values (1.6) range from roughly 15,000–22,000. Since we have a relatively high ESS (after taking into account burnin and adaption), we can say that our MCMC algorithm is efficient.

ACF plots (1.7) drop off steeply, suggesting there is low autocorrelation and good sample independence. Therefore, we can not only conclude that our samples for each sector are independent, but are efficiently being drawn to the true posterior.

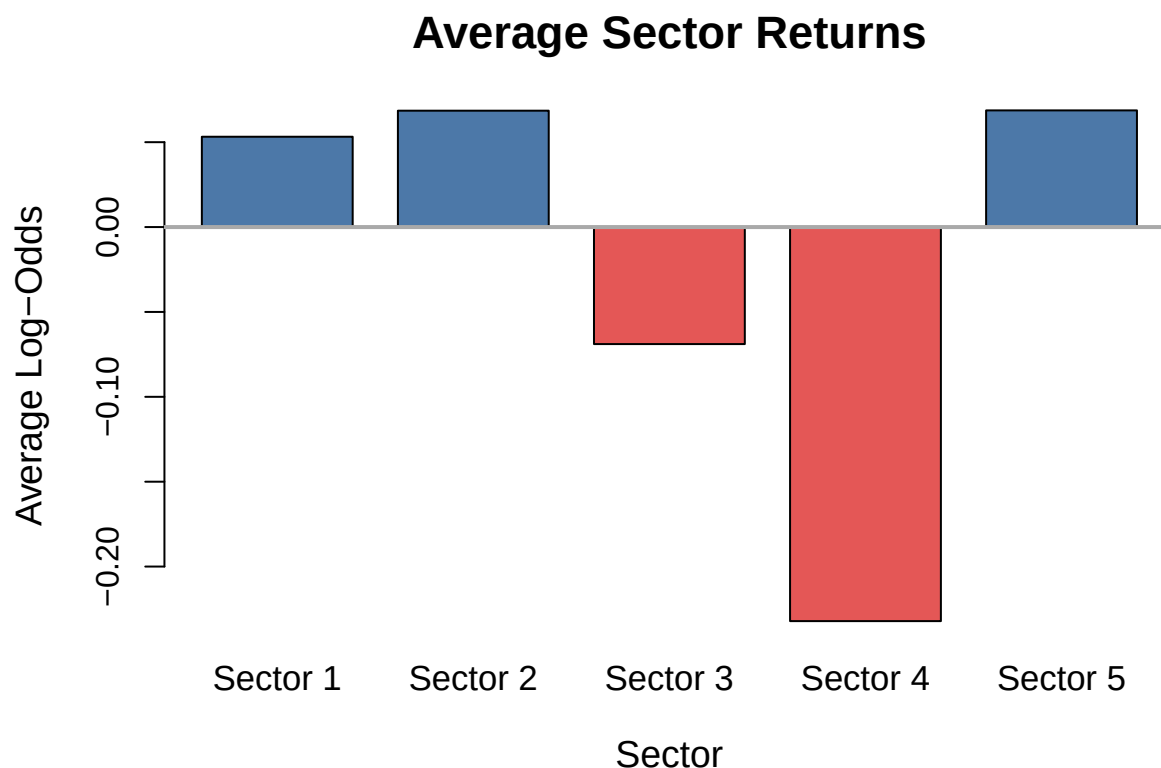
Summary Interpretations

Looking at the summary (1.3), we can observe the mean, standard deviation, and corresponding confidence intervals for each η_{sc} , μ_s , and τ_s .

However, we are most interested in the mean, as returns of individual stocks are typically modeled as varying around a common sector mean.

μ_s can give us insights on which sector preforms the best on average, as it represents the log-odds of a positive return for each sector. Here, it seems like we should invest in sectors 1, 2, and 5, as the average log odds of company stock return a net positive.

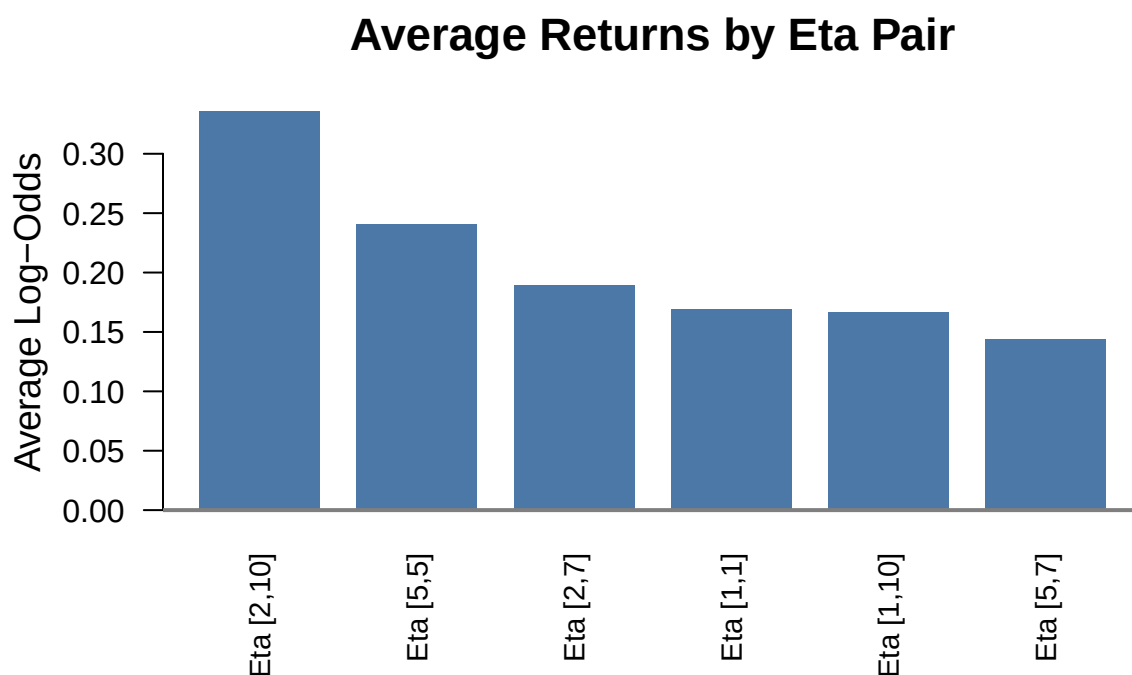
```
##          mu[1]          mu[2]          mu[3]          mu[4]          mu[5]
## 0.05320714  0.06855078 -0.06893301 -0.23213381  0.06873257
```



Taking a closer look at the mean of each η_{sc} , this represents the average log-odds of a positive return for each company c in a given sector s . We can use this information to look at which

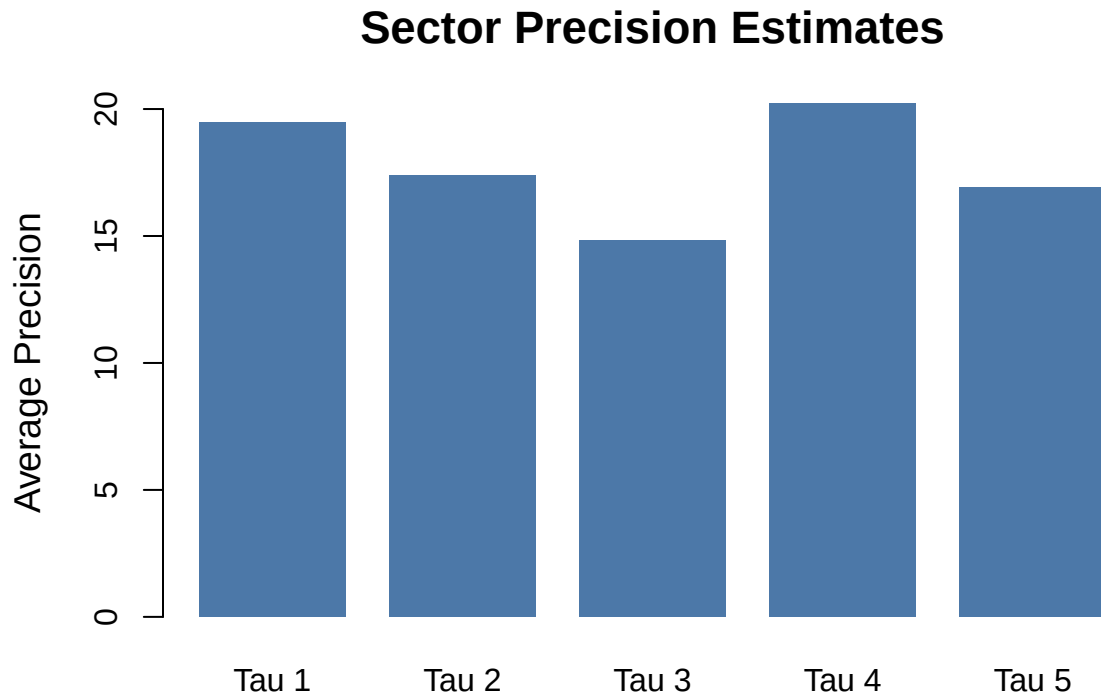
companies in sector s are worth investing in. Here are some of the highest return companies and their sectors. Note: Full η_{sc} listed in Appendix.

```
## eta[2,10] eta[5,5] eta[2,7] eta[1,1] eta[1,10] eta[5,7]
## 0.3360736 0.2403633 0.1895844 0.1691693 0.1665085 0.1441129
```



Meanwhile, the means of τ_s tells us the average precision of company c 's log-odds across sector s . A larger precision in sector s indicates that the companies within these sectors and their stocks have more similar behaviors. Likewise, a smaller precision indicates that companies within sector s have more different behaviors. Given that sectors 1, 2, and 5 return an average log odds net positive, sector 1 has the highest consistency.

```
## tau[1] tau[2] tau[3] tau[4] tau[5]
## 19.48815 17.40777 14.84679 20.21426 16.91223
```



Appendix

1.1 - Initialization of variables.

```
set.seed(614)

S <- length(unique(stock_file$sector)) # S = 5; 5 total sectors
C <- length(unique(stock_file$stock)) # C = 10; 10 total companies and their stocks
F <- 30 # 30 total observations for each stock
N <- 60000 # Total number of iterations, including burn-in and adaptation
nburn <- 20000
nadapt <- 20000

y_array <- array(rbinom(S * C * F, size = 1, prob = 0.5), dim = c(S, C, F)) #Array to h
```

1.2 - JAGS code.

```
#Install packages needed

library(rjags)
library(coda)

mymodel <- "model {
  #likelihood
  for (s in 1:S) {
    for (c in 1:C) {
      for (f in 1:F) {
        #likelihood - loop through each sector, company, and flips to get likelihood
        y[s, c, f] ~ dbern(theta[s, c])
      }
      eta[s, c] ~ dnorm(mu[s], tau[s])    # Eta represents company log-odds of positive flips
      theta[s, c] <- ilogit(eta[s, c])    # Convert log-odds to probability
    }
  }

  #prior
  for (s in 1:S) {
    mu[s] ~ dnorm(0, 1/20)                # mean log-odds across sectors
    tau[s] ~ dgamma(1, 0.1)              # precision across sectors
  }
}
"
```

1.3 - JAGS model initialization.

```

data_jags <- list(
  y = y_array,
  S = S,
  C = C,
  F = F
)

set.seed(614)

myinit <- list(
  .RNG.name = "base::Super-Duper",
  .RNG.seed = 614,
  mu = rnorm(S, 0, sqrt(20)),
  tau = rgamma(S, 1, 0.1),
  eta = array(rnorm(S * C), dim = c(S, C))
)

fitModel= jags.model(textConnection(mymodel),data=data_jags,inits=myinit,n.chains=1, n.burnin=1000,
n.thin=100,verbose=FALSE)

## Compiling model graph
##   Resolving undeclared variables
##   Allocating nodes
## Graph information:
##   Observed stochastic nodes: 1500
##   Unobserved stochastic nodes: 60
##   Total graph size: 1618
##
## Initializing model

```

```

update(fitModel, nburn)

fit.samples=coda.samples(fitModel, c("mu", "tau", "eta"), n.iter=N)

summary(fit.samples)

```

```

##
## Iterations = 40001:1e+05
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 60000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean          SD Naive SE Time-series SE
## eta[1,1]  0.169169  0.2355 0.0009614      0.001724
## eta[2,1]  0.045623  0.2388 0.0009749      0.001654
## eta[3,1] -0.042629  0.2474 0.0010101      0.001635
## eta[4,1] -0.201945  0.2287 0.0009336      0.001758
## eta[5,1] -0.102071  0.2470 0.0010083      0.001793
## eta[1,2]  0.032363  0.2303 0.0009402      0.001647
## eta[2,2] -0.100184  0.2472 0.0010091      0.001869
## eta[3,2]  0.013408  0.2498 0.0010196      0.001662
## eta[4,2] -0.073446  0.2357 0.0009621      0.001815
## eta[5,2]  0.142762  0.2436 0.0009944      0.001702
## eta[1,3]  0.079257  0.2308 0.0009421      0.001669
## eta[2,3]  0.092301  0.2379 0.0009714      0.001627
## eta[3,3] -0.364427  0.2808 0.0011462      0.002266

```

## eta[4,3]	-0.333260	0.2328	0.0009503	0.001742
## eta[5,3]	0.043995	0.2423	0.0009891	0.001677
## eta[1,4]	-0.099011	0.2374	0.0009691	0.001810
## eta[2,4]	-0.003397	0.2397	0.0009784	0.001681
## eta[3,4]	0.064108	0.2521	0.0010294	0.001715
## eta[4,4]	-0.288381	0.2320	0.0009469	0.001705
## eta[5,4]	-0.201061	0.2628	0.0010730	0.002093
## eta[1,5]	0.036691	0.2306	0.0009415	0.001671
## eta[2,5]	0.092999	0.2375	0.0009697	0.001682
## eta[3,5]	0.119136	0.2589	0.0010570	0.001827
## eta[4,5]	-0.202164	0.2284	0.0009324	0.001669
## eta[5,5]	0.240363	0.2490	0.0010164	0.001832
## eta[1,6]	-0.054732	0.2345	0.0009572	0.001712
## eta[2,6]	0.089927	0.2386	0.0009739	0.001663
## eta[3,6]	-0.096886	0.2486	0.0010148	0.001644
## eta[4,6]	-0.246195	0.2292	0.0009356	0.001704
## eta[5,6]	0.094132	0.2400	0.0009799	0.001610
## eta[1,7]	-0.007564	0.2328	0.0009503	0.001666
## eta[2,7]	0.189584	0.2451	0.0010006	0.001765
## eta[3,7]	-0.150077	0.2497	0.0010194	0.001655
## eta[4,7]	-0.331453	0.2331	0.0009514	0.001702
## eta[5,7]	0.144113	0.2406	0.0009824	0.001632
## eta[1,8]	0.079599	0.2307	0.0009417	0.001626
## eta[2,8]	-0.050611	0.2432	0.0009930	0.001769
## eta[3,8]	0.116377	0.2593	0.0010584	0.001817
## eta[4,8]	-0.201143	0.2307	0.0009418	0.001731
## eta[5,8]	0.042153	0.2406	0.0009823	0.001636


```

## eta[1,9]    0.125435  0.2342 0.0009563      0.001729
## eta[2,9]   -0.003886  0.2394 0.0009772      0.001726
## eta[3,9]   -0.310690  0.2686 0.0010964      0.002065
## eta[4,9]   -0.201441  0.2292 0.0009356      0.001647
## eta[5,9]    0.141957  0.2420 0.0009879      0.001650
## eta[1,10]   0.166509  0.2359 0.0009632      0.001723
## eta[2,10]   0.336074  0.2642 0.0010788      0.002060
## eta[3,10]  -0.041401  0.2500 0.0010205      0.001650
## eta[4,10]  -0.242185  0.2293 0.0009361      0.001649
## eta[5,10]   0.139566  0.2409 0.0009835      0.001661
## mu[1]       0.053207  0.1457 0.0005948      0.001506
## mu[2]       0.068551  0.1495 0.0006105      0.001459
## mu[3]      -0.068933  0.1569 0.0006404      0.001412
## mu[4]      -0.232134  0.1446 0.0005902      0.001531
## mu[5]       0.068733  0.1513 0.0006177      0.001448
## tau[1]     19.488152 12.0457 0.0491762      0.094392
## tau[2]     17.407766 11.5051 0.0469694      0.096563
## tau[3]     14.846790 10.3935 0.0424314      0.094396
## tau[4]     20.214255 12.2466 0.0499964      0.095866
## tau[5]     16.912234 11.1410 0.0454830      0.094562

```

```
##
```

```
## 2. Quantiles for each variable:
```

```
##
```

```

##           2.5%       25%       50%       75%      97.5%
## eta[1,1] -0.2780  0.012846  0.1622457  0.31938  0.65569
## eta[2,1] -0.4281 -0.112647  0.0464580  0.20336  0.51703
## eta[3,1] -0.5265 -0.204853 -0.0451427  0.11709  0.45012

```

```

## eta[4,1] -0.6526 -0.351433 -0.2024069 -0.05123 0.25056
## eta[5,1] -0.6185 -0.258030 -0.0922494 0.06531 0.35660
## eta[1,2] -0.4218 -0.119624 0.0328904 0.18290 0.48765
## eta[2,2] -0.6196 -0.255784 -0.0899600 0.06776 0.35877
## eta[3,2] -0.4663 -0.152727 0.0088292 0.17462 0.52086
## eta[4,2] -0.5135 -0.235204 -0.0820224 0.07694 0.42064
## eta[5,2] -0.3259 -0.019801 0.1375851 0.29997 0.63762
## eta[1,3] -0.3734 -0.073143 0.0767962 0.23014 0.53987
## eta[2,3] -0.3730 -0.064617 0.0903257 0.24767 0.56585
## eta[3,3] -0.9709 -0.537323 -0.3442112 -0.17020 0.13229
## eta[4,3] -0.8110 -0.483097 -0.3262469 -0.17604 0.10457
## eta[5,3] -0.4384 -0.113078 0.0447332 0.20330 0.51841
## eta[1,4] -0.5907 -0.251587 -0.0902213 0.06073 0.34729
## eta[2,4] -0.4879 -0.159510 0.0001312 0.15673 0.45855
## eta[3,4] -0.4135 -0.105641 0.0559918 0.22381 0.58678
## eta[4,4] -0.7589 -0.437369 -0.2843155 -0.13410 0.15693
## eta[5,4] -0.7632 -0.366349 -0.1840077 -0.02081 0.27279
## eta[1,5] -0.4236 -0.113130 0.0373731 0.18885 0.48664
## eta[2,5] -0.3734 -0.061940 0.0918734 0.24783 0.56922
## eta[3,5] -0.3618 -0.057440 0.1082988 0.28327 0.65853
## eta[4,5] -0.6534 -0.351831 -0.2020101 -0.05337 0.25342
## eta[5,5] -0.2235 0.073499 0.2297641 0.39628 0.76122
## eta[1,6] -0.5352 -0.204704 -0.0480116 0.10211 0.39522
## eta[2,6] -0.3765 -0.066617 0.0876376 0.24454 0.56985
## eta[3,6] -0.5924 -0.257842 -0.0938743 0.06702 0.38714
## eta[4,6] -0.7029 -0.395801 -0.2437371 -0.09500 0.20152
## eta[5,6] -0.3729 -0.064308 0.0913335 0.24882 0.57816

```

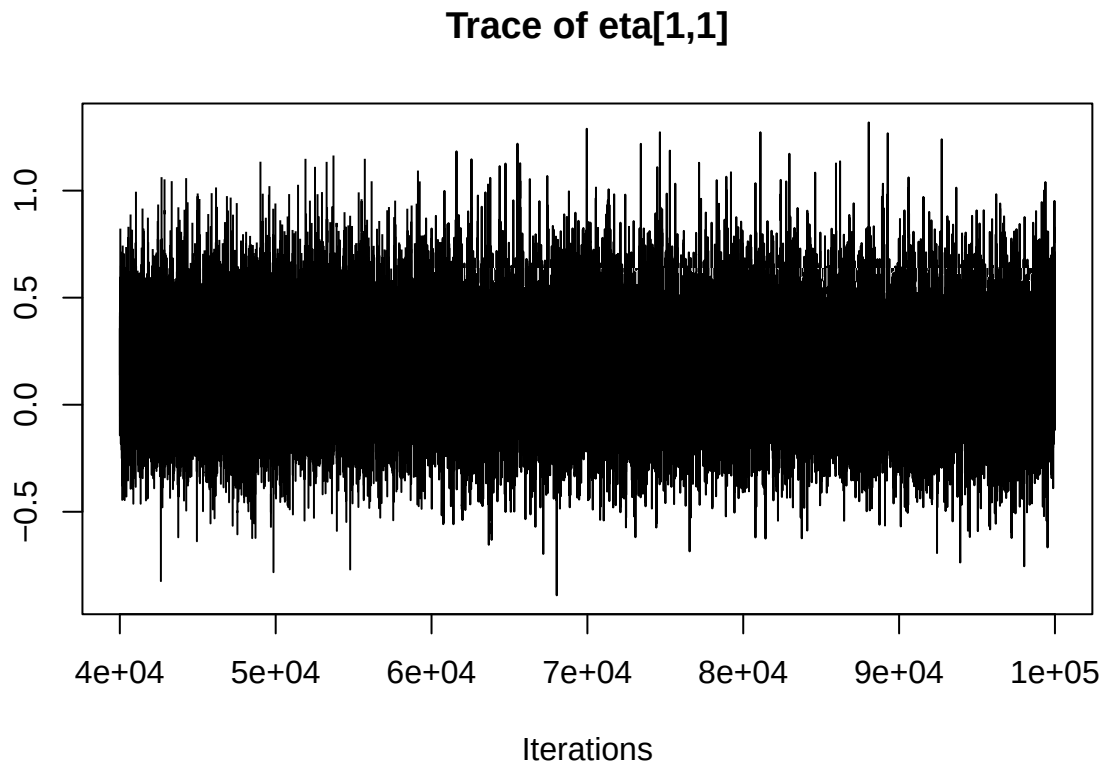
```

## eta[1,7]  -0.4747 -0.158986 -0.0039847  0.14680  0.44008
## eta[2,7]  -0.2748  0.025835  0.1827646  0.34315  0.69344
## eta[3,7]  -0.6619 -0.310351 -0.1445475  0.01670  0.32693
## eta[4,7]  -0.8103 -0.479040 -0.3251353 -0.17731  0.11067
## eta[5,7]  -0.3191 -0.015946  0.1383831  0.29993  0.63120
## eta[1,8]  -0.3702 -0.071812  0.0773606  0.22919  0.54551
## eta[2,8]  -0.5555 -0.206399 -0.0440354  0.11191  0.41031
## eta[3,8]  -0.3608 -0.059210  0.1049959  0.27969  0.65848
## eta[4,8]  -0.6517 -0.353067 -0.2033757 -0.05143  0.25953
## eta[5,8]  -0.4377 -0.113999  0.0432869  0.19966  0.51960
## eta[1,9]  -0.3260 -0.029568  0.1203325  0.27584  0.60202
## eta[2,9]  -0.4882 -0.158764 -0.0001299  0.15462  0.46210
## eta[3,9]  -0.8812 -0.478137 -0.2954101 -0.12635  0.17517
## eta[4,9]  -0.6484 -0.353249 -0.2022341 -0.05079  0.25330
## eta[5,9]  -0.3222 -0.019319  0.1380830  0.29715  0.63169
## eta[1,10] -0.2807  0.008521  0.1594075  0.31781  0.65149
## eta[2,10] -0.1367  0.154942  0.3189235  0.49993  0.91062
## eta[3,10] -0.5292 -0.206443 -0.0424861  0.12296  0.45402
## eta[4,10] -0.7007 -0.390473 -0.2409468 -0.09255  0.20757
## eta[5,10] -0.3232 -0.020765  0.1355140  0.29455  0.63068
## mu[1]      -0.2328 -0.043855  0.0532982  0.15013  0.33786
## mu[2]      -0.2240 -0.030114  0.0690845  0.16751  0.36239
## mu[3]      -0.3799 -0.170663 -0.0688476  0.03386  0.24054
## mu[4]      -0.5180 -0.328140 -0.2323028 -0.13653  0.05262
## mu[5]      -0.2295 -0.030484  0.0681103  0.16713  0.36886
## tau[1]      4.4263 10.805367 16.7067376 25.20210 49.96791
## tau[2]      3.5993  9.225070 14.6317351 22.49547 46.91983

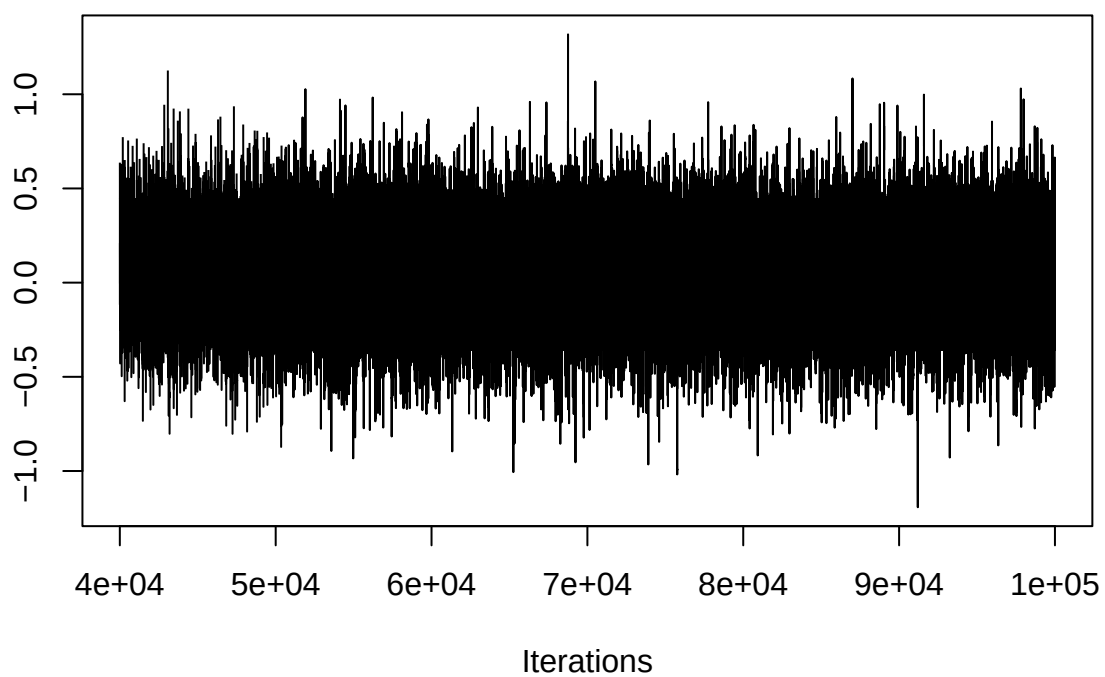
```

```
## tau[3]      2.8710  7.470397 12.1191034 19.28118 42.13783
## tau[4]      4.6978 11.446335 17.4857880 26.02343 51.23190
## tau[5]      3.5557  8.951192 14.1895107 21.91636 45.57875
```

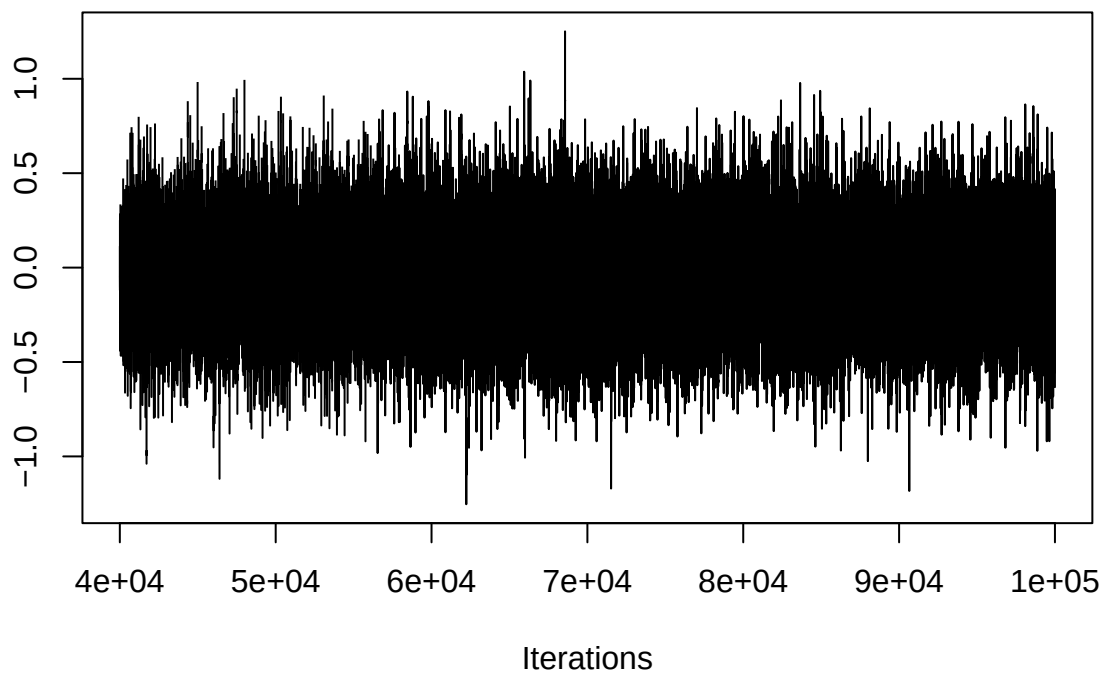
```
traceplot(fit.samples)
```



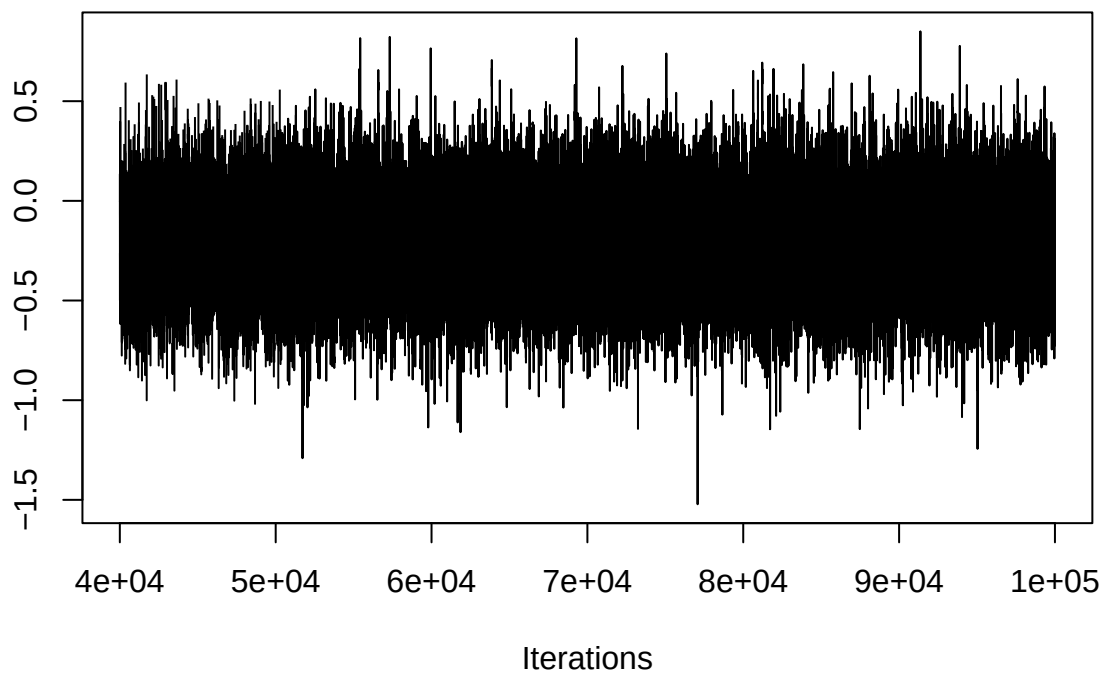
Trace of eta[2,1]



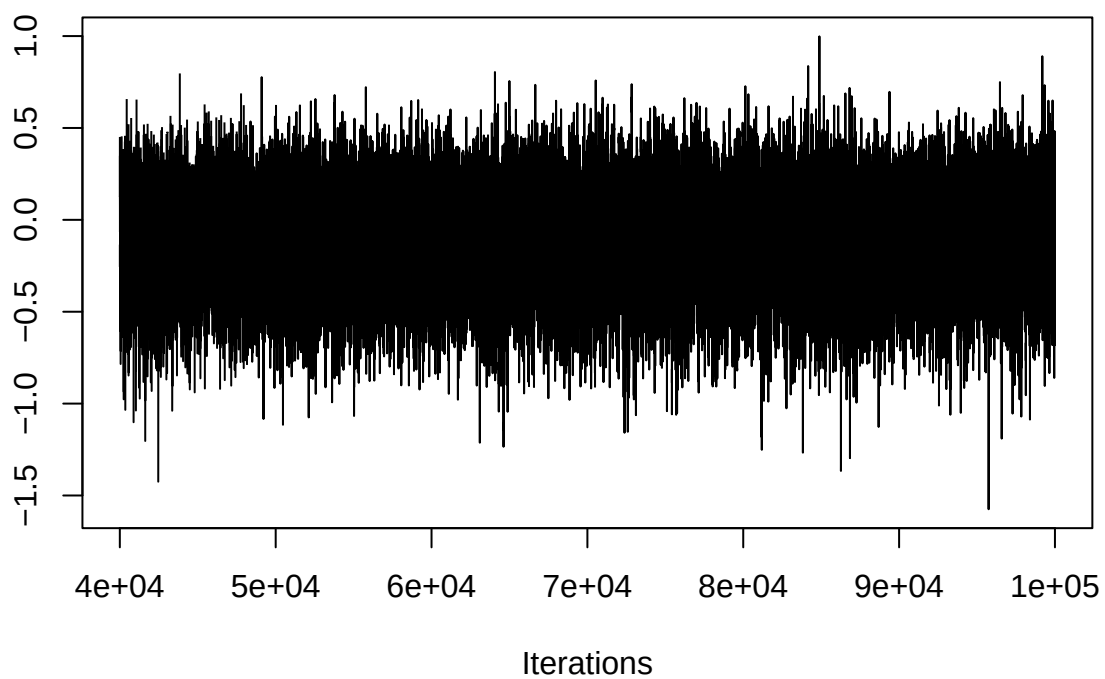
Trace of eta[3,1]



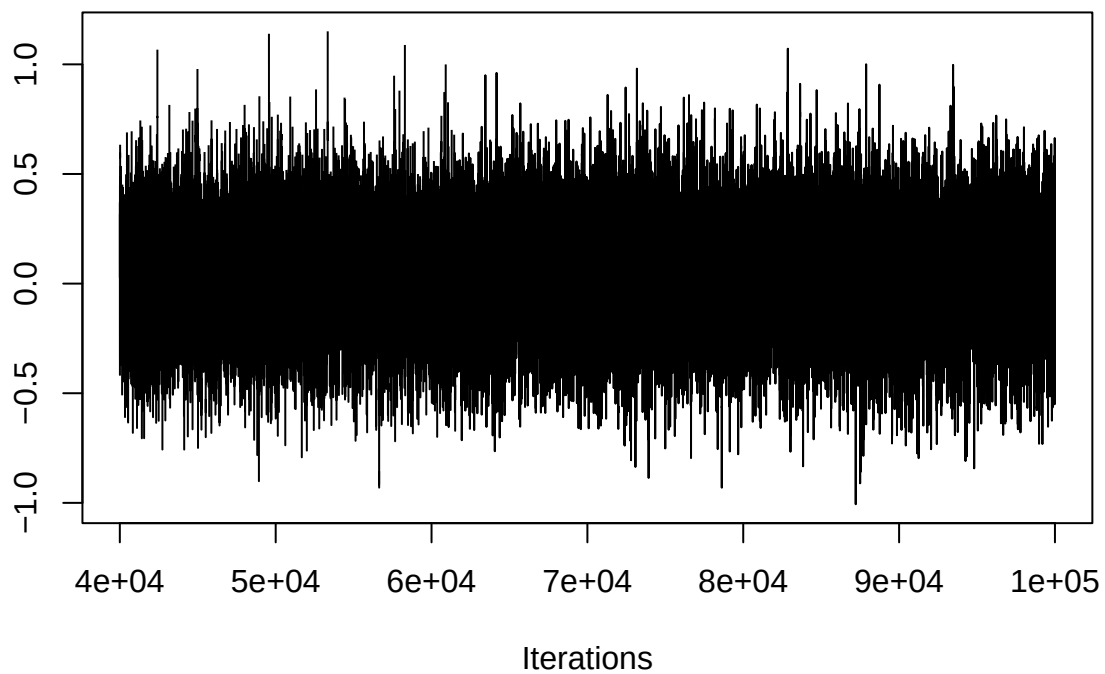
Trace of eta[4,1]



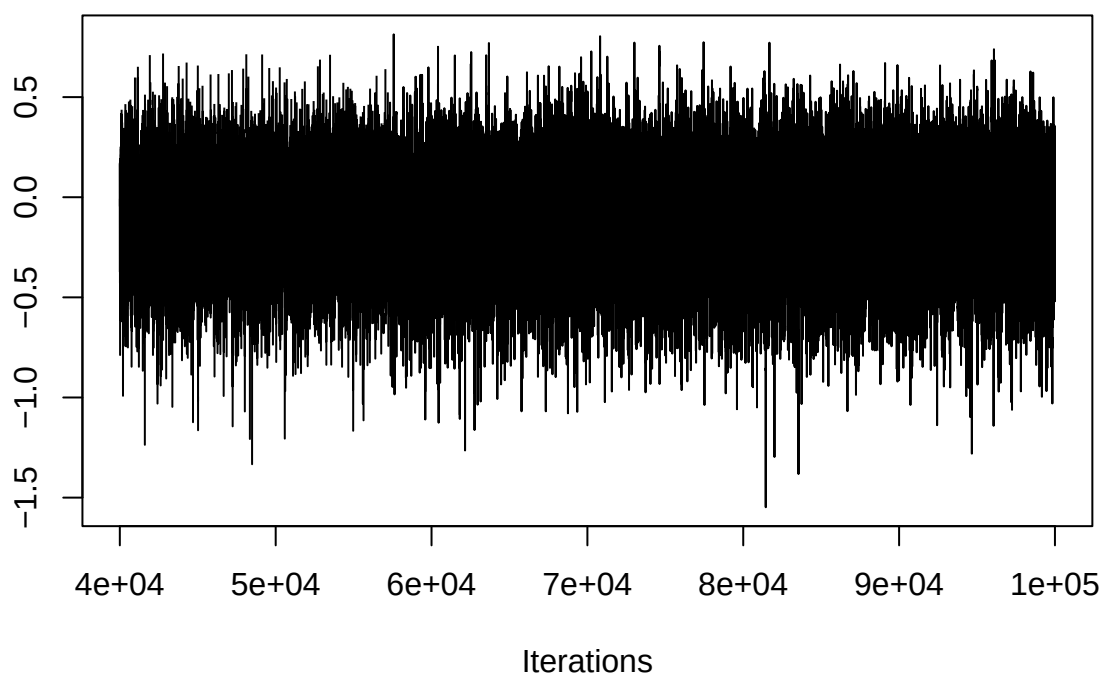
Trace of eta[5,1]



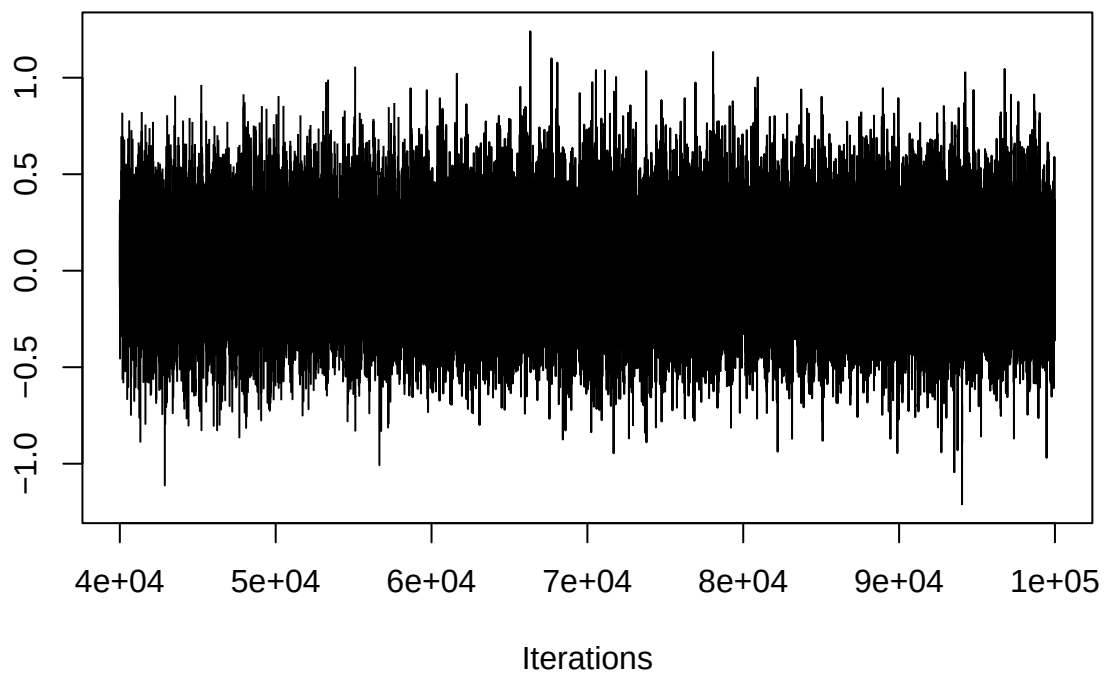
Trace of eta[1,2]



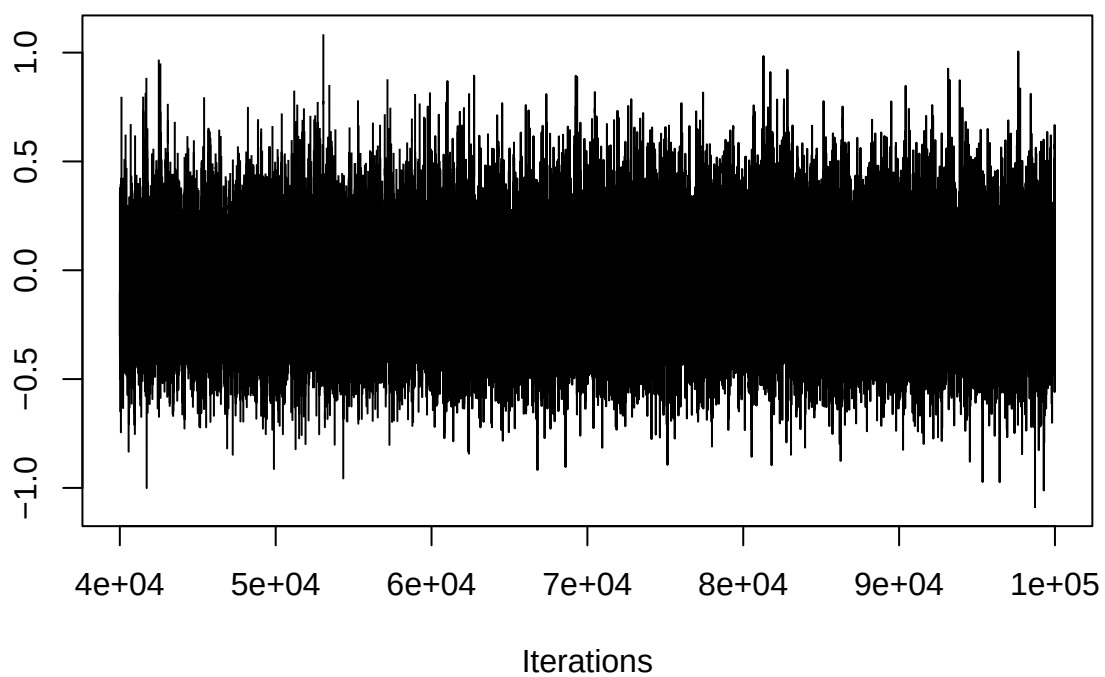
Trace of eta[2,2]



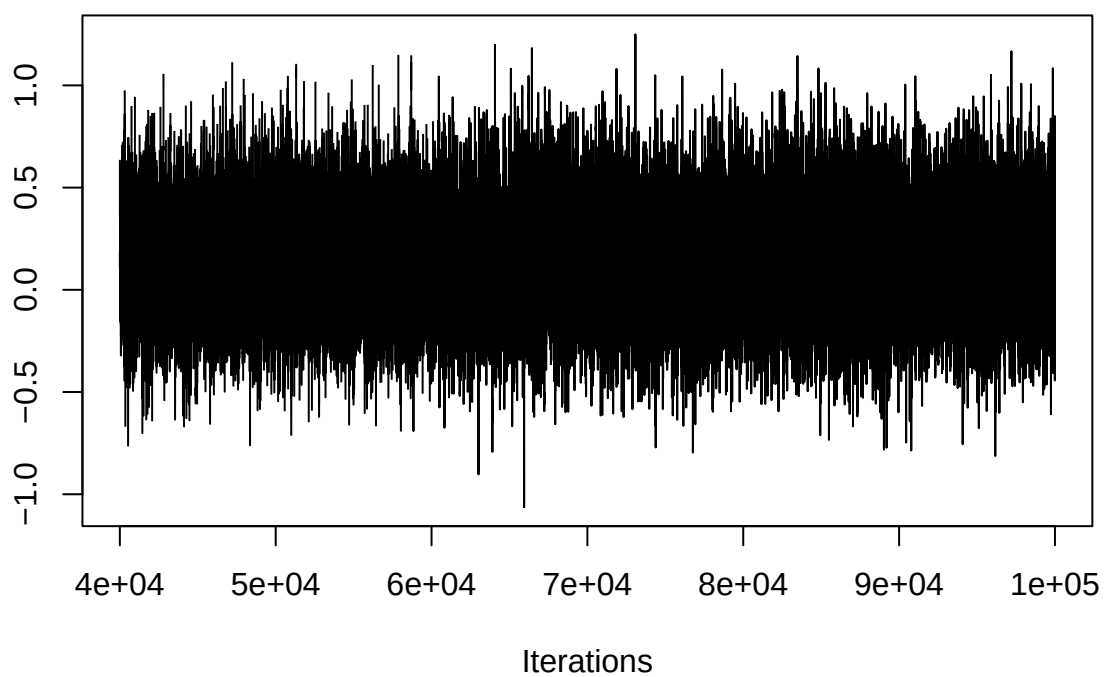
Trace of eta[3,2]



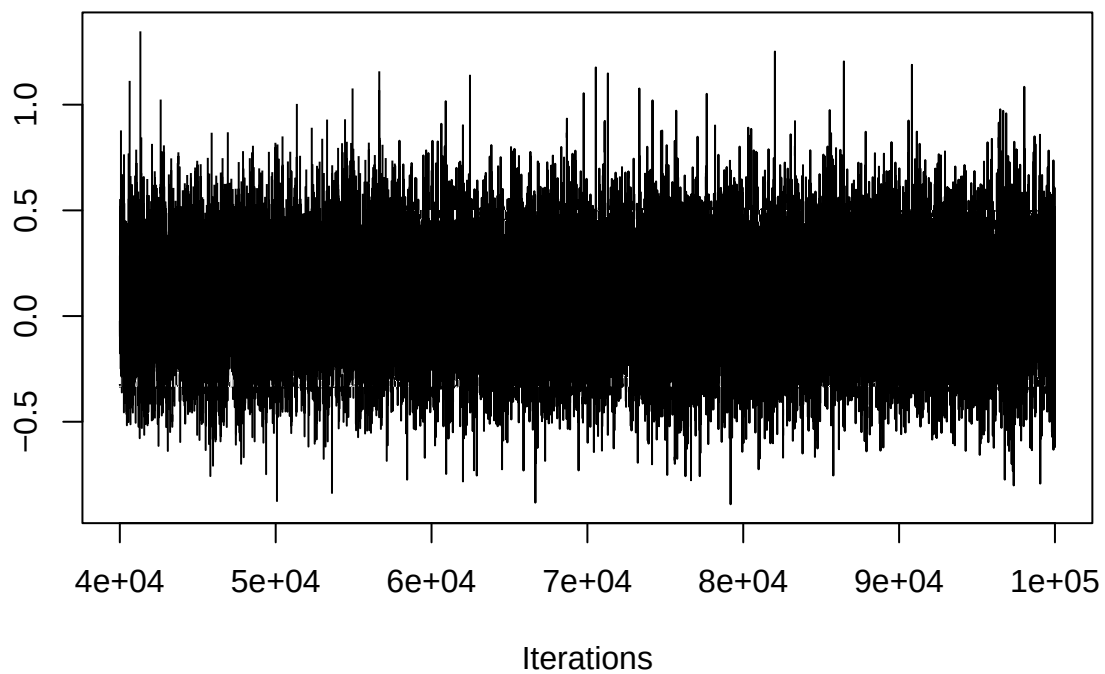
Trace of eta[4,2]



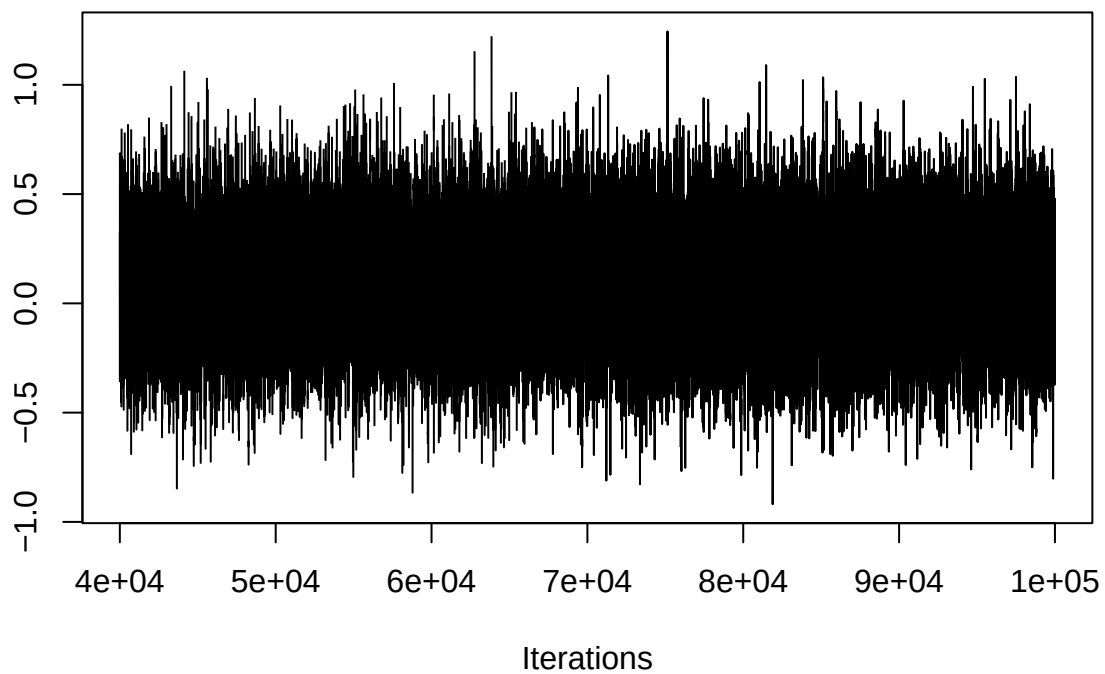
Trace of eta[5,2]



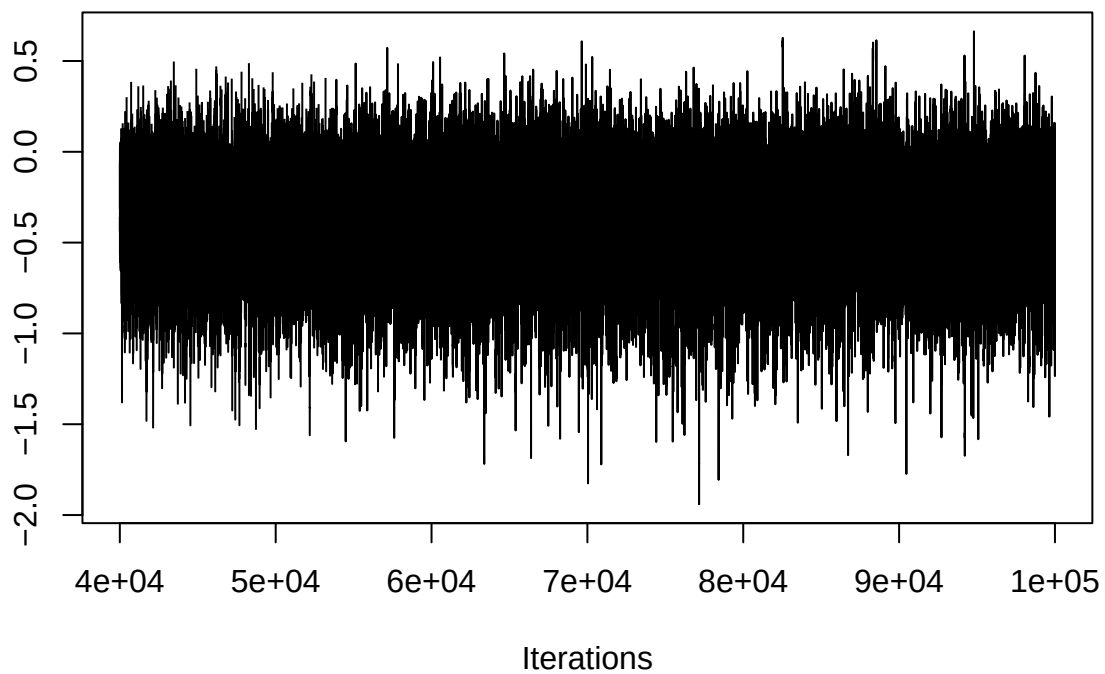
Trace of eta[1,3]



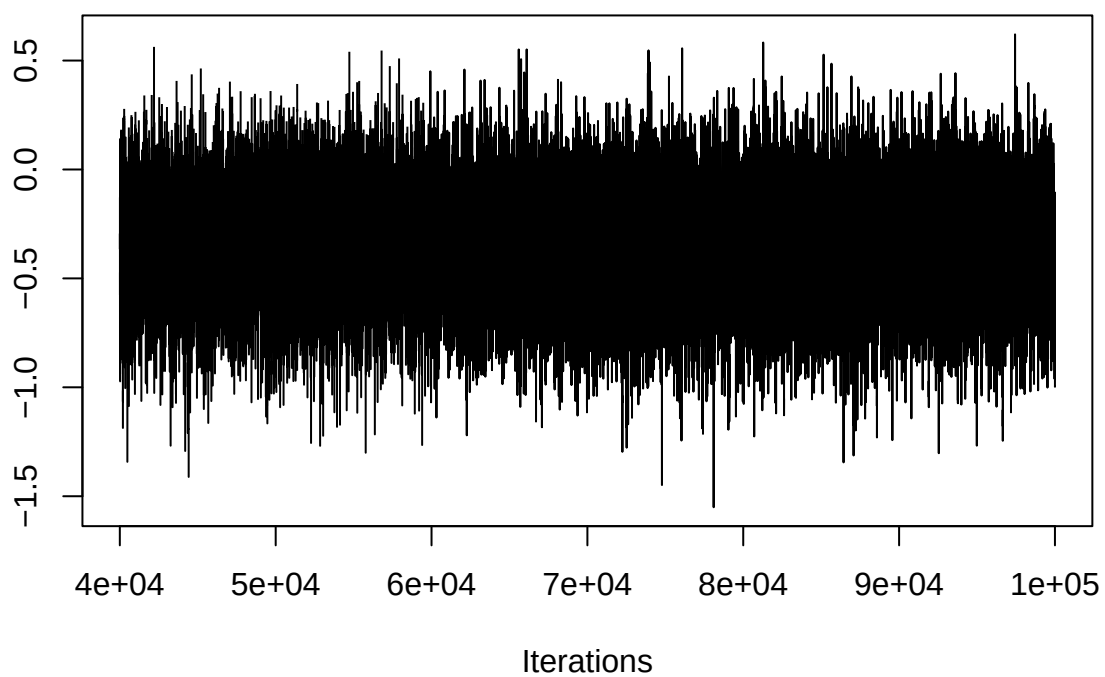
Trace of eta[2,3]



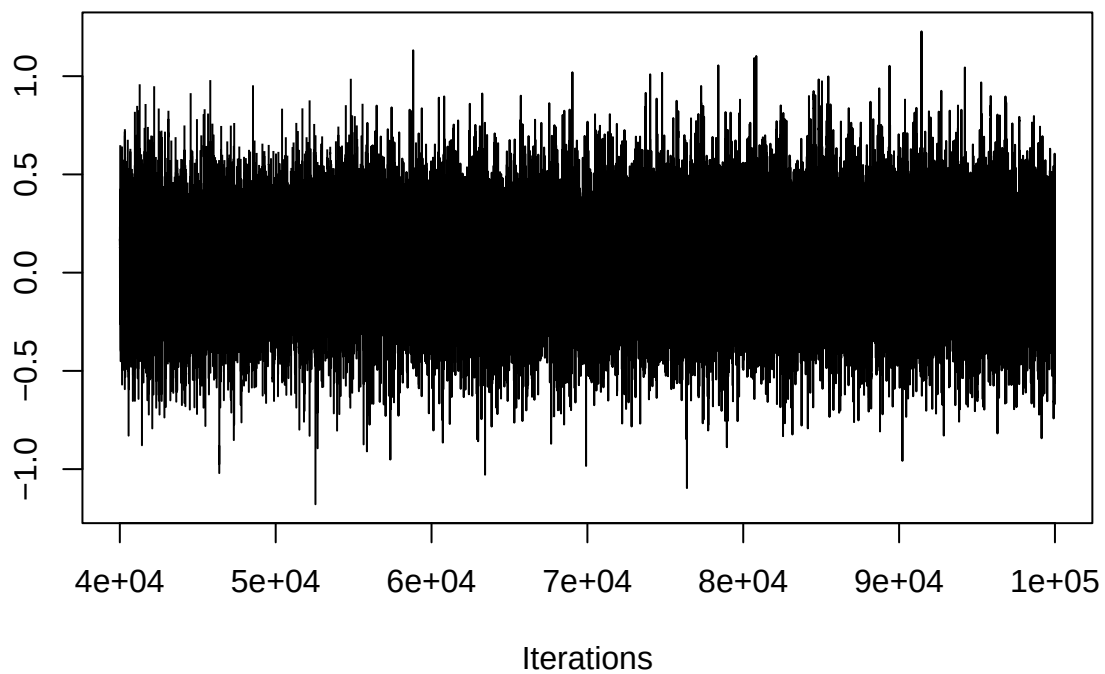
Trace of eta[3,3]



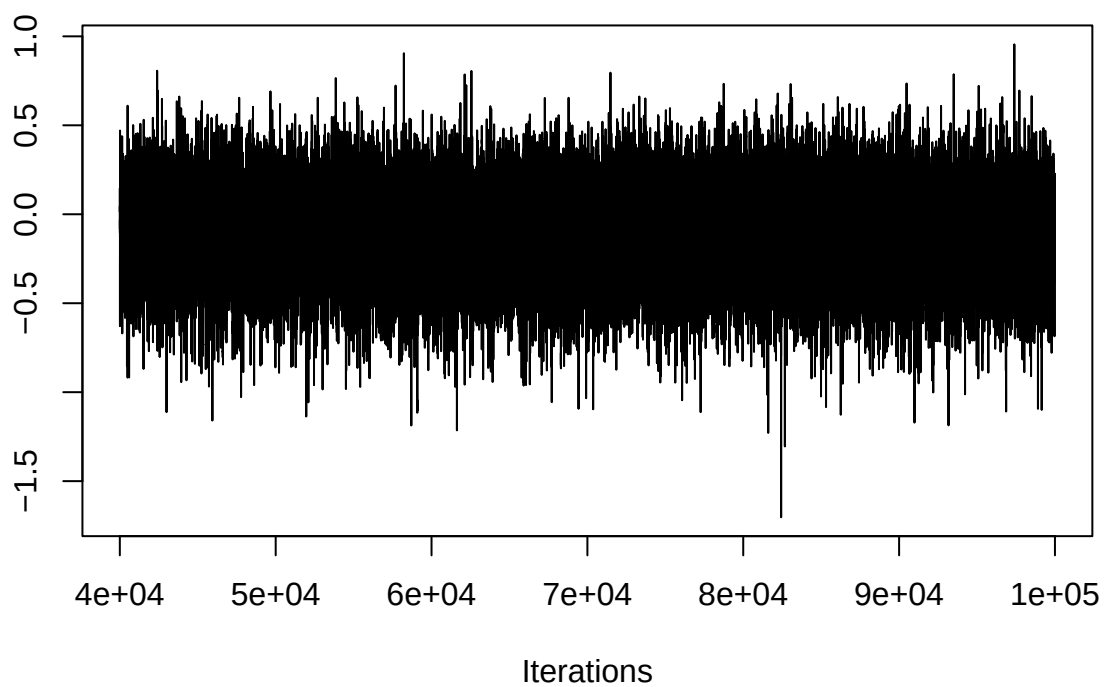
Trace of eta[4,3]



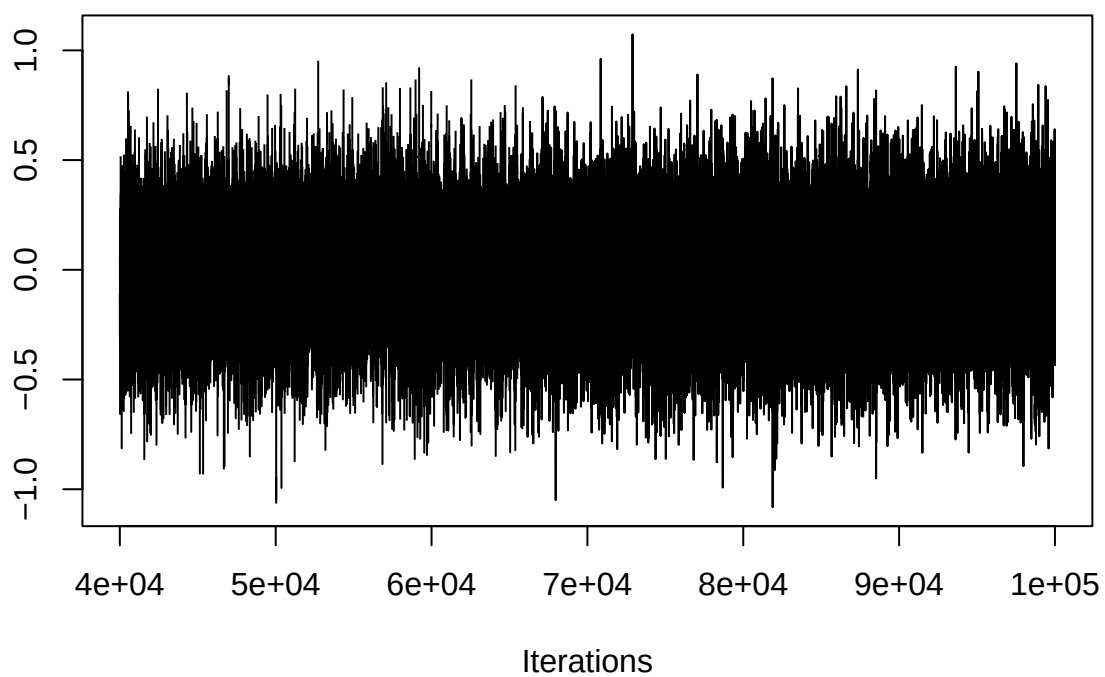
Trace of eta[5,3]



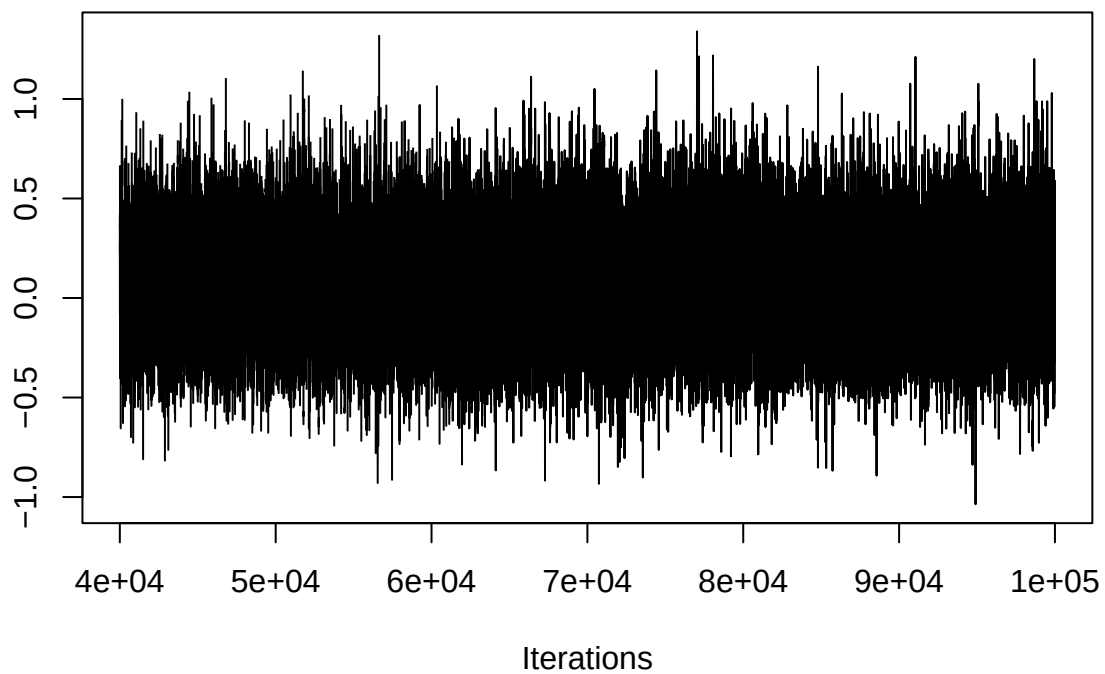
Trace of eta[1,4]



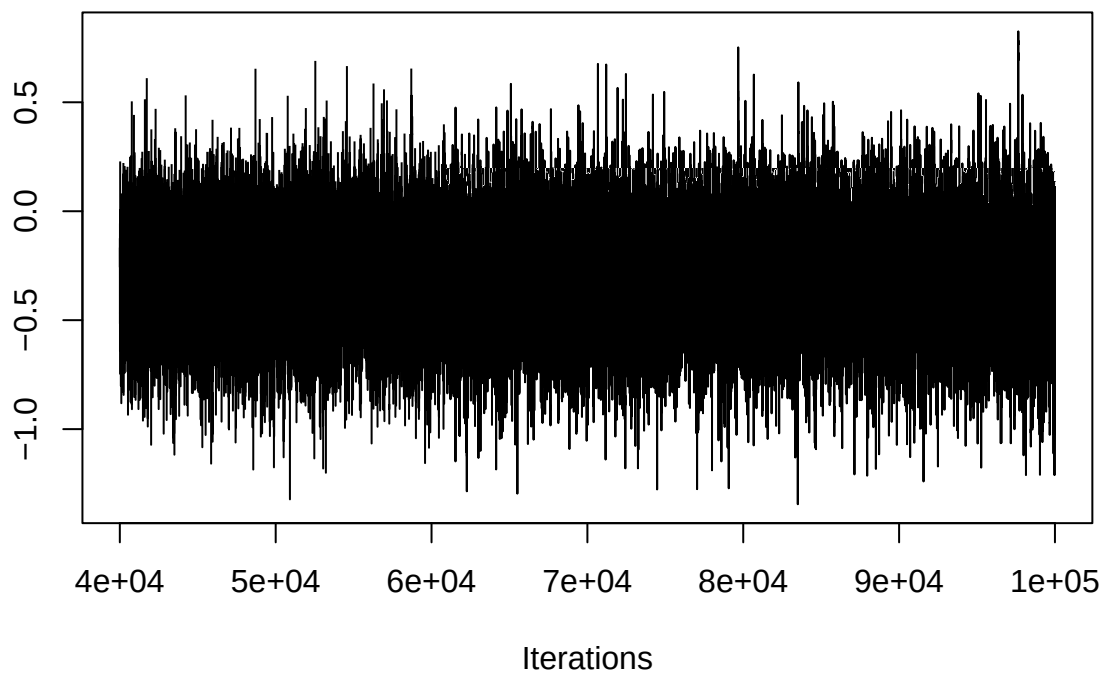
Trace of eta[2,4]



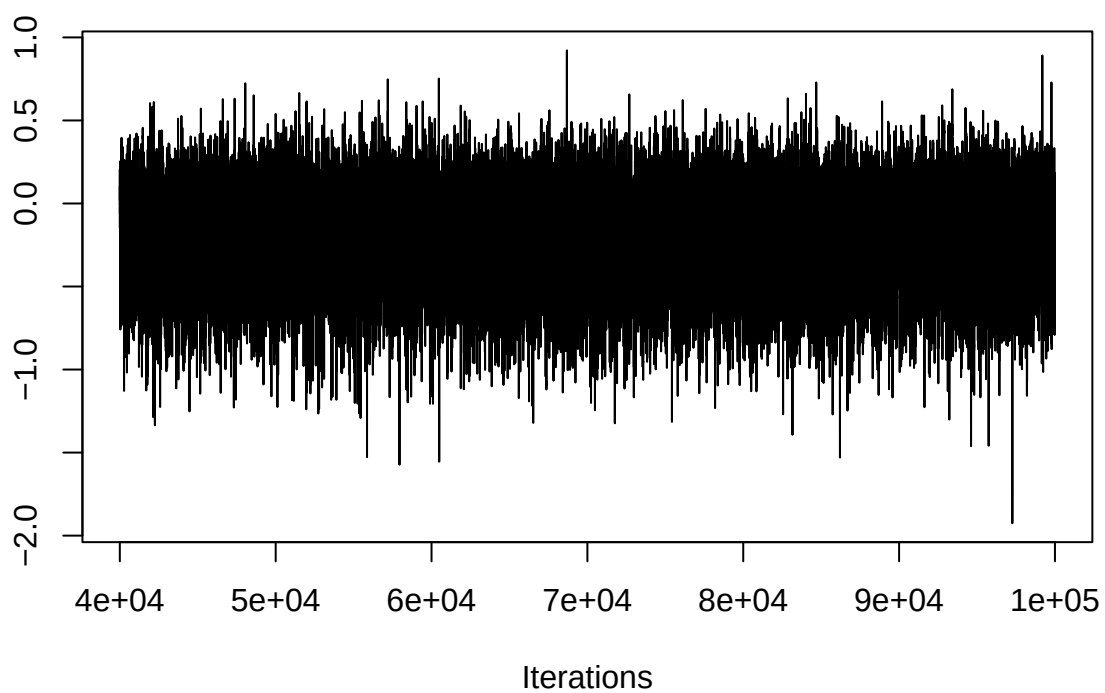
Trace of eta[3,4]



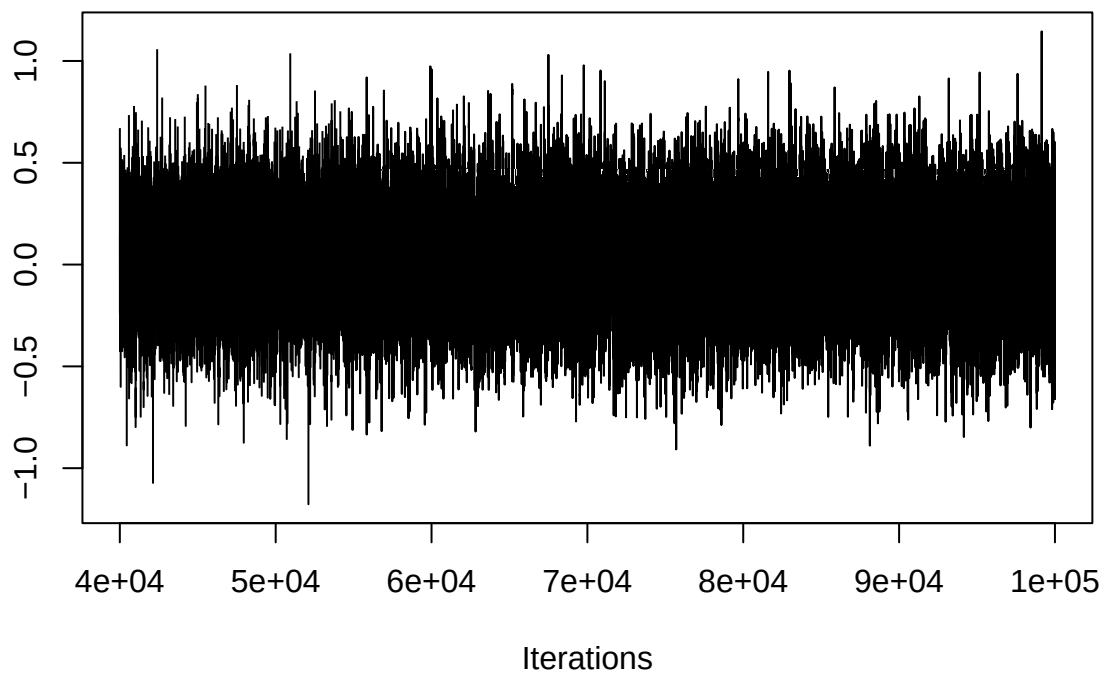
Trace of eta[4,4]



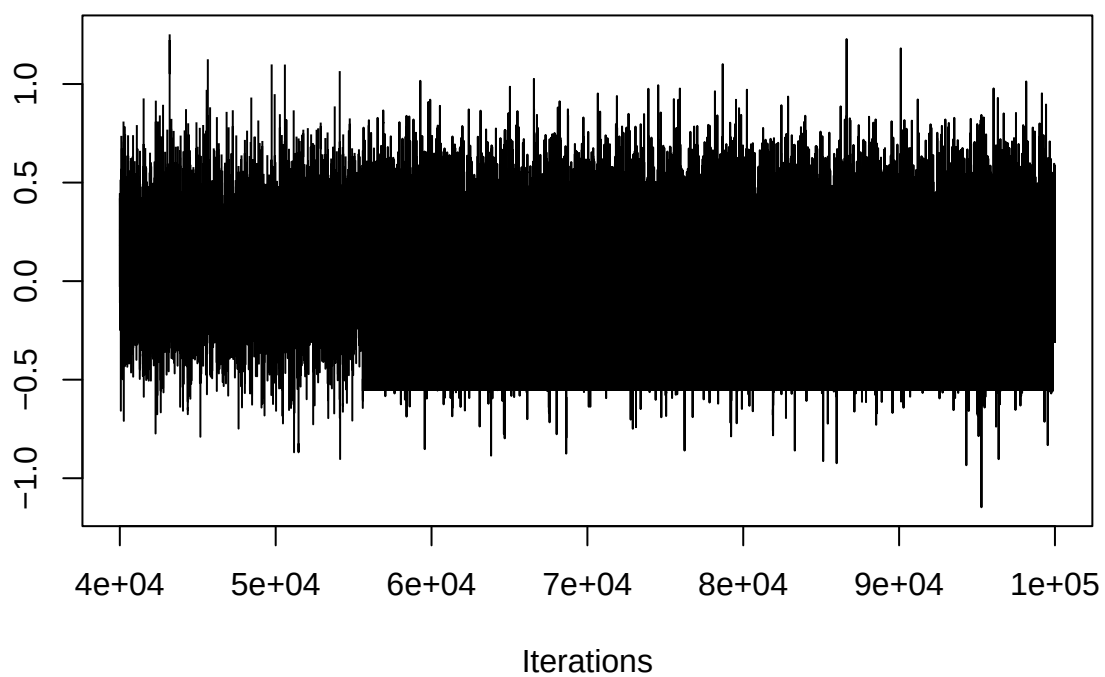
Trace of eta[5,4]



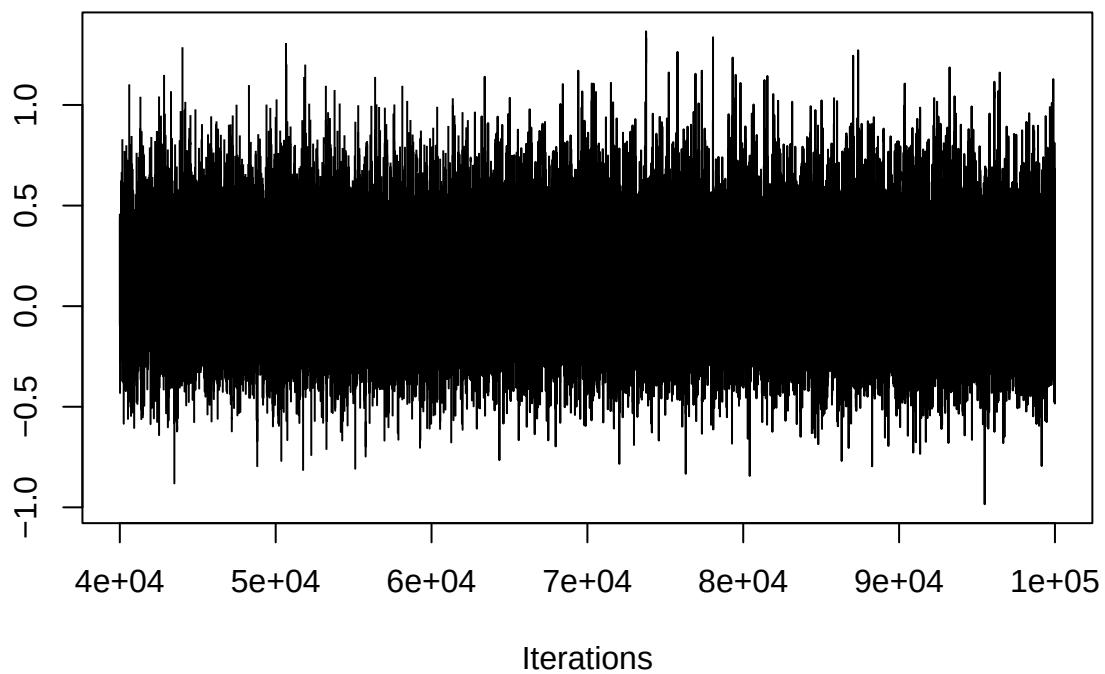
Trace of eta[1,5]



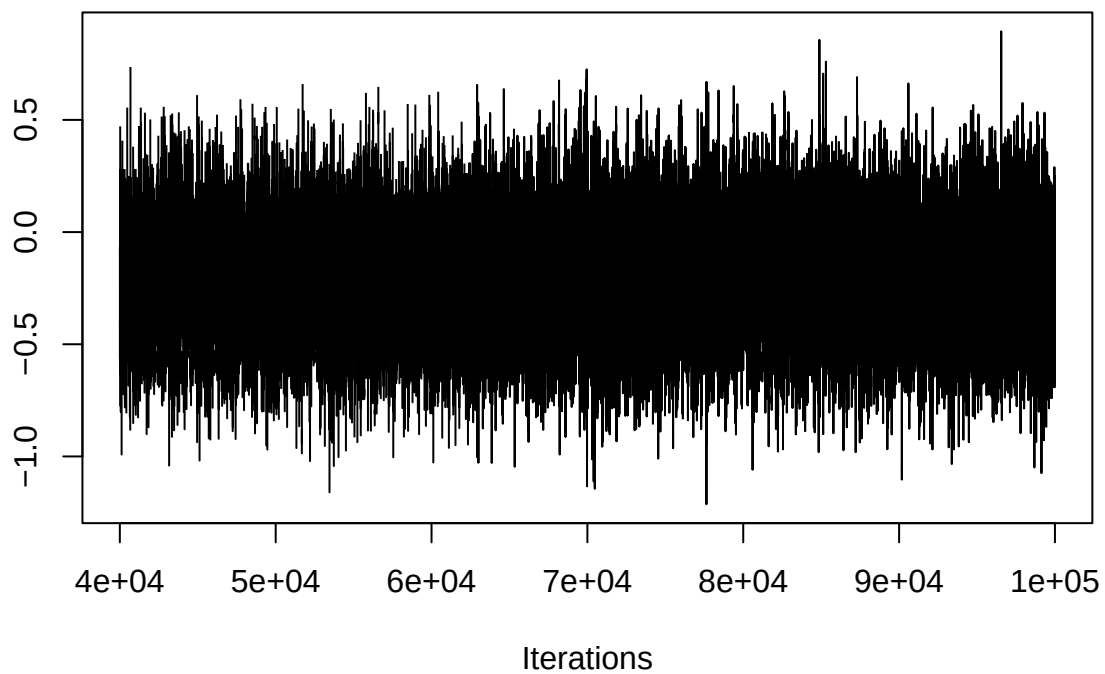
Trace of eta[2,5]



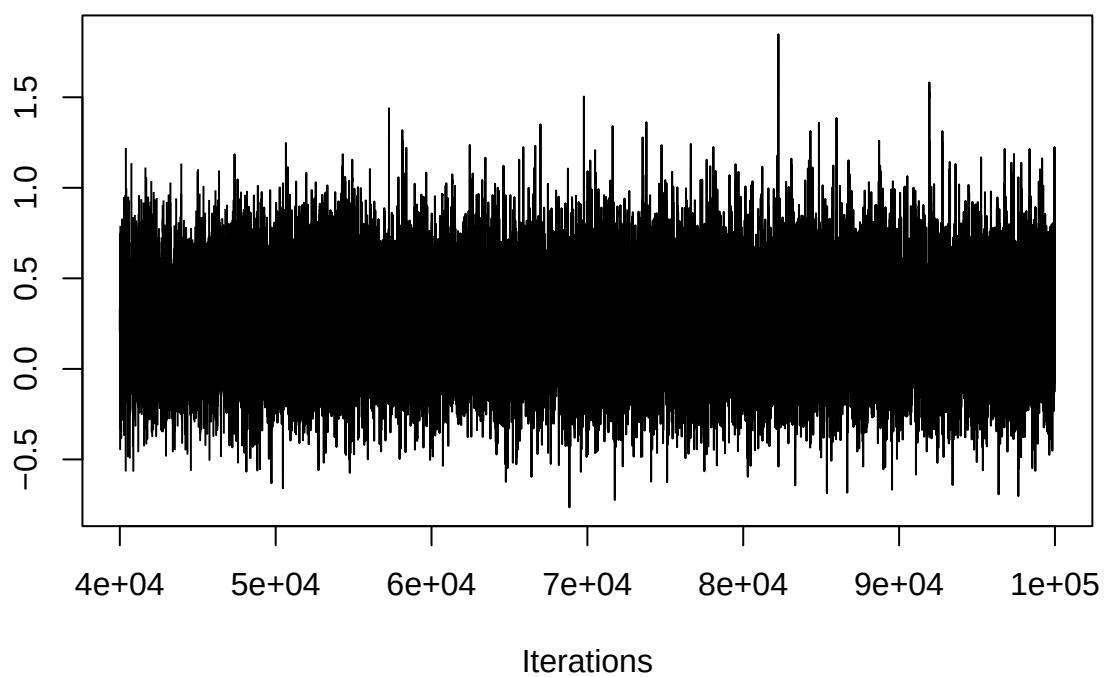
Trace of eta[3,5]



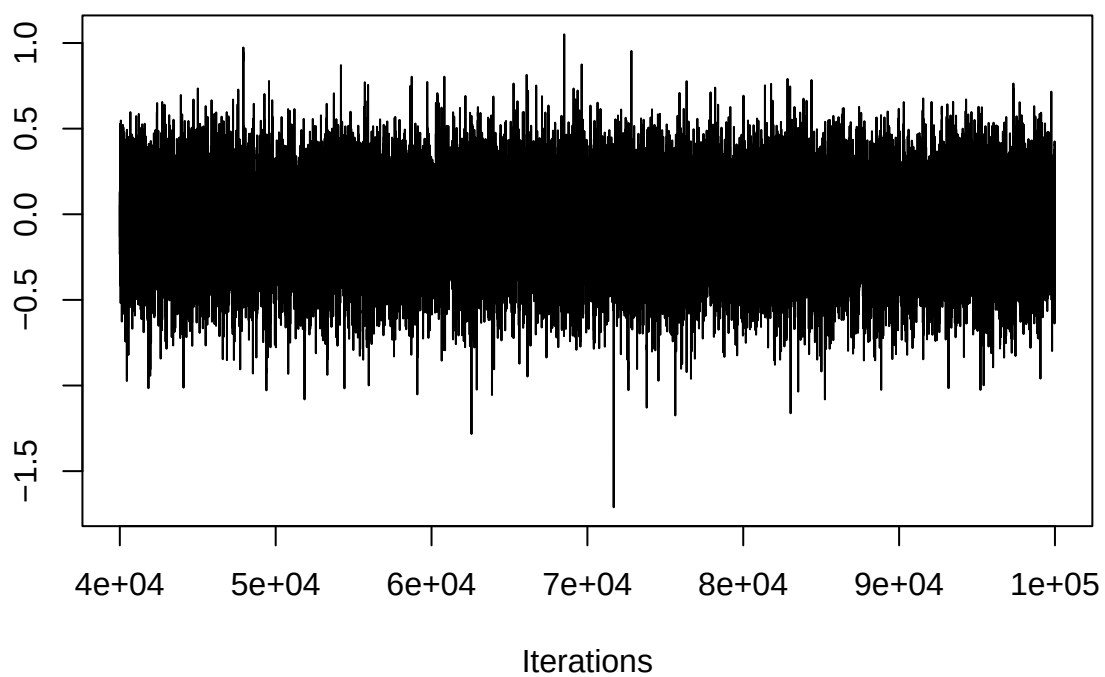
Trace of eta[4,5]



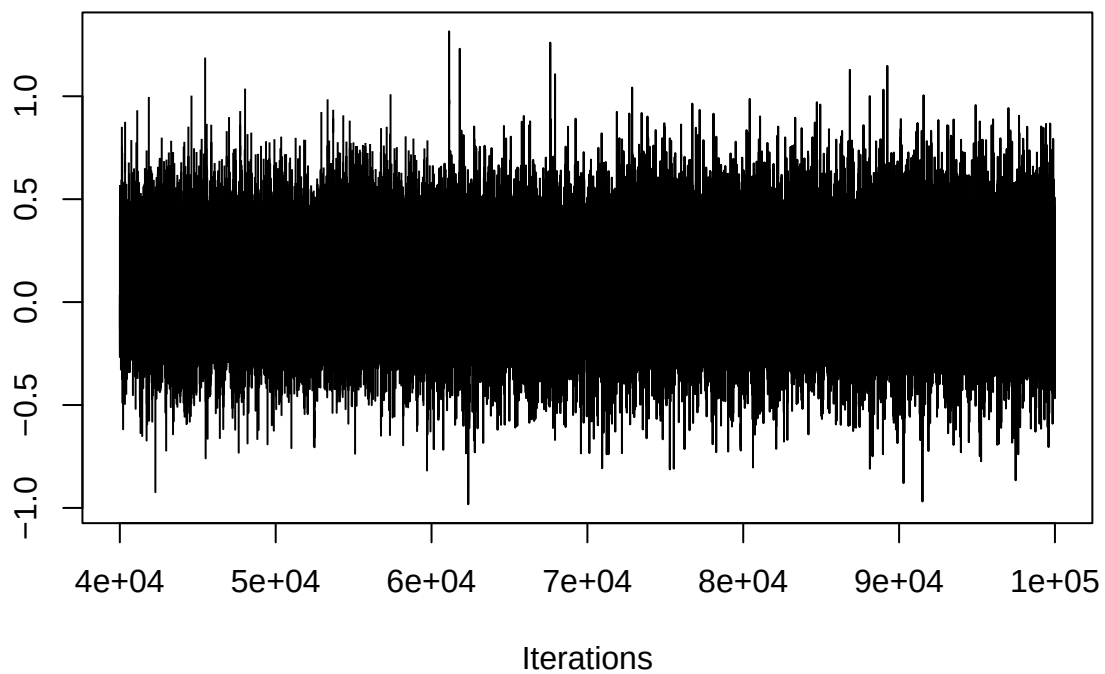
Trace of eta[5,5]



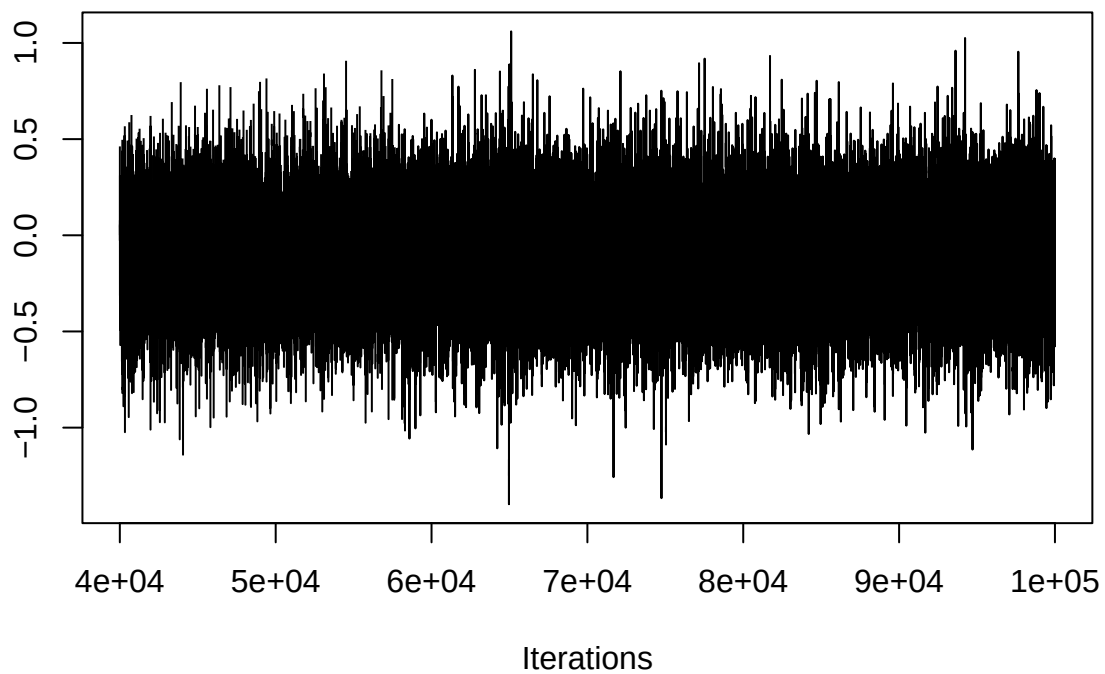
Trace of eta[1,6]



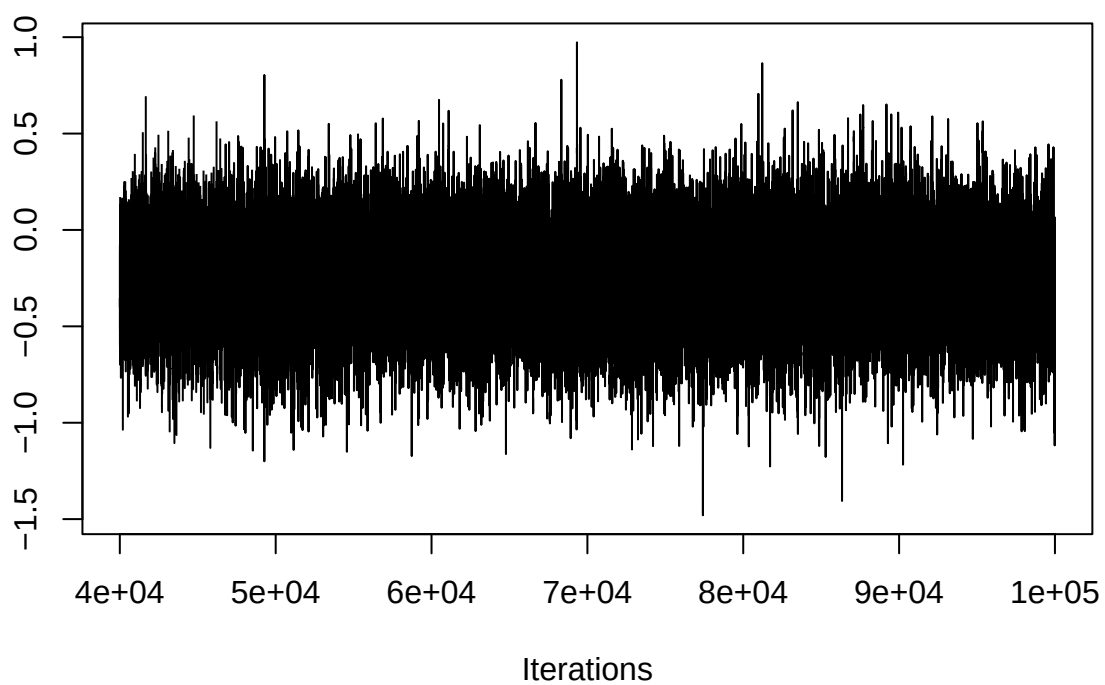
Trace of eta[2,6]



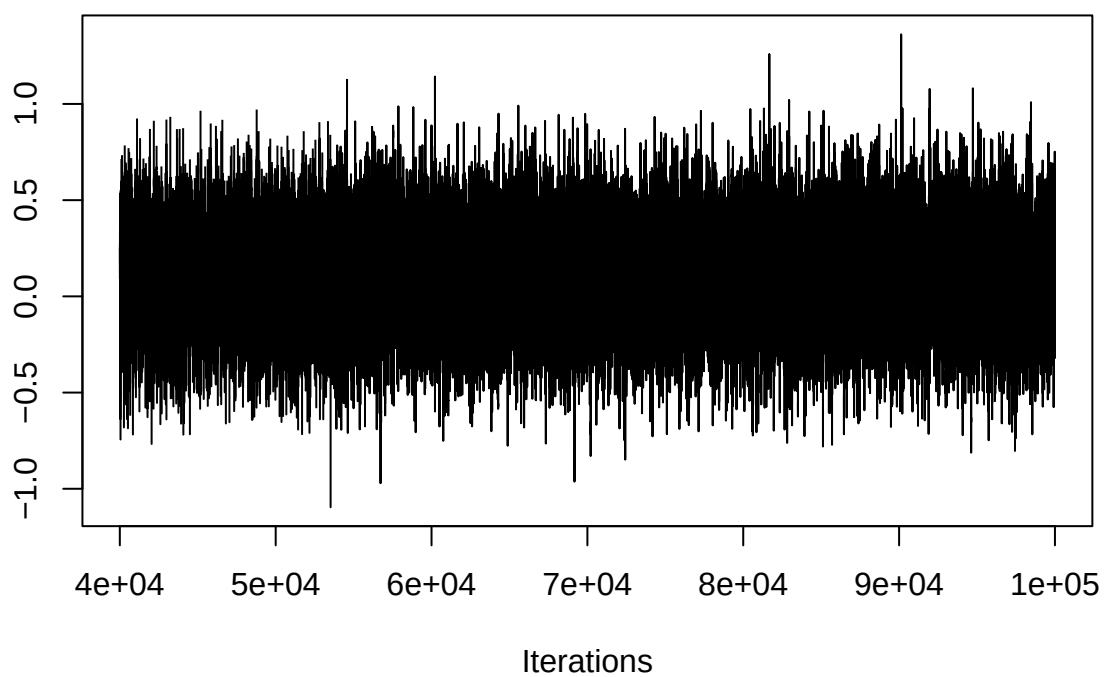
Trace of eta[3,6]



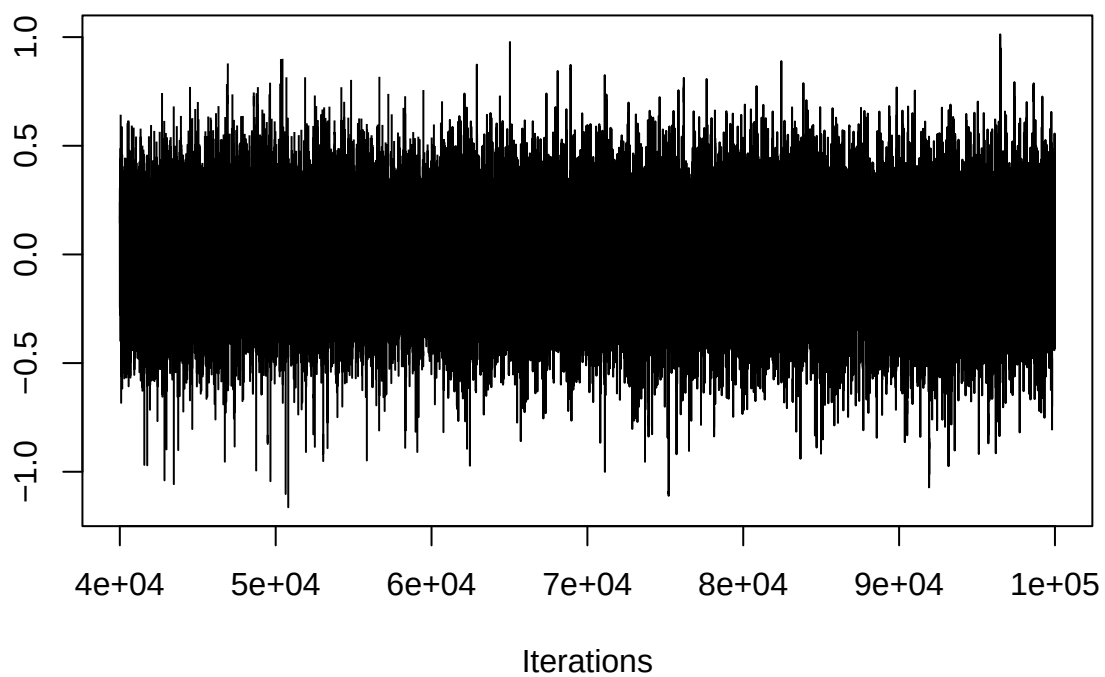
Trace of eta[4,6]



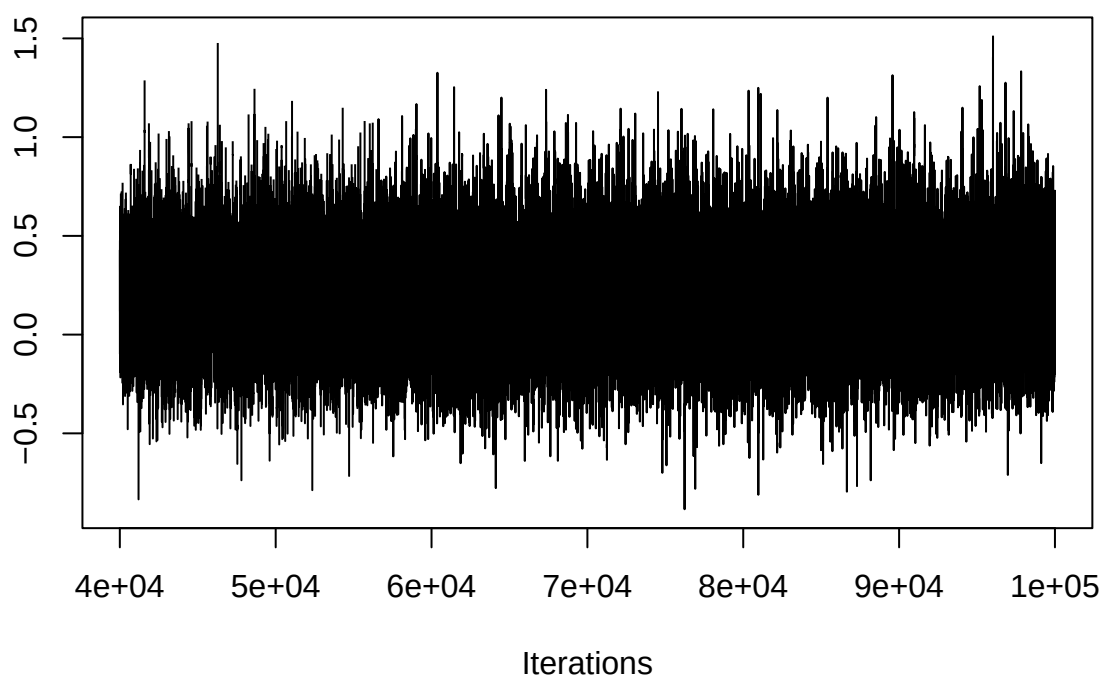
Trace of eta[5,6]



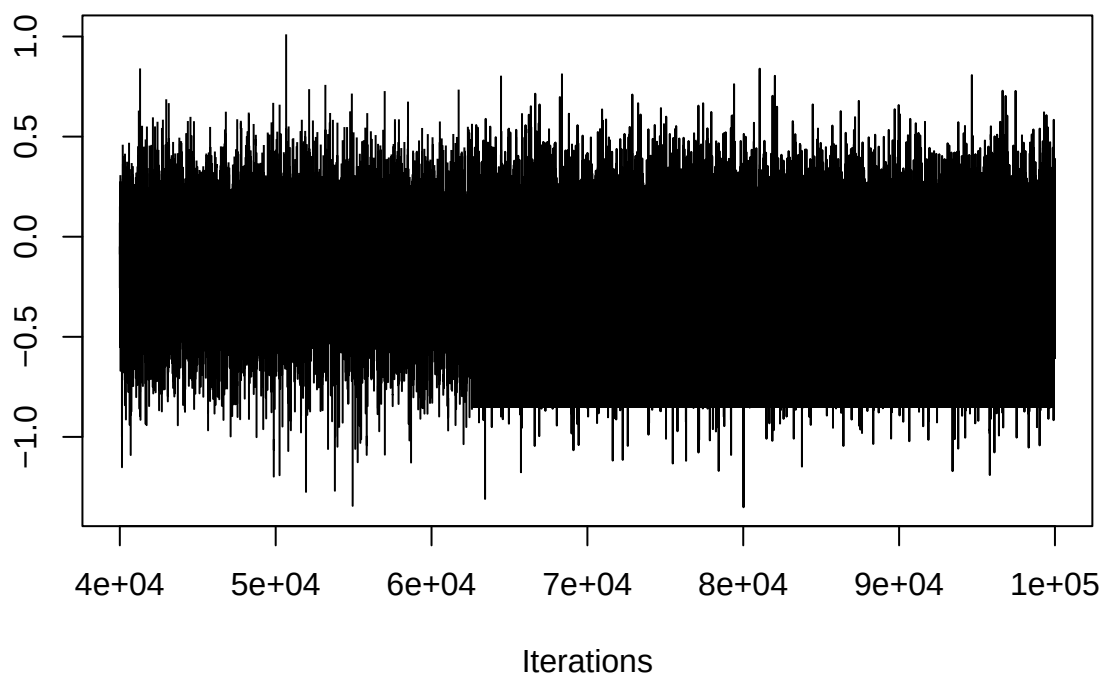
Trace of eta[1,7]



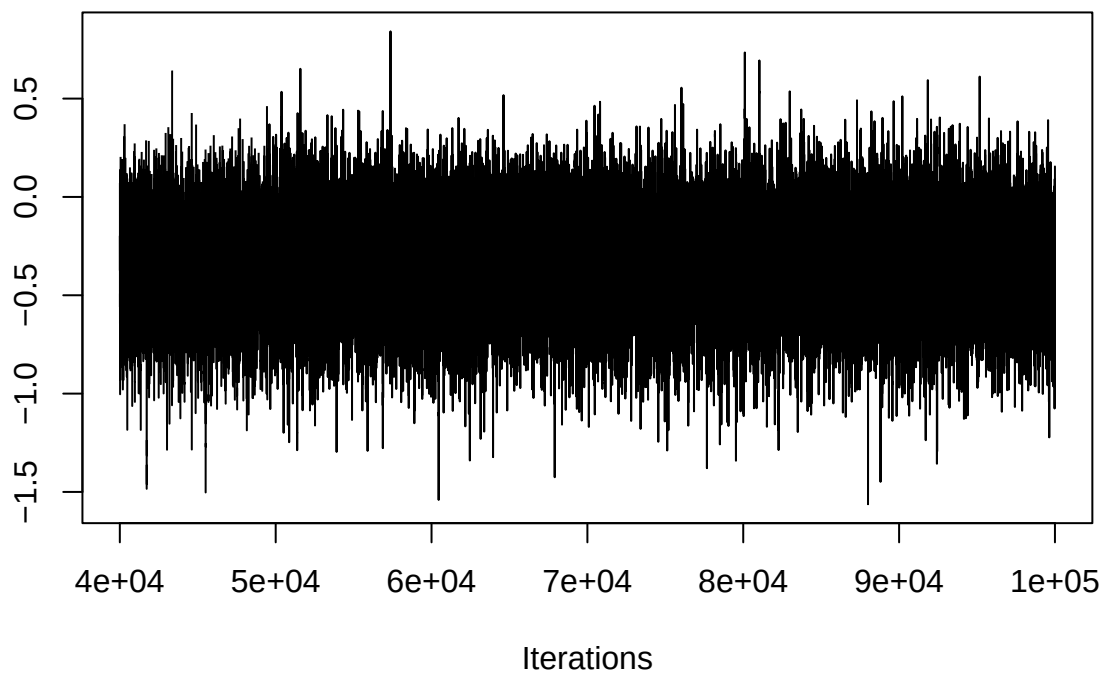
Trace of eta[2,7]



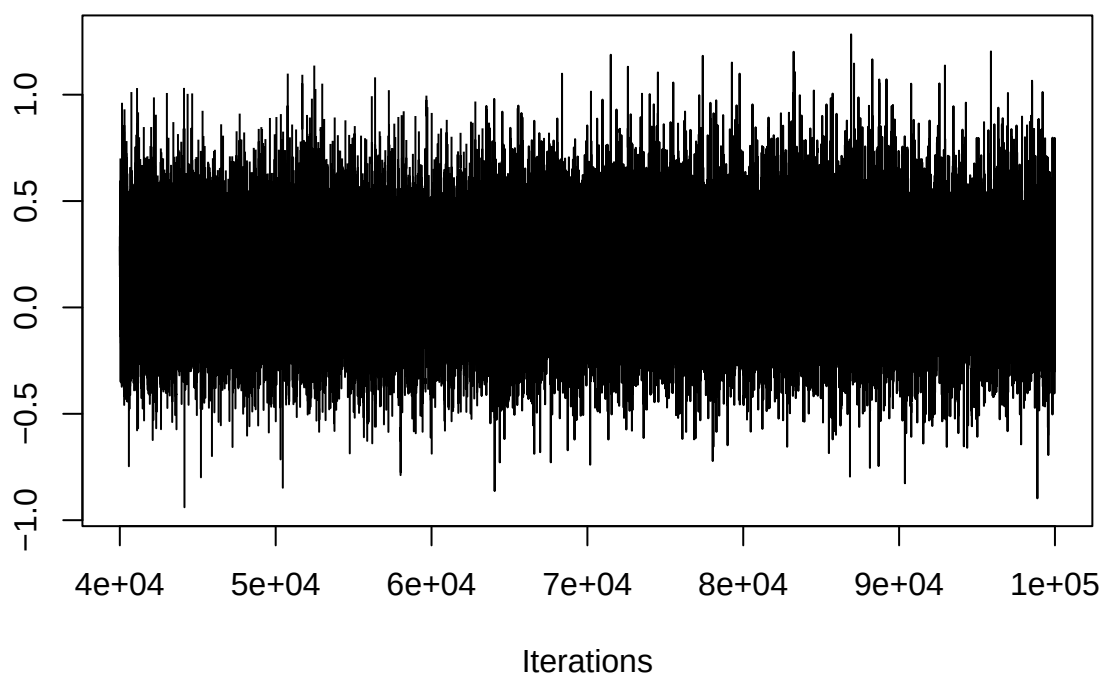
Trace of eta[3,7]



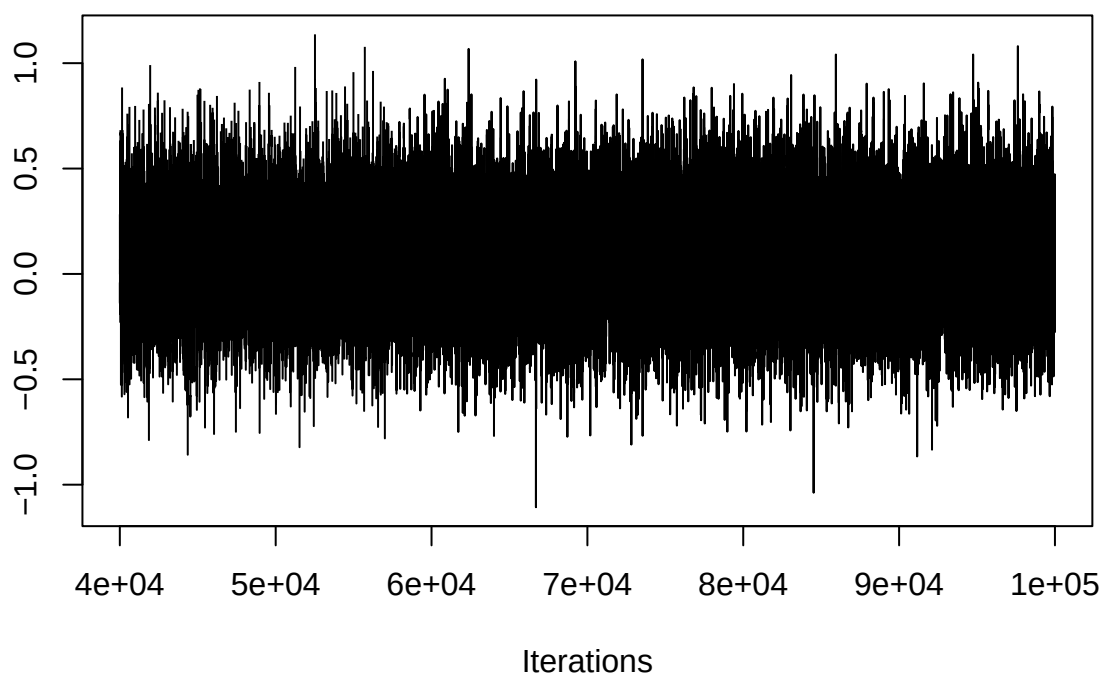
Trace of eta[4,7]



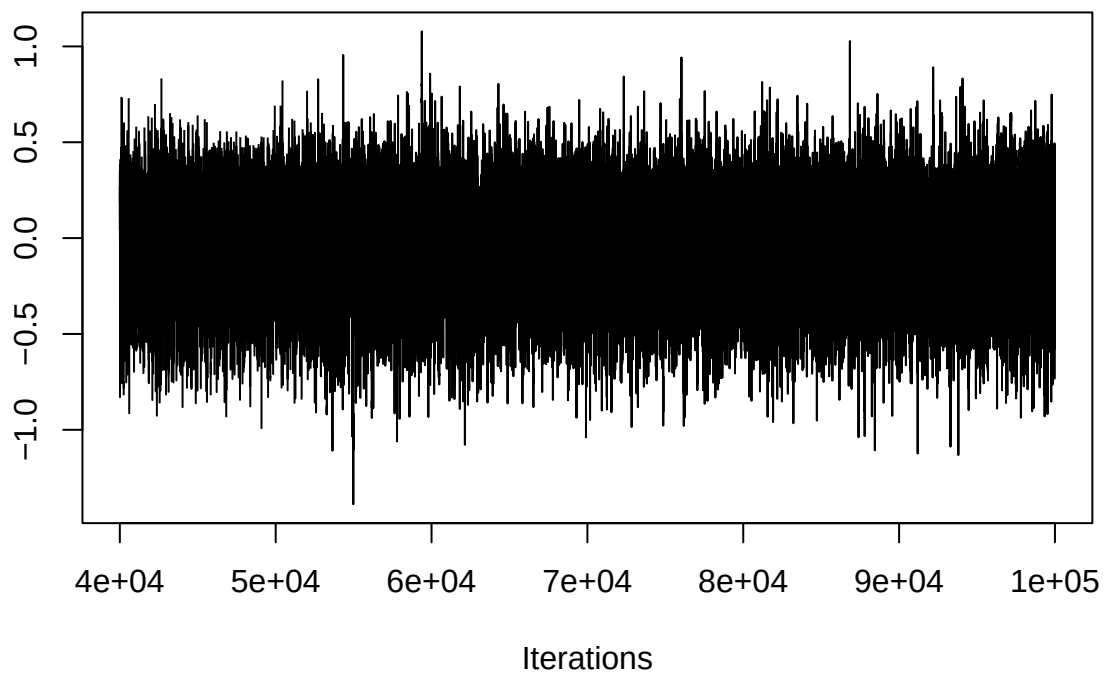
Trace of eta[5,7]



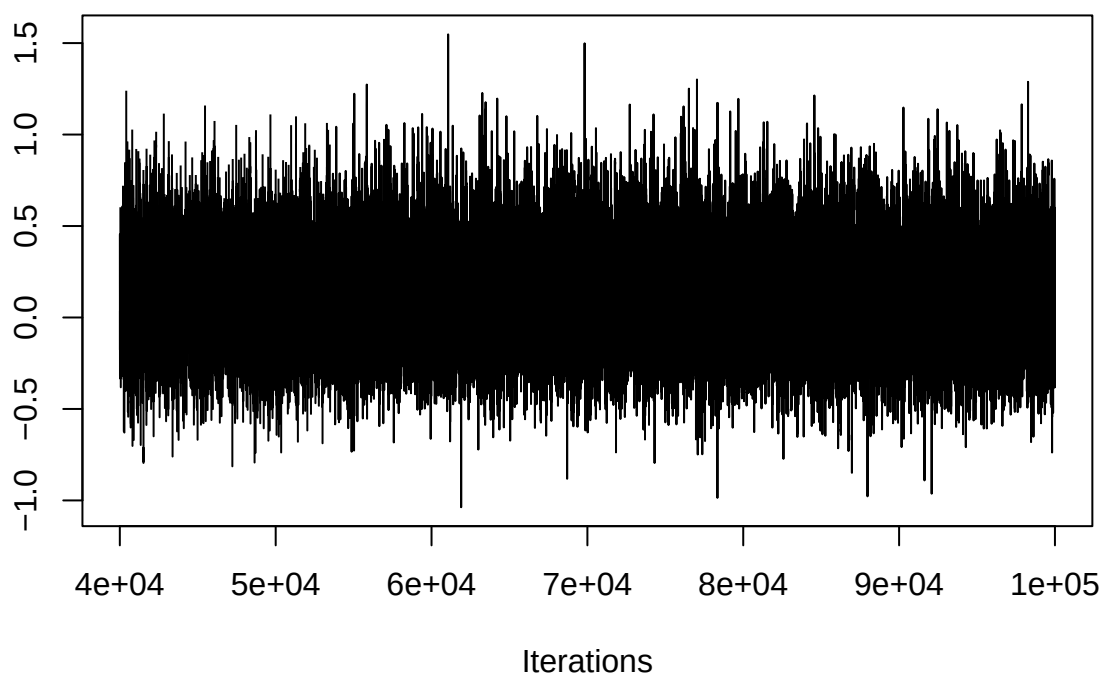
Trace of eta[1,8]



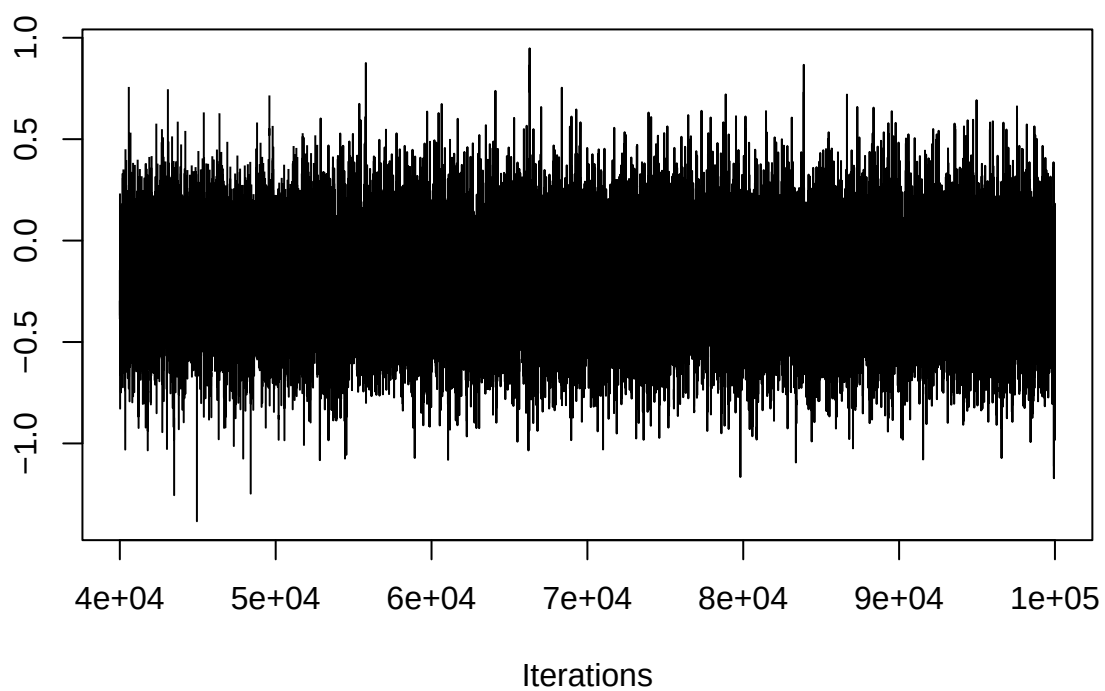
Trace of eta[2,8]



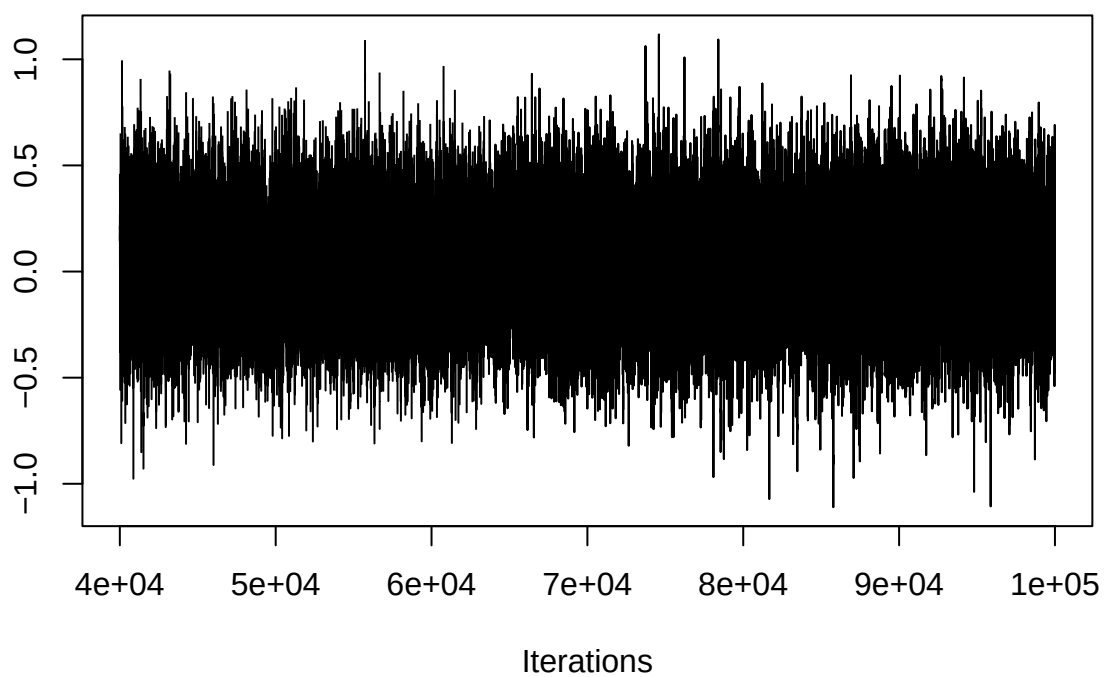
Trace of eta[3,8]



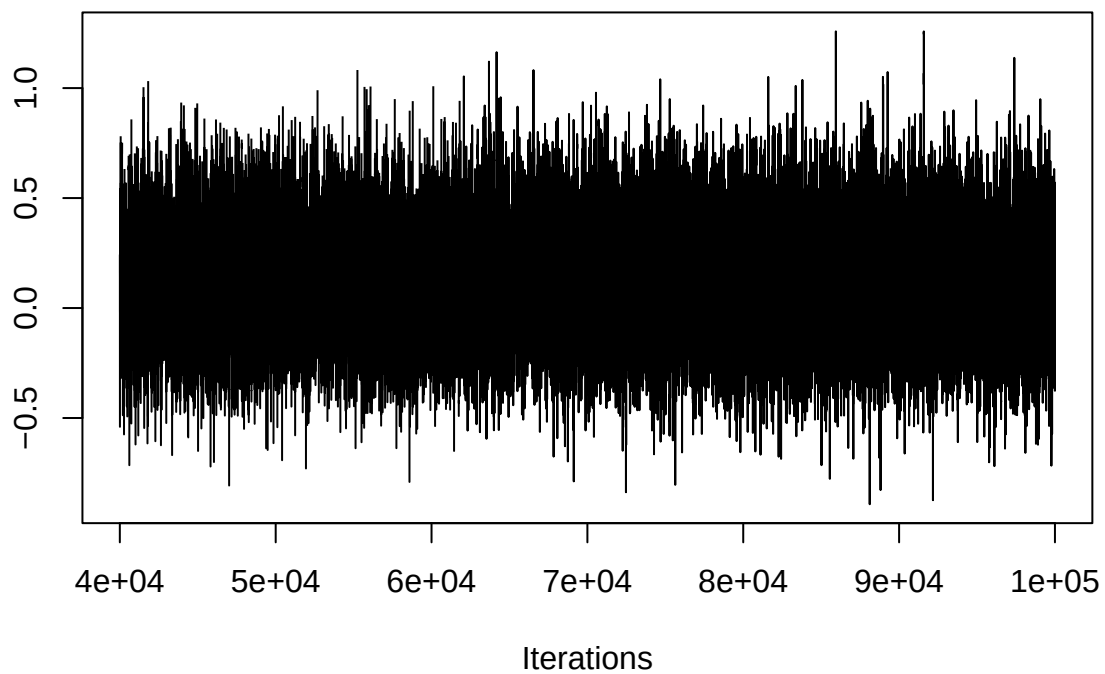
Trace of eta[4,8]



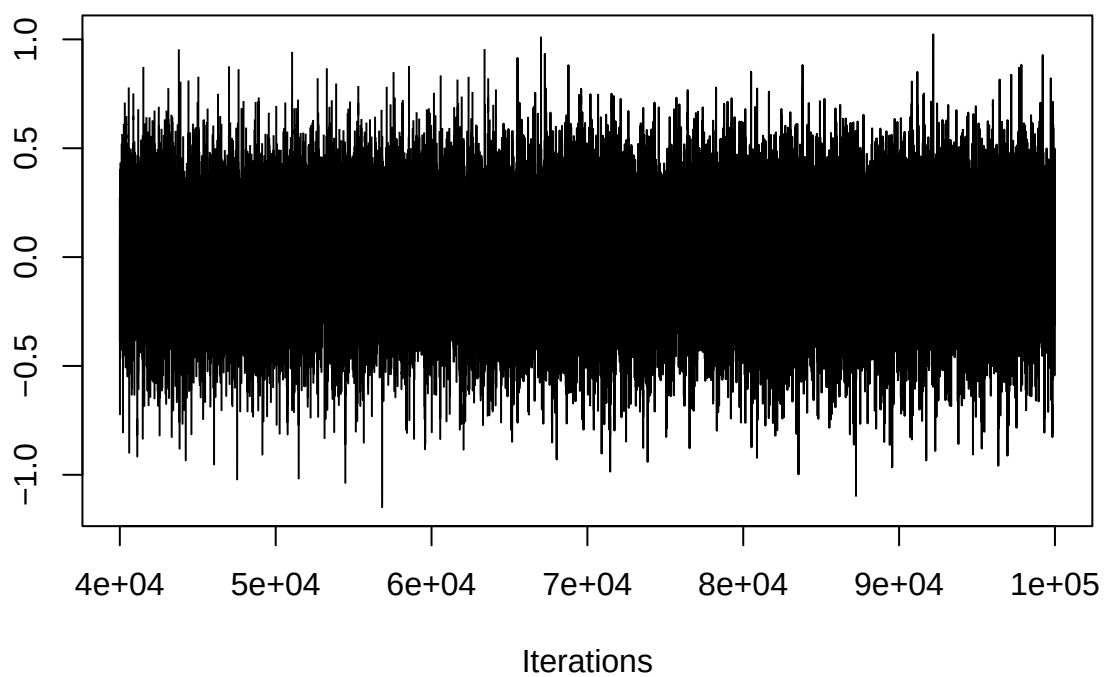
Trace of eta[5,8]



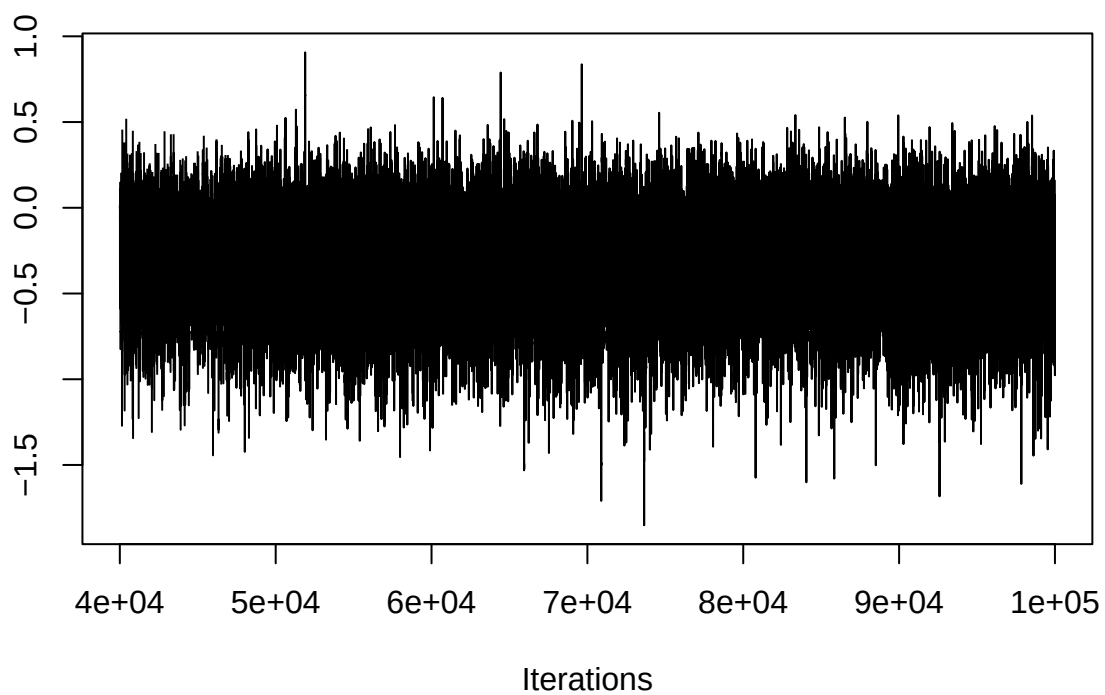
Trace of eta[1,9]



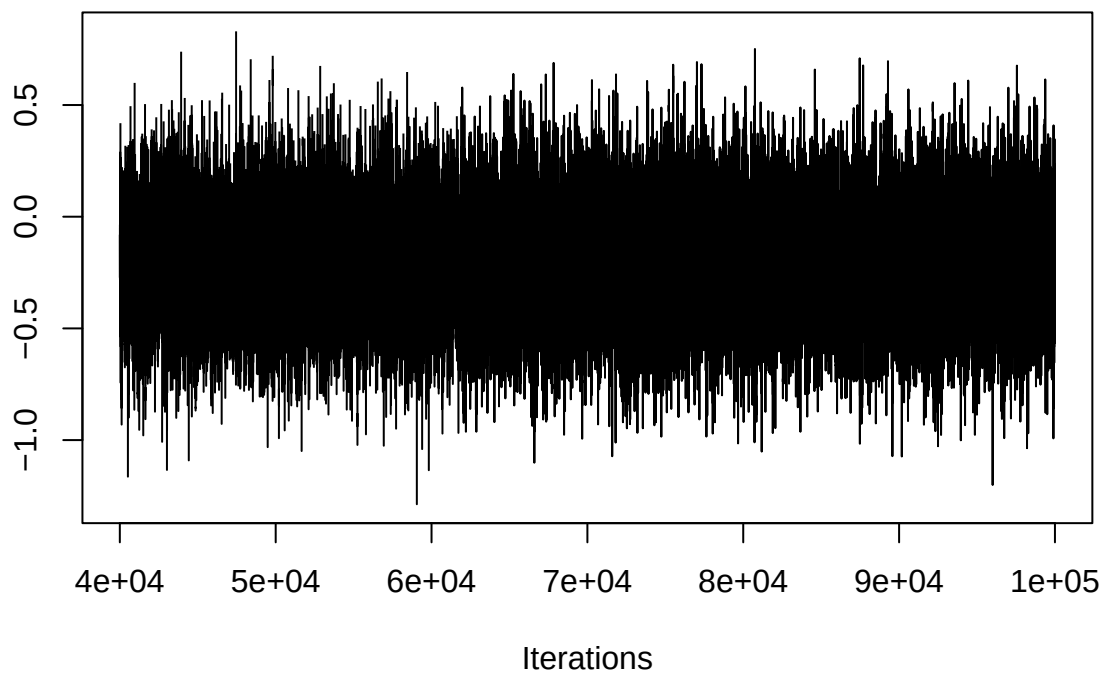
Trace of eta[2,9]



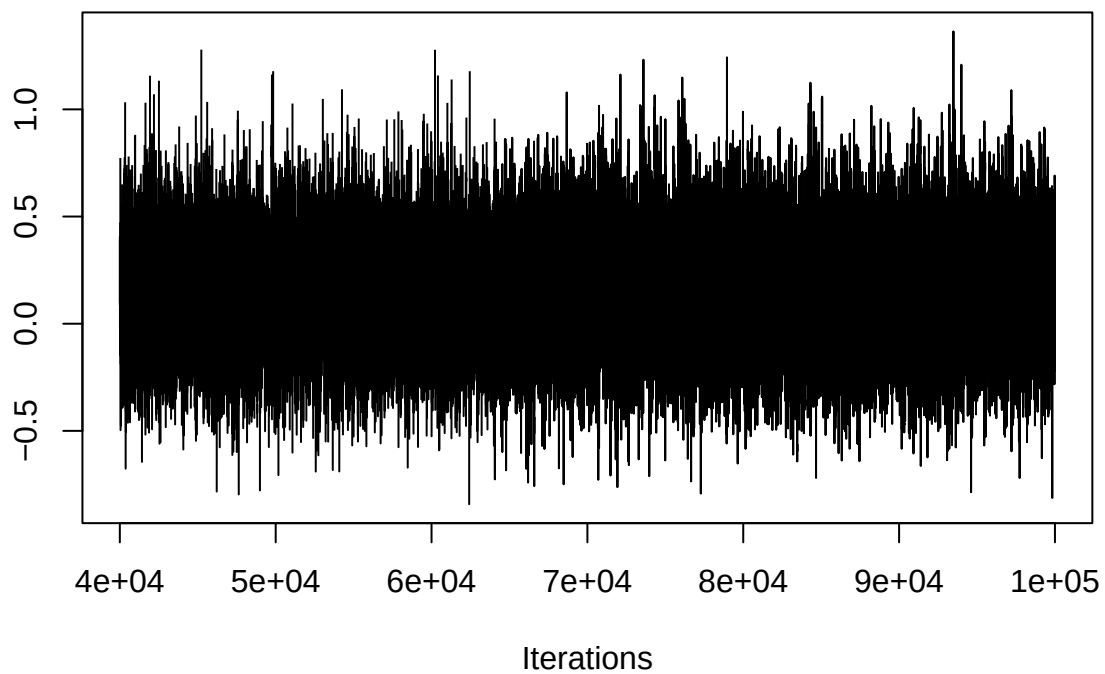
Trace of eta[3,9]



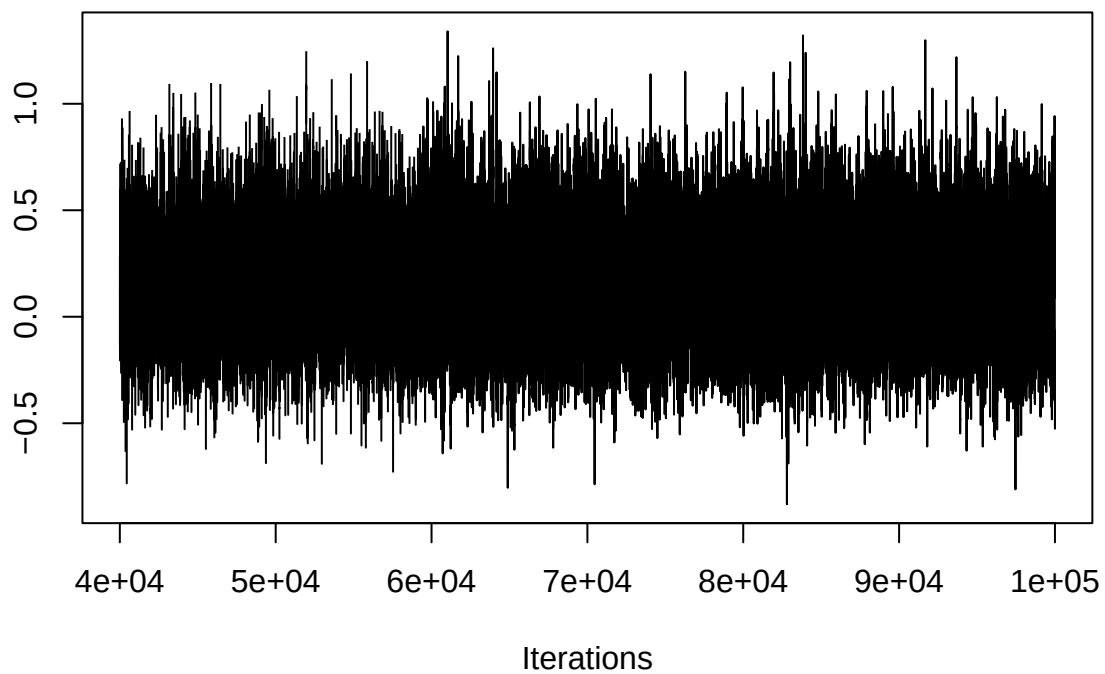
Trace of eta[4,9]



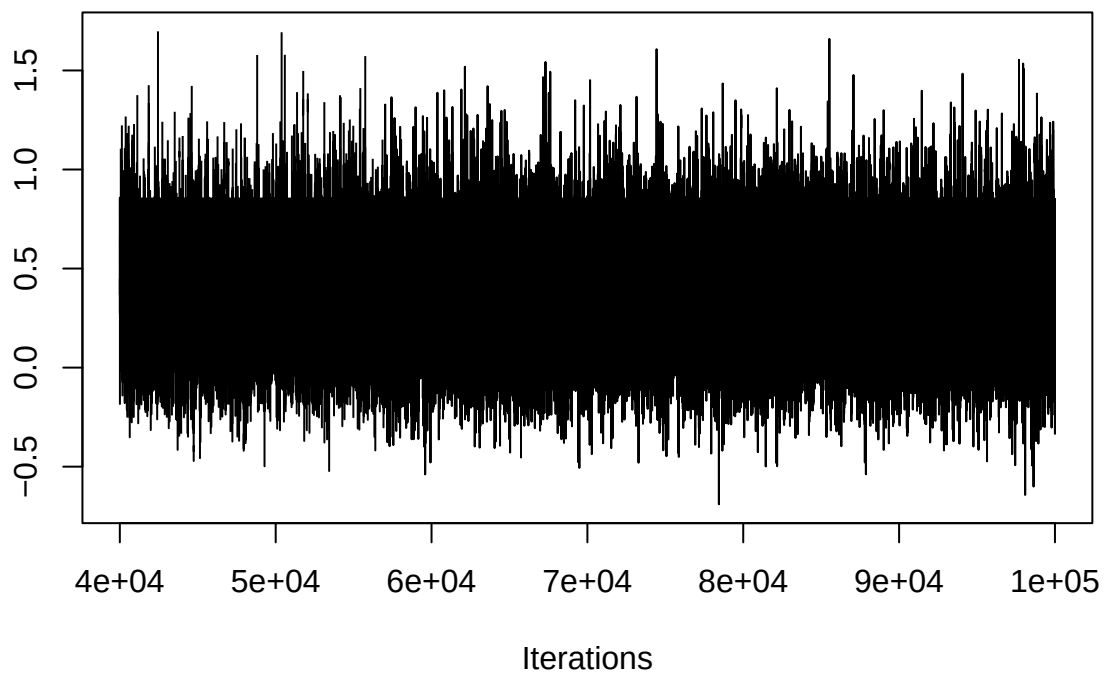
Trace of eta[5,9]



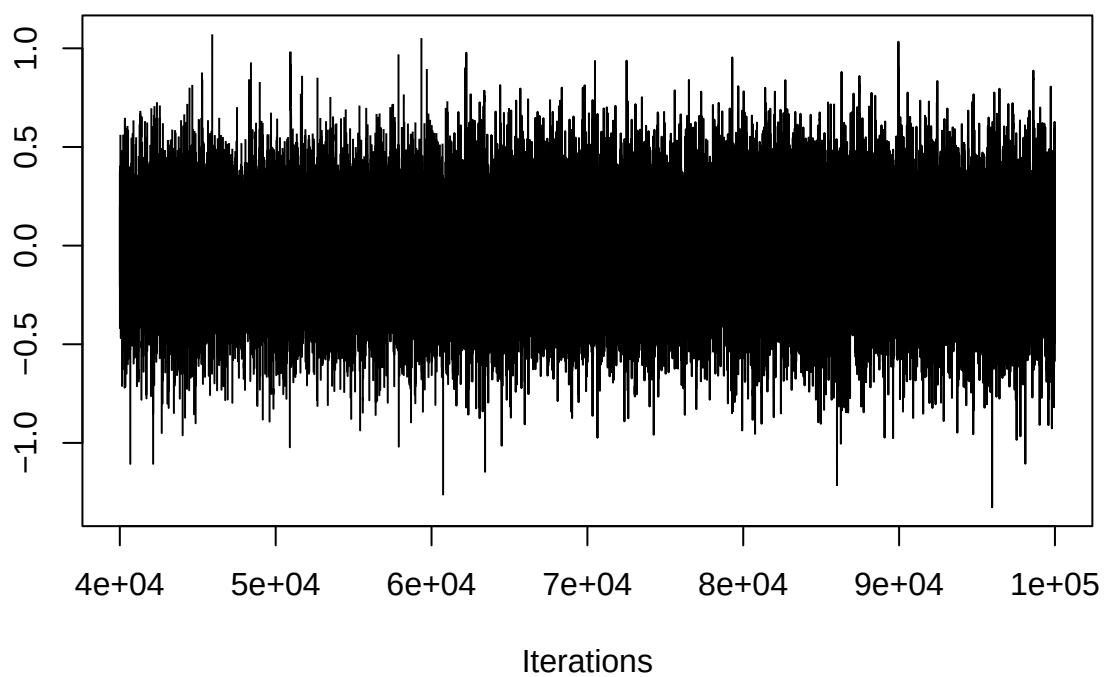
Trace of eta[1,10]



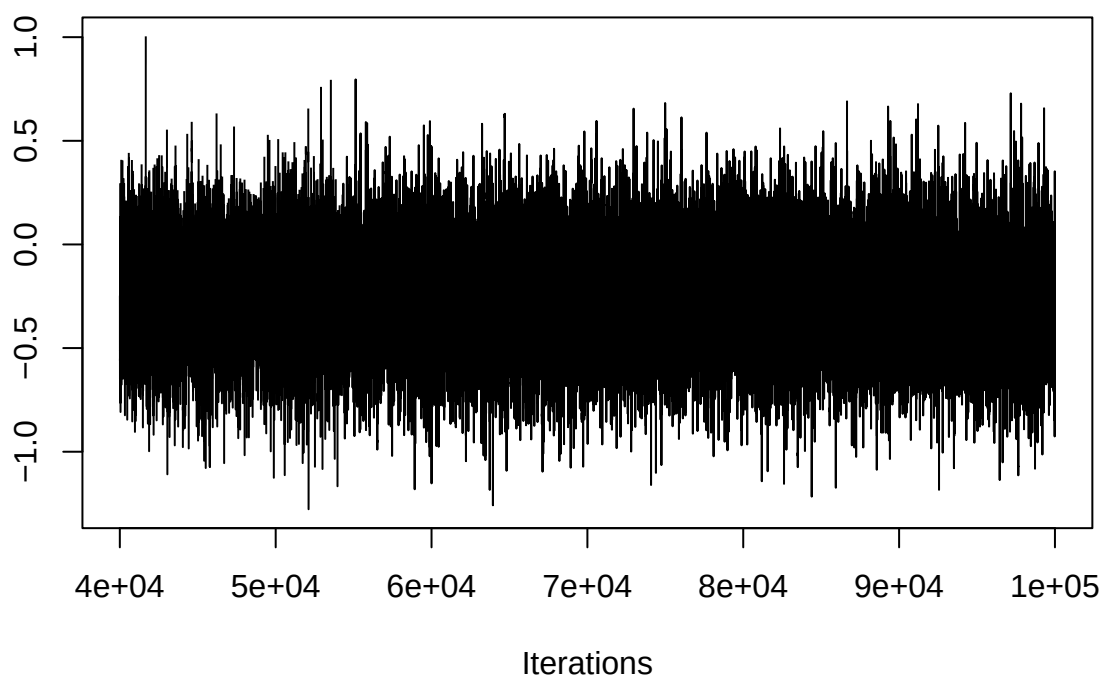
Trace of eta[2,10]



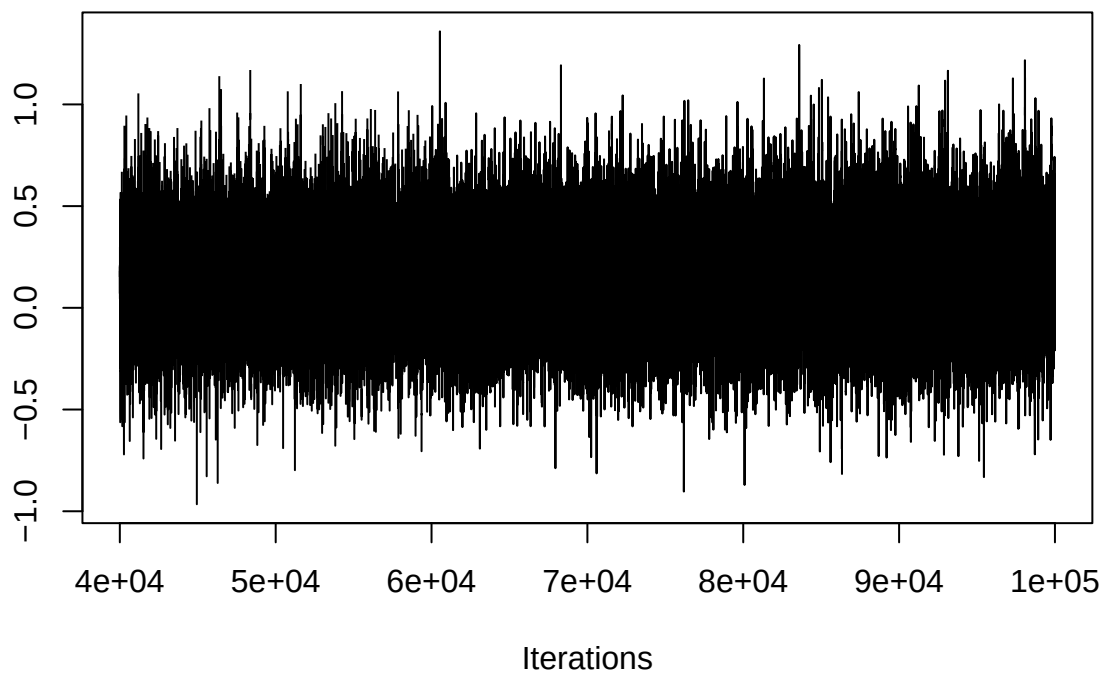
Trace of eta[3,10]



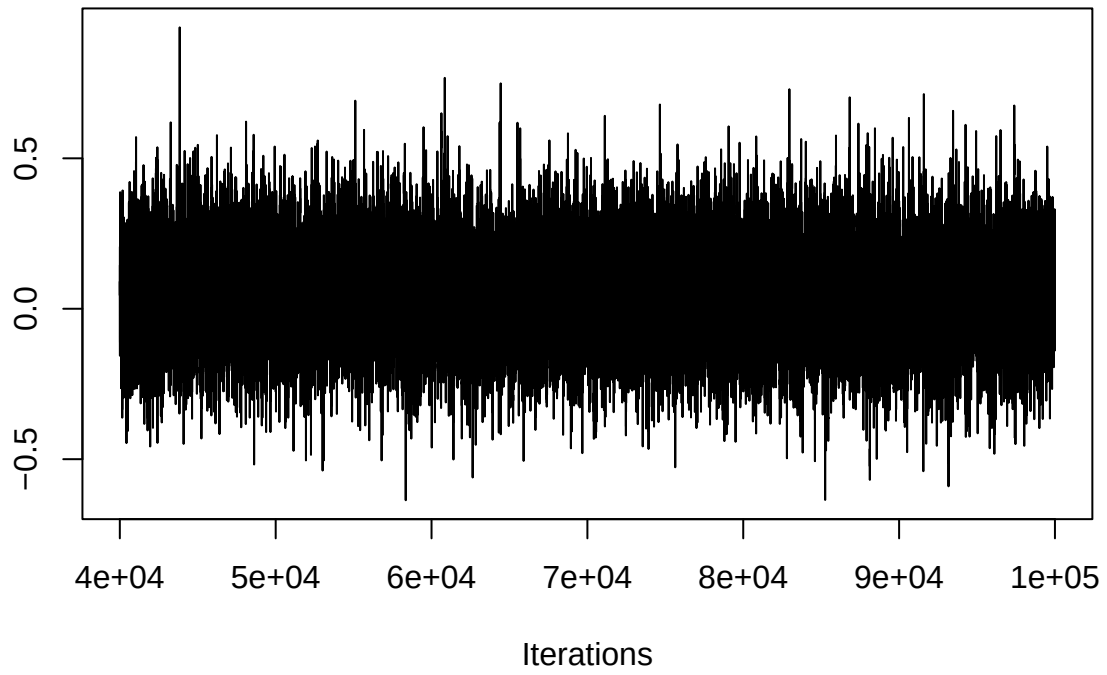
Trace of eta[4,10]



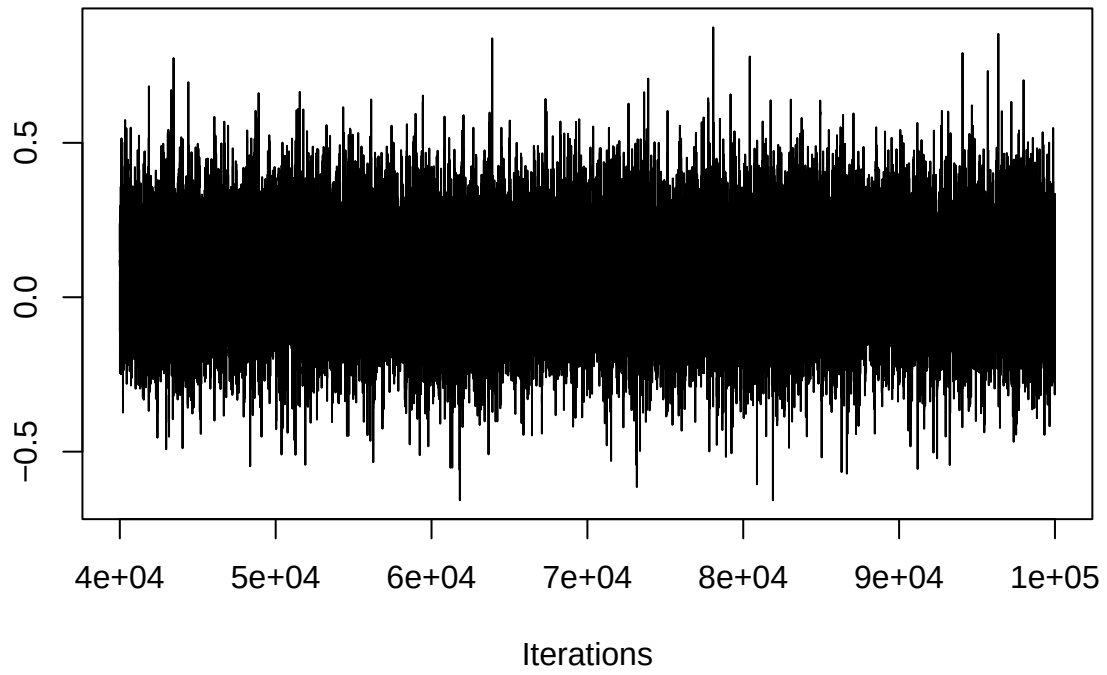
Trace of eta[5,10]



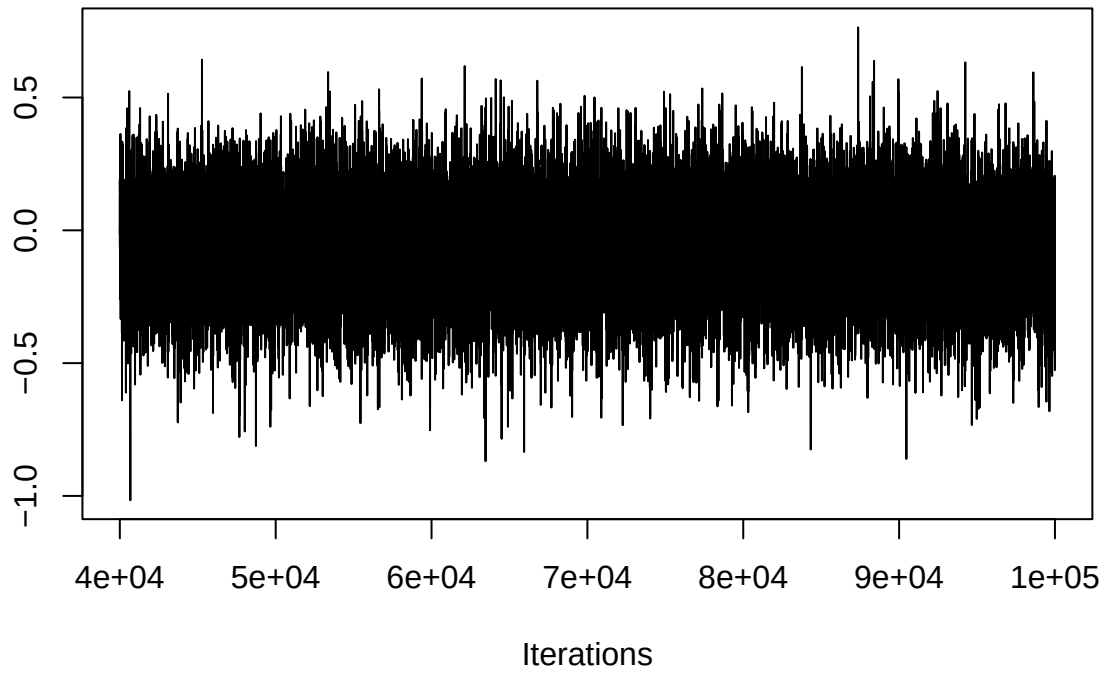
Trace of mu[1]



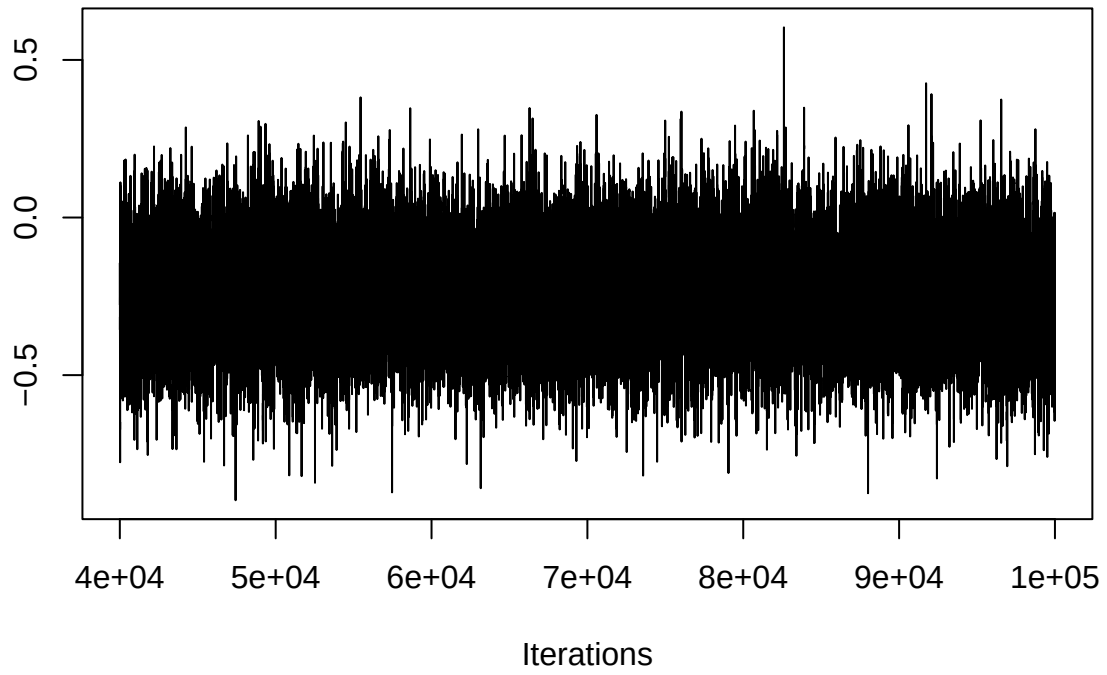
Trace of mu[2]



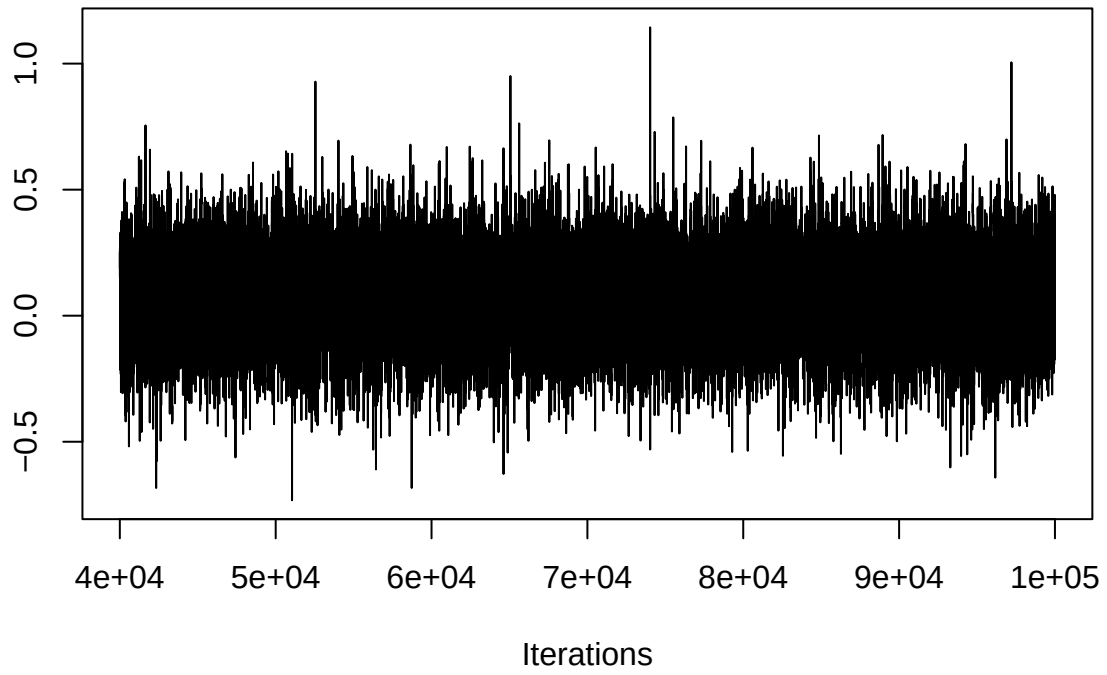
Trace of mu[3]

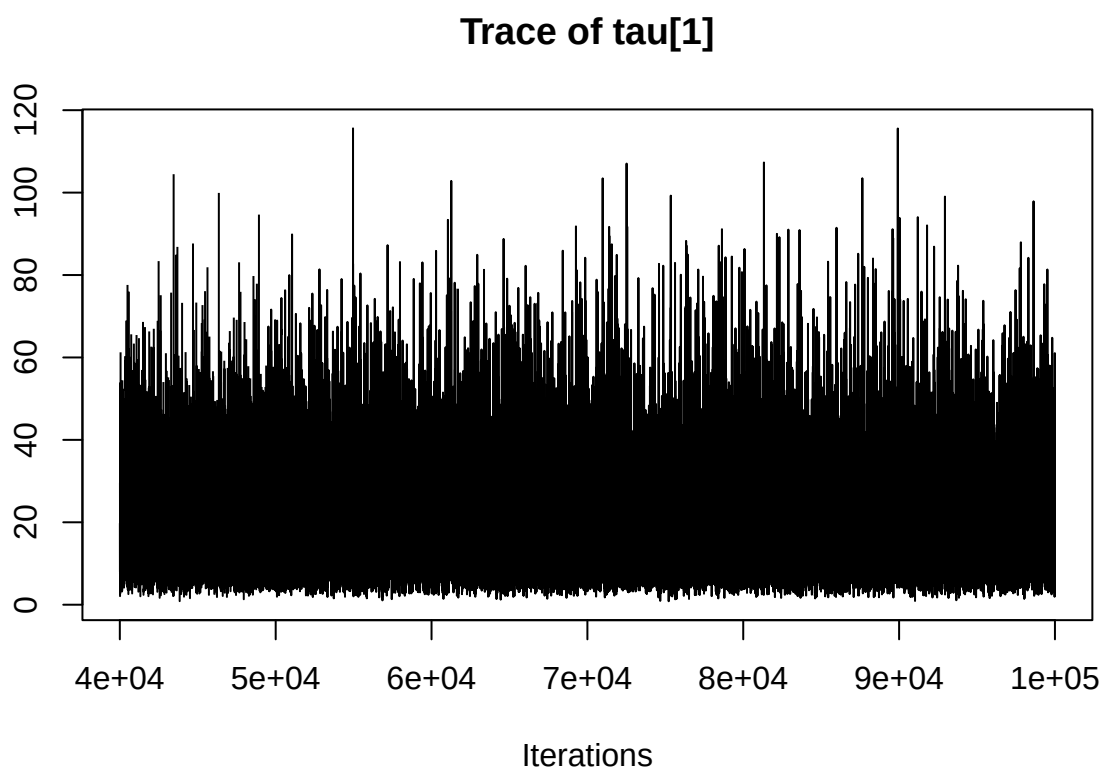


Trace of mu[4]

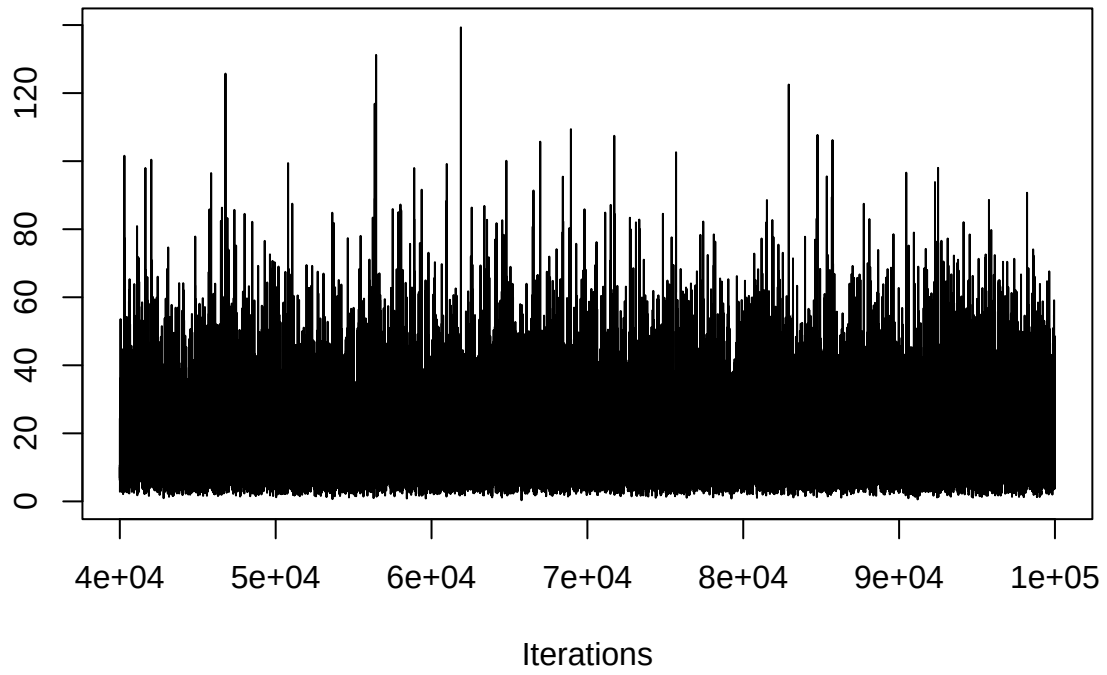


Trace of mu[5]

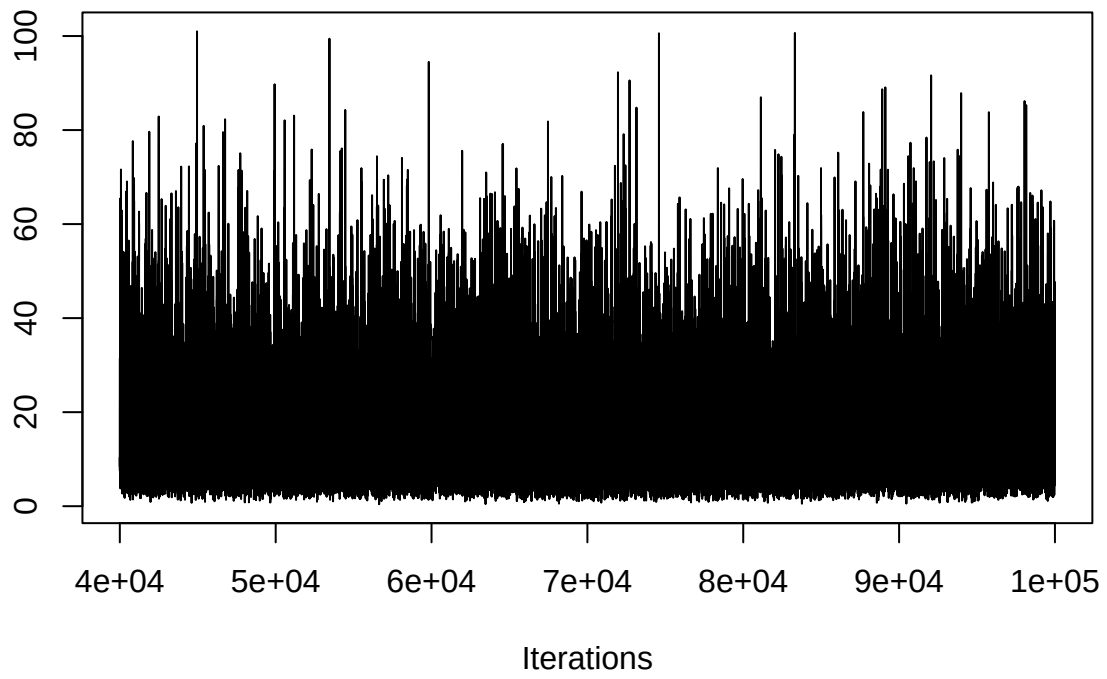




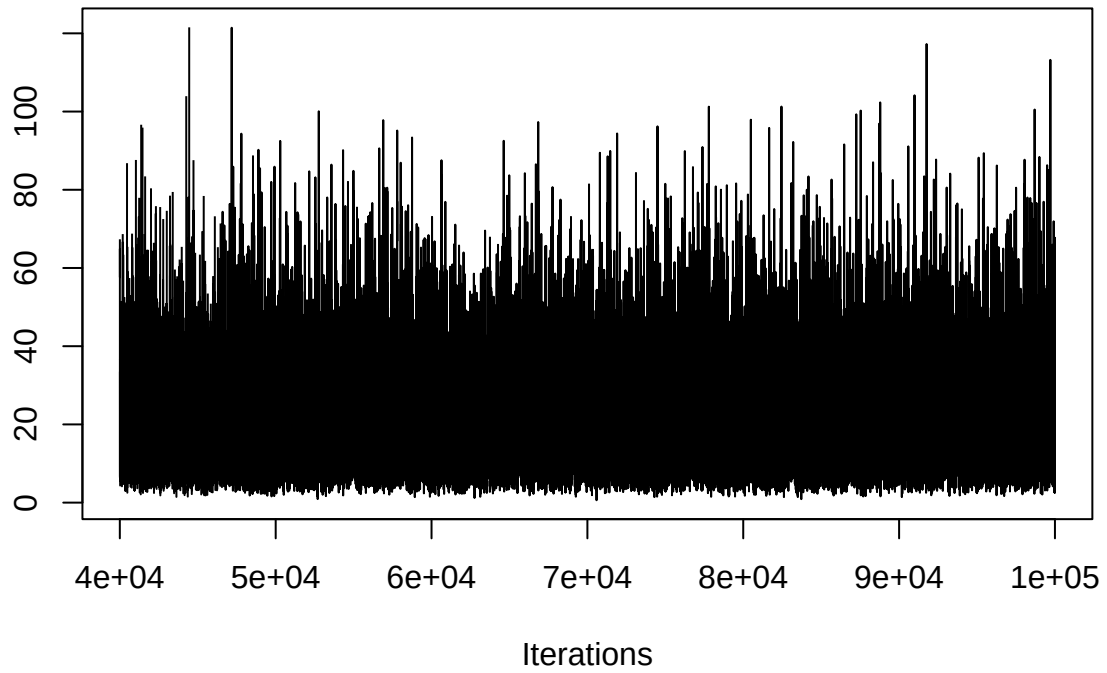
Trace of tau[2]

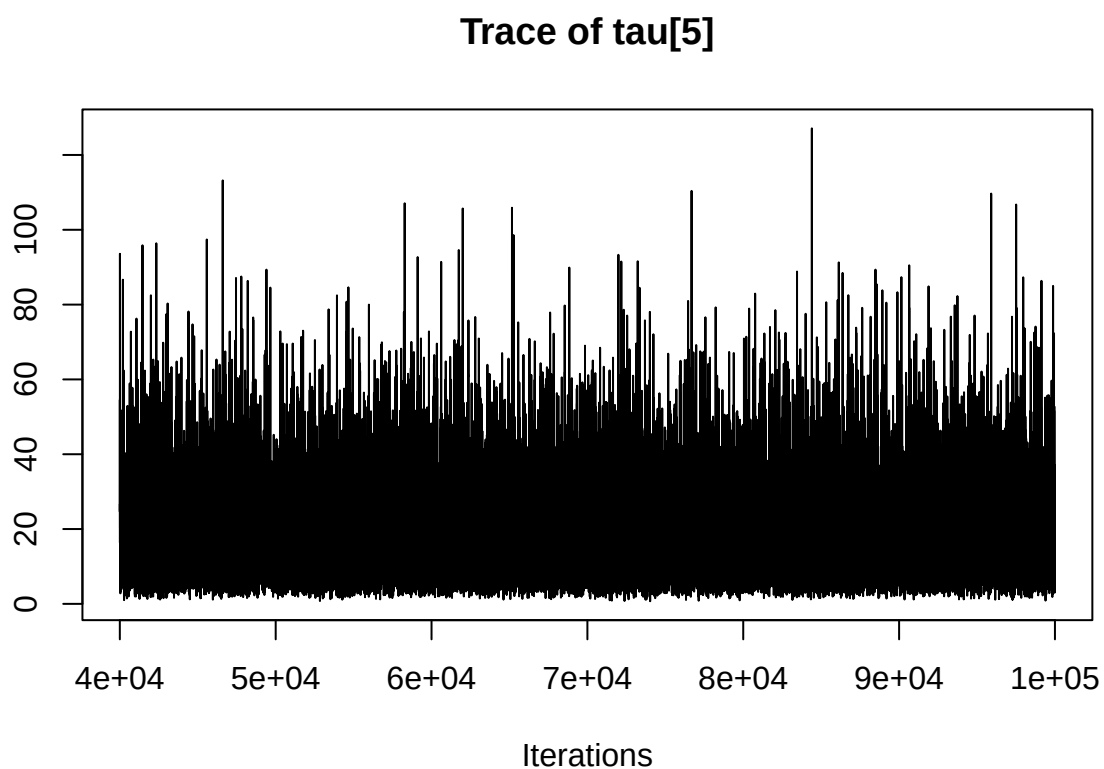


Trace of tau[3]



Trace of tau[4]

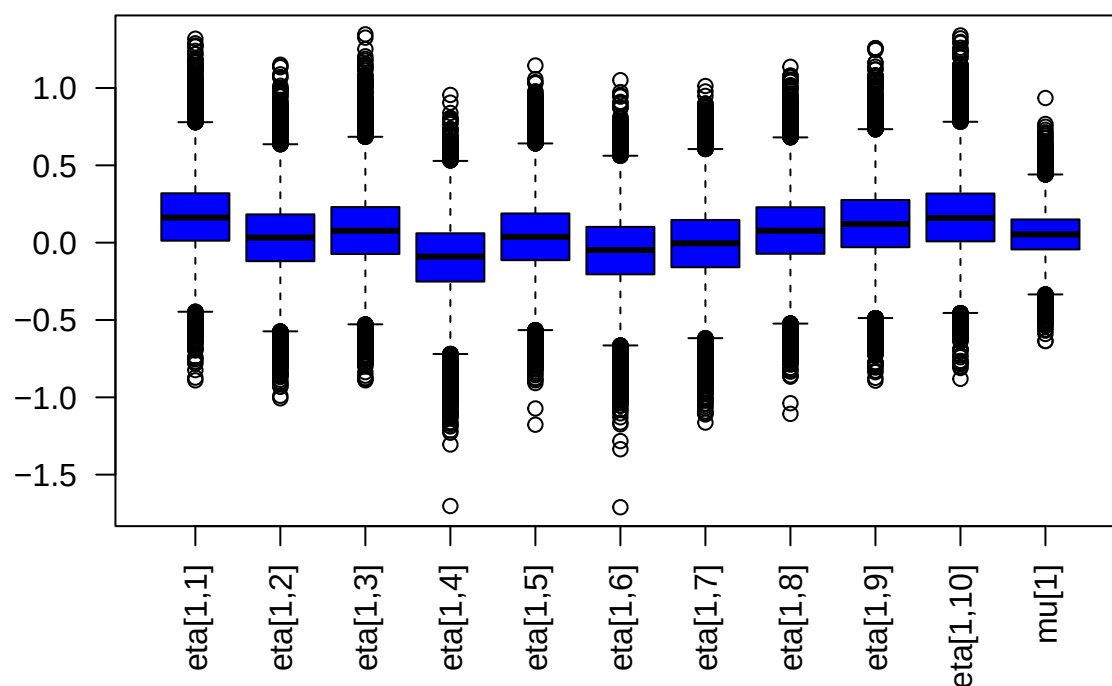




1.4 - Posterior boxplots of all sectors and companies stocks.

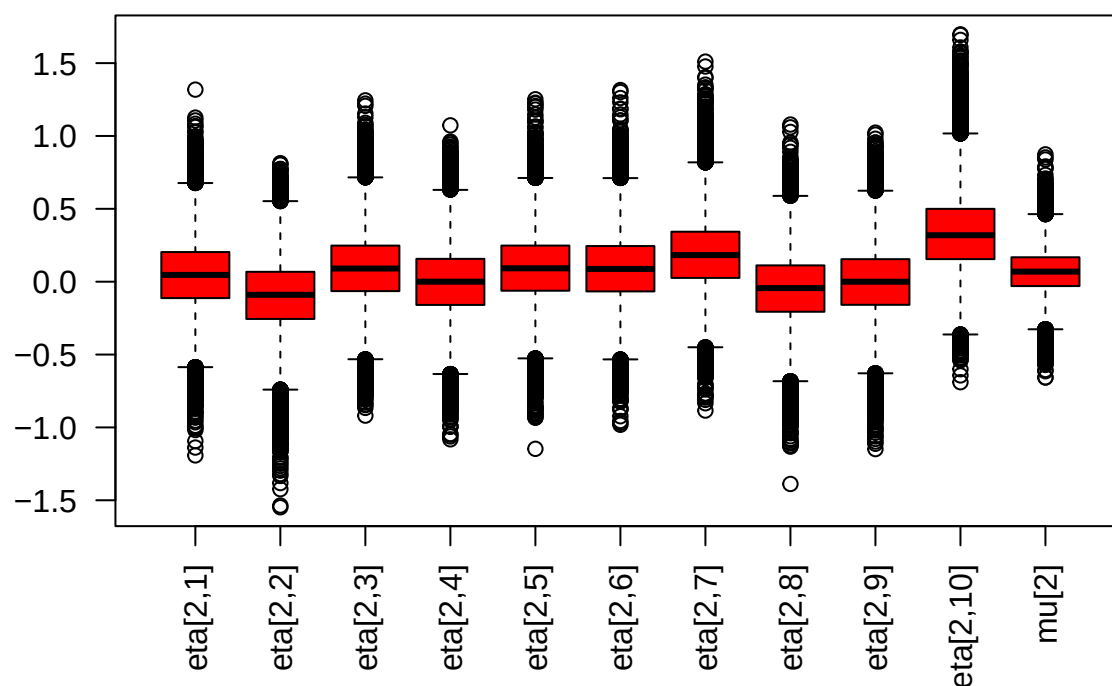
```
#Looking at each sector seperately
sector1_matrix <- as.matrix(fit.samples)
boxplot(sector1_matrix[, c("eta[1,1]", "eta[1,2]", "eta[1,3]", "eta[1,4]", "eta[1,5]", "
    main = "Posterior Boxplots for sector 1 and mu",
    col = "blue",
    las = 2) # Rotates axis labels for readability
```

Posterior Boxplots for sector 1 and mu



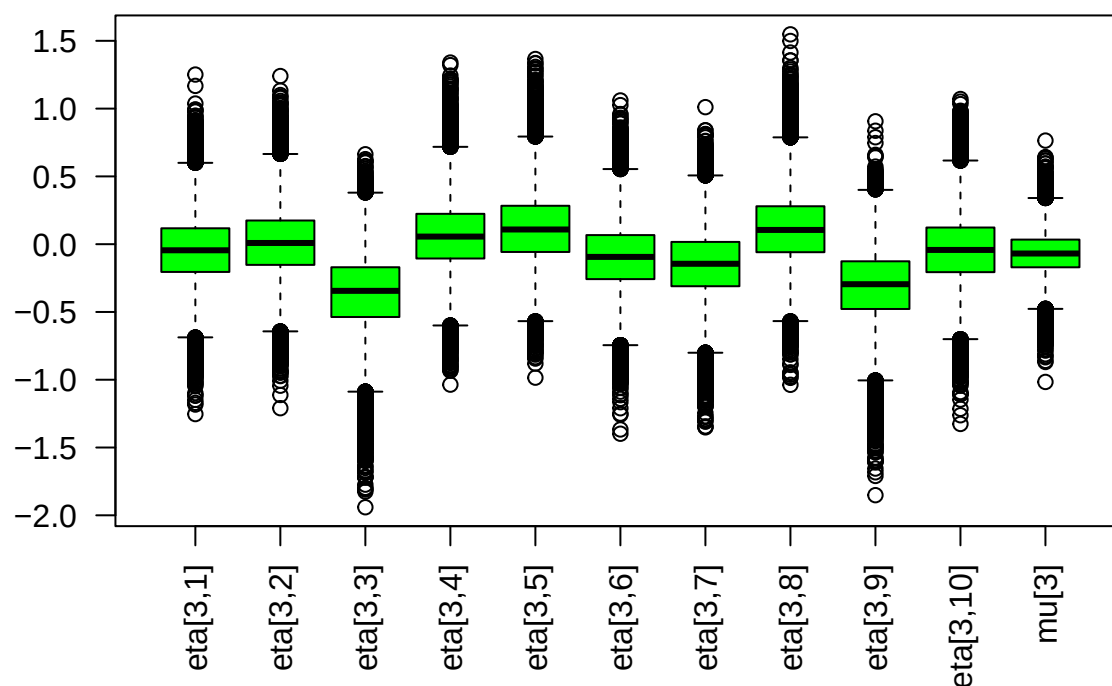
```
sector2_matrix <- as.matrix(fit.samples)
boxplot(sector2_matrix[, c("eta[2,1]", "eta[2,2]", "eta[2,3]", "eta[2,4]", "eta[2,5]", "
    main = "Posterior Boxplots for sector 2 and mu",
    col = "red",
    las = 2) # Rotates axis labels for readability
```


Posterior Boxplots for sector 2 and mu



```
sector3_matrix <- as.matrix(fit.samples)
boxplot(sector3_matrix[, c("eta[3,1]", "eta[3,2]", "eta[3,3]", "eta[3,4]", "eta[3,5]", "
    main = "Posterior Boxplots for sector 3 and mu",
    col = "green",
    las = 2) # Rotates axis labels for readability
```

Posterior Boxplots for sector 3 and mu



```
sector4_matrix <- as.matrix(fit.samples)
```

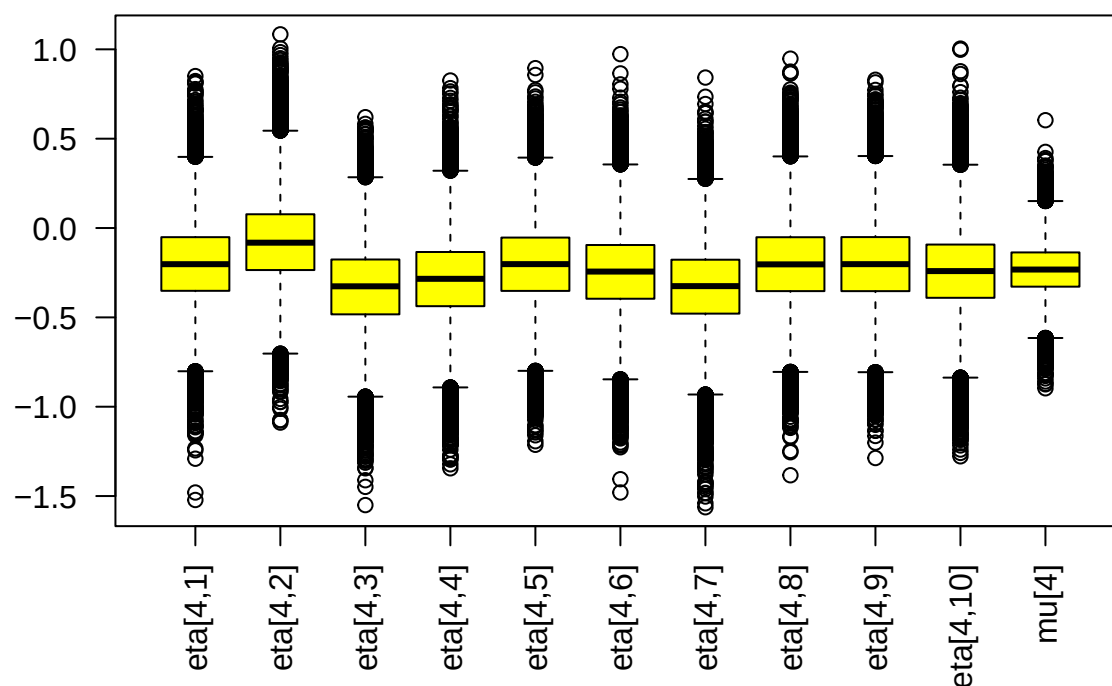
```
boxplot(sector4_matrix[, c("eta[4,1]", "eta[4,2]", "eta[4,3]", "eta[4,4]", "eta[4,5]", "
```

```
    main = "Posterior Boxplots for sector 4 and mu",
```

```
    col = "yellow",
```

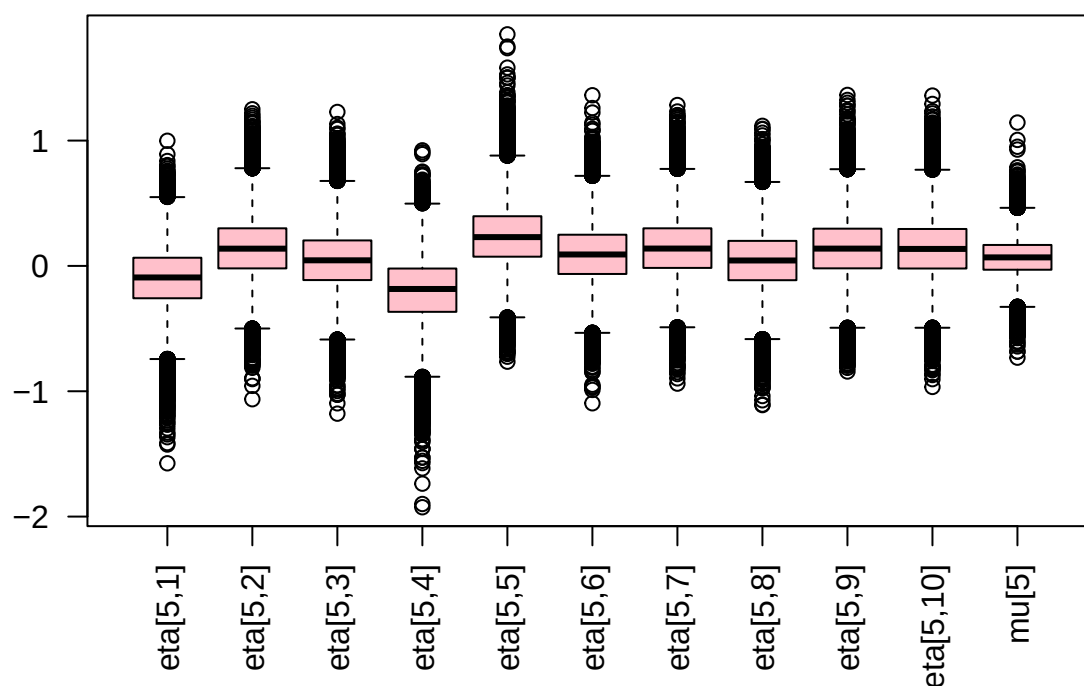
```
    las = 2) # Rotates axis labels for readability
```

Posterior Boxplots for sector 4 and mu



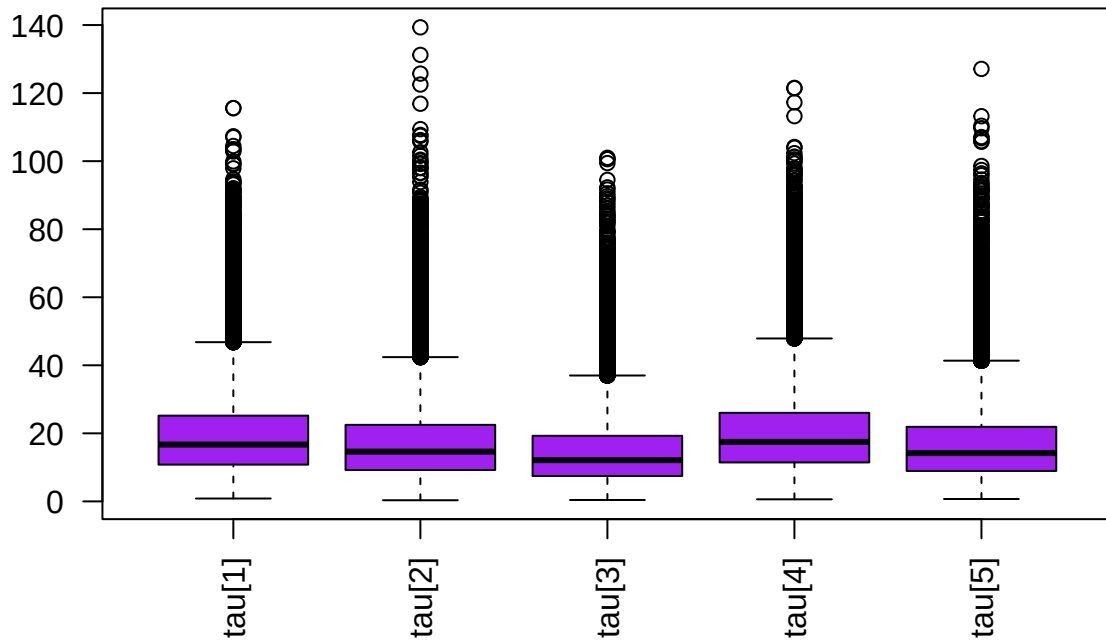
```
sector5_matrix <- as.matrix(fit.samples)
boxplot(sector5_matrix[, c("eta[5,1]", "eta[5,2]", "eta[5,3]", "eta[5,4]", "eta[5,5]", "
    main = "Posterior Boxplots for sector 5 and mu",
    col = "pink",
    las = 2) # Rotates axis labels for readability
```

Posterior Boxplots for sector 5 and mu



```
tau_matrix <- as.matrix(fit.samples)
boxplot(tau_matrix[, c("tau[1]", "tau[2]", "tau[3]", "tau[4]", "tau[5]")],
        main = "Posterior Boxplots for tau",
        col = "purple",
        las = 2) # Rotates axis labels for readability
```

Posterior Boxplots for tau



1.5 - Starting values.

```
head(fit.samples[[1]])
```

```
## Markov Chain Monte Carlo (MCMC) output:
```

```
## Start = 40001
```

```
## End = 40007
```

```
## Thinning interval = 1
```

```
##      eta[1,1]  eta[2,1]  eta[3,1]  eta[4,1]  eta[5,1]  eta[1,2]
## [1,]  0.35607454  0.4484118 -0.28788500  0.134079140 -0.13659163 -0.13110408
## [2,]  0.28557720  0.6370086 -0.03548565 -0.004877163 -0.25519317  0.37098506
## [3,] -0.14307983  0.5269718  0.11375803 -0.004368831  0.45097924  0.06498030
## [4,]  0.03286165 -0.1166573 -0.05713897 -0.063188768  0.12545476  0.02662207
## [5,]  0.04642056 -0.0624354 -0.11998460 -0.088364477  0.34861370  0.31633689
```

```

## [6,] 0.43132704 0.2101203 0.02321539 -0.233585261 0.06991775 -0.41653452
## [7,] 0.13763916 0.1661327 0.25141273 -0.499104454 -0.06603201 -0.13736921
##      eta[2,2]    eta[3,2]    eta[4,2]    eta[5,2]    eta[1,3]    eta[2,3]
## [1,] 0.156361689 0.16478980 -0.24164633 0.1861146 -0.33454302 0.32325546
## [2,] 0.168786828 0.26270258 -0.09859674 0.1119393 -0.32147068 -0.35690461
## [3,] -0.373242186 0.29849662 -0.30439997 0.1176616 -0.32670703 0.53795842
## [4,] 0.172781927 -0.08474496 -0.22682645 0.2096637 -0.02246403 0.69011137
## [5,] -0.045736260 0.16087364 -0.09576958 0.5342387 -0.06746285 0.63025738
## [6,] -0.009132745 -0.06727342 0.35585103 0.6358261 -0.09665978 0.14916739
## [7,] -0.007479420 0.36770766 -0.38869600 0.3638445 0.04681075 0.05828348
##      eta[3,3]    eta[4,3]    eta[5,3]    eta[1,4]    eta[2,4]    eta[3,4]
## [1,] -0.35976041 -0.2954019 0.1277524 -0.10538591 -0.10995990 0.41695030
## [2,] -0.43802162 -0.3656350 0.1258979 0.14558083 -0.17521874 0.36061557
## [3,] -0.05942347 -0.3603114 0.1485425 0.01256256 -0.30562895 0.04970229
## [4,] -0.31459437 -0.2087429 0.1716347 0.04490021 -0.20386681 -0.07771007
## [5,] -0.24008503 -0.1522849 0.1601915 -0.03944300 -0.19726078 0.27829635
## [6,] 0.02747872 -0.1276724 0.1737187 -0.08466377 0.23584643 0.22441420
## [7,] -0.56445780 -0.1727983 0.1390591 -0.03143397 -0.03395584 0.37600307
##      eta[4,4]    eta[5,4]    eta[1,5]    eta[2,5]    eta[3,5]    eta[4,5]
## [1,] -0.4666410 0.0164429978 -0.15432823 0.18054478 0.45570784 -0.5323616
## [2,] -0.1238161 0.2048812660 0.64072935 0.44545029 -0.09778889 -0.3253283
## [3,] -0.1267351 0.1003690256 0.67011374 -0.02867462 -0.03770447 -0.3199259
## [4,] -0.2567192 -0.1475480999 0.18291841 0.42025949 0.01616558 -0.0620413
## [5,] -0.1667841 0.1073217297 -0.24051614 0.40594739 0.07289597 -0.3060847
## [6,] -0.2950208 0.0009462354 -0.07012506 0.37847334 0.17846665 -0.4497985
## [7,] -0.5162499 -0.0613170435 -0.04092105 -0.11030680 0.43358557 -0.5647724
##      eta[5,5]    eta[1,6]    eta[2,6]    eta[3,6]    eta[4,6]    eta[5,6]

```

```

## [1,] 0.1982575 0.12576571 0.03549977 0.0439912993 -0.35503522 0.23672043
## [2,] 0.3290724 -0.23405114 -0.10800250 0.4337666960 -0.40081289 0.25318589
## [3,] 0.2151964 0.08908084 -0.26679843 0.4620542345 -0.32501906 0.21793496
## [4,] 0.2401479 0.13263108 0.09411520 -0.0277926745 -0.25867201 0.13179120
## [5,] 0.2763162 -0.13005227 0.27844690 0.0559860744 -0.07803855 0.09037533
## [6,] 0.2350850 -0.11211226 0.41933614 0.0009291662 -0.60107341 0.18864810
## [7,] 0.1901782 0.05459726 -0.13477699 0.0781557089 -0.33147265 0.27771383
##      eta[1,7]    eta[2,7]    eta[3,7]    eta[4,7]    eta[5,7]    eta[1,8]
## [1,] 0.3044501 0.42732790 -0.050125081 -0.47758779 0.1441429 -0.041827824
## [2,] 0.1442364 0.34818568 -0.087770614 -0.19736468 0.1821794 -0.139460041
## [3,] 0.2164758 0.31392829 0.070402624 -0.30606740 0.3334677 0.055162547
## [4,] 0.2411158 0.14000817 -0.256265136 -0.37403503 0.1972110 0.282785015
## [5,] 0.1888026 -0.09851436 -0.100883628 0.07106120 0.2882242 0.254500419
## [6,] -0.2462421 0.47701266 -0.004280038 -0.09931913 0.1780927 -0.001392565
## [7,] 0.1794596 0.46365833 0.151048699 -0.55925309 0.1435143 0.014192434
##      eta[2,8]    eta[3,8]    eta[4,8]    eta[5,8]    eta[1,9]    eta[2,9]
## [1,] 0.26793614 0.45388882 -0.2972167 0.2121377 0.24176241 0.27455271
## [2,] 0.17548929 0.44857603 -0.3862640 0.2159191 0.16787615 0.27116686
## [3,] 0.35354300 0.22131398 -0.2235201 0.1876959 0.13910065 0.10251342
## [4,] 0.04324918 0.19287389 -0.1453379 0.1450347 -0.01326474 -0.38638645
## [5,] 0.15015041 0.11249220 -0.3085030 0.2693894 -0.02387594 -0.33476669
## [6,] 0.17715419 -0.17720627 -0.3482270 0.1411969 -0.51444449 -0.08509074
## [7,] 0.34476351 -0.09908483 -0.3687490 0.1726126 -0.54015962 0.08845848
##      eta[3,9]    eta[4,9]    eta[5,9]    eta[1,10]    eta[2,10]    eta[3,10]
## [1,] -0.720333617 -0.08181609 0.13695846 -0.05988190 0.8583565 -0.1279178
## [2,] -0.353278740 -0.13962952 0.40633476 0.22201497 0.2730248 0.3085115
## [3,] 0.013979278 -0.25004916 0.35307820 0.24095309 0.4501842 0.3789618

```

```
## [4,] 0.015524002 -0.27233894 0.23267601 0.29157926 0.3685139 0.1984944
## [5,] -0.004502389 -0.21107982 0.08457120 0.28820030 0.8537473 -0.2052617
## [6,] 0.010774083 -0.03577247 0.38790519 0.08656791 0.6587549 0.2036519
## [7,] -0.065618502 -0.05412270 0.09828955 -0.09440874 -0.1835160 -0.2310840
##      eta[4,10] eta[5,10]      mu[1]      mu[2]      mu[3]      mu[4]
## [1,] -0.5108280 0.20206142 0.20546258 0.1469524 0.142551758 -0.2770635
## [2,] -0.4541145 0.07074229 0.07350130 0.2387788 0.190902586 -0.2641582
## [3,] -0.3938383 0.21107064 0.04430793 0.1000766 -0.014362223 -0.2495437
## [4,] -0.2870663 0.15225775 0.09054686 0.1178082 -0.001873608 -0.1441343
## [5,] -0.2468107 0.18172359 -0.11457014 0.1122469 0.034295190 -0.1583033
## [6,] -0.3340422 0.05004565 -0.15631163 0.1217587 0.133382502 -0.3248353
## [7,] -0.4573515 0.04469543 0.14490997 -0.1097817 -0.062693138 -0.3562210
##      mu[5]      tau[1]      tau[2]      tau[3]      tau[4]      tau[5]
## [1,] 0.1866079 7.612794 9.967218 8.416524 36.627026 26.66342
## [2,] 0.1993025 15.413107 6.191353 10.410056 64.912245 24.64812
## [3,] 0.2134449 19.789292 10.692802 7.514568 57.641618 34.65726
## [4,] 0.2145031 19.480041 9.645875 10.398124 67.341164 54.55376
## [5,] 0.2540986 12.831482 6.837950 31.426988 6.436782 30.36686
## [6,] 0.2119256 10.268812 24.390037 23.192512 19.577641 16.34341
## [7,] 0.1528650 8.798299 14.357747 13.282304 33.495153 93.58097
```

1.6 - Effective Sample Size of all sectors and companies stocks.

```
#Sector 1
```

```
cat("ESS of sector 1 company 1: ", effectiveSize(sector1_matrix[, c("eta[1,1]")]), "\n")
```

```
## ESS of sector 1 company 1: 18666.01
```



```

cat("ESS of sector 1 company 2: ", effectiveSize(sector1_matrix[, c("eta[1,2]")]), "\n")

## ESS of sector 1 company 2: 19542.85

cat("ESS of sector 1 company 3: ", effectiveSize(sector1_matrix[, c("eta[1,3]")]), "\n")

## ESS of sector 1 company 3: 19109.78

cat("ESS of sector 1 company 4: ", effectiveSize(sector1_matrix[, c("eta[1,4]")]), "\n")

## ESS of sector 1 company 4: 17194.58

cat("ESS of sector 1 company 5: ", effectiveSize(sector1_matrix[, c("eta[1,5]")]), "\n")

## ESS of sector 1 company 5: 19039.26

cat("ESS of sector 1 company 6: ", effectiveSize(sector1_matrix[, c("eta[1,6]")]), "\n")

## ESS of sector 1 company 6: 18766.03

cat("ESS of sector 1 company 7: ", effectiveSize(sector1_matrix[, c("eta[1,7]")]), "\n")

## ESS of sector 1 company 7: 19514.78

cat("ESS of sector 1 company 8: ", effectiveSize(sector1_matrix[, c("eta[1,8]")]), "\n")

## ESS of sector 1 company 8: 20119.71

cat("ESS of sector 1 company 9: ", effectiveSize(sector1_matrix[, c("eta[1,9]")]), "\n")

## ESS of sector 1 company 9: 18351.11

cat("ESS of sector 1 company 10: ", effectiveSize(sector1_matrix[, c("eta[1,10]")]), "\n")

## ESS of sector 1 company 10: 18757.2

#Sector 2

```

```

cat("ESS of sector 2 company 1: ", effectiveSize(sector2_matrix[, c("eta[2,1]")]), "\n")

## ESS of sector 2 company 1: 20841.14

cat("ESS of sector 2 company 2: ", effectiveSize(sector2_matrix[, c("eta[2,2]")]), "\n")

## ESS of sector 2 company 2: 17483.89

cat("ESS of sector 2 company 3: ", effectiveSize(sector2_matrix[, c("eta[2,3]")]), "\n")

## ESS of sector 2 company 3: 21380.88

cat("ESS of sector 2 company 4: ", effectiveSize(sector2_matrix[, c("eta[2,4]")]), "\n")

## ESS of sector 2 company 4: 20333.51

cat("ESS of sector 2 company 5: ", effectiveSize(sector2_matrix[, c("eta[2,5]")]), "\n")

## ESS of sector 2 company 5: 19953

cat("ESS of sector 2 company 6: ", effectiveSize(sector2_matrix[, c("eta[2,6]")]), "\n")

## ESS of sector 2 company 6: 20570.1

cat("ESS of sector 2 company 7: ", effectiveSize(sector2_matrix[, c("eta[2,7]")]), "\n")

## ESS of sector 2 company 7: 19281.65

cat("ESS of sector 2 company 8: ", effectiveSize(sector2_matrix[, c("eta[2,8]")]), "\n")

## ESS of sector 2 company 8: 18900.15

cat("ESS of sector 2 company 9: ", effectiveSize(sector2_matrix[, c("eta[2,9]")]), "\n")

## ESS of sector 2 company 9: 19228.93

cat("ESS of sector 2 company 10: ", effectiveSize(sector2_matrix[, c("eta[2,10]")]), "\n")

## ESS of sector 2 company 10: 16455.19

```

```
#Sector 3
```

```
cat("ESS of sector 3 company 1: ", effectiveSize(sector3_matrix[, c("eta[3,1]")]), "\n")
```

```
## ESS of sector 3 company 1: 22889.75
```

```
cat("ESS of sector 3 company 2: ", effectiveSize(sector3_matrix[, c("eta[3,2]")]), "\n")
```

```
## ESS of sector 3 company 2: 22572.03
```

```
cat("ESS of sector 3 company 3: ", effectiveSize(sector3_matrix[, c("eta[3,3]")]), "\n")
```

```
## ESS of sector 3 company 3: 15346.98
```

```
cat("ESS of sector 3 company 4: ", effectiveSize(sector3_matrix[, c("eta[3,4]")]), "\n")
```

```
## ESS of sector 3 company 4: 21617.9
```

```
cat("ESS of sector 3 company 5: ", effectiveSize(sector3_matrix[, c("eta[3,5]")]), "\n")
```

```
## ESS of sector 3 company 5: 20081.37
```

```
cat("ESS of sector 3 company 6: ", effectiveSize(sector3_matrix[, c("eta[3,6]")]), "\n")
```

```
## ESS of sector 3 company 6: 22865.94
```

```
cat("ESS of sector 3 company 7: ", effectiveSize(sector3_matrix[, c("eta[3,7]")]), "\n")
```

```
## ESS of sector 3 company 7: 22776.69
```

```
cat("ESS of sector 3 company 8: ", effectiveSize(sector3_matrix[, c("eta[3,8]")]), "\n")
```

```
## ESS of sector 3 company 8: 20352.31
```

```
cat("ESS of sector 3 company 9: ", effectiveSize(sector3_matrix[, c("eta[3,9]")]), "\n")
```

```
## ESS of sector 3 company 9: 16919.21
```

```

cat("ESS of sector 3 company 10: ", effectiveSize(sector3_matrix[, c("eta[3,10]")]), "\n")

## ESS of sector 3 company 10: 22950.06

#Sector 4

cat("ESS of sector 4 company 1: ", effectiveSize(sector4_matrix[, c("eta[4,1]")]), "\n")

## ESS of sector 4 company 1: 16925.31

cat("ESS of sector 4 company 2: ", effectiveSize(sector4_matrix[, c("eta[4,2]")]), "\n")

## ESS of sector 4 company 2: 16864.02

cat("ESS of sector 4 company 3: ", effectiveSize(sector4_matrix[, c("eta[4,3]")]), "\n")

## ESS of sector 4 company 3: 17852.89

cat("ESS of sector 4 company 4: ", effectiveSize(sector4_matrix[, c("eta[4,4]")]), "\n")

## ESS of sector 4 company 4: 18504.41

cat("ESS of sector 4 company 5: ", effectiveSize(sector4_matrix[, c("eta[4,5]")]), "\n")

## ESS of sector 4 company 5: 18735.69

cat("ESS of sector 4 company 6: ", effectiveSize(sector4_matrix[, c("eta[4,6]")]), "\n")

## ESS of sector 4 company 6: 18093.78

cat("ESS of sector 4 company 7: ", effectiveSize(sector4_matrix[, c("eta[4,7]")]), "\n")

## ESS of sector 4 company 7: 18742.57

cat("ESS of sector 4 company 8: ", effectiveSize(sector4_matrix[, c("eta[4,8]")]), "\n")

## ESS of sector 4 company 8: 17756.82

```

```
cat("ESS of sector 4 company 9: ", effectiveSize(sector4_matrix[, c("eta[4,9]")]), "\n")
```

```
## ESS of sector 4 company 9: 19359.74
```

```
cat("ESS of sector 4 company 10: ", effectiveSize(sector4_matrix[, c("eta[4,10]")]), "\n")
```

```
## ESS of sector 4 company 10: 19325.21
```

```
#sector 5
```

```
cat("ESS of sector 5 company 1: ", effectiveSize(sector5_matrix[, c("eta[5,1]")]), "\n")
```

```
## ESS of sector 5 company 1: 18969.19
```

```
cat("ESS of sector 5 company 2: ", effectiveSize(sector5_matrix[, c("eta[5,2]")]), "\n")
```

```
## ESS of sector 5 company 2: 20471.45
```

```
cat("ESS of sector 5 company 3: ", effectiveSize(sector5_matrix[, c("eta[5,3]")]), "\n")
```

```
## ESS of sector 5 company 3: 20869.53
```

```
cat("ESS of sector 5 company 4: ", effectiveSize(sector5_matrix[, c("eta[5,4]")]), "\n")
```

```
## ESS of sector 5 company 4: 15762.23
```

```
cat("ESS of sector 5 company 5: ", effectiveSize(sector5_matrix[, c("eta[5,5]")]), "\n")
```

```
## ESS of sector 5 company 5: 18459.43
```

```
cat("ESS of sector 5 company 6: ", effectiveSize(sector5_matrix[, c("eta[5,6]")]), "\n")
```

```
## ESS of sector 5 company 6: 22217.63
```

```
cat("ESS of sector 5 company 7: ", effectiveSize(sector5_matrix[, c("eta[5,7]")]), "\n")
```

```
## ESS of sector 5 company 7: 21740.11
```

```
cat("ESS of sector 5 company 8: ", effectiveSize(sector5_matrix[, c("eta[5,8]")]), "\n")
```

```
## ESS of sector 5 company 8: 21629.1
```

```
cat("ESS of sector 5 company 9: ", effectiveSize(sector5_matrix[, c("eta[5,9]")]), "\n")
```

```
## ESS of sector 5 company 9: 21496.73
```

```
cat("ESS of sector 5 company 10: ", effectiveSize(sector5_matrix[, c("eta[5,10]")]), "\n")
```

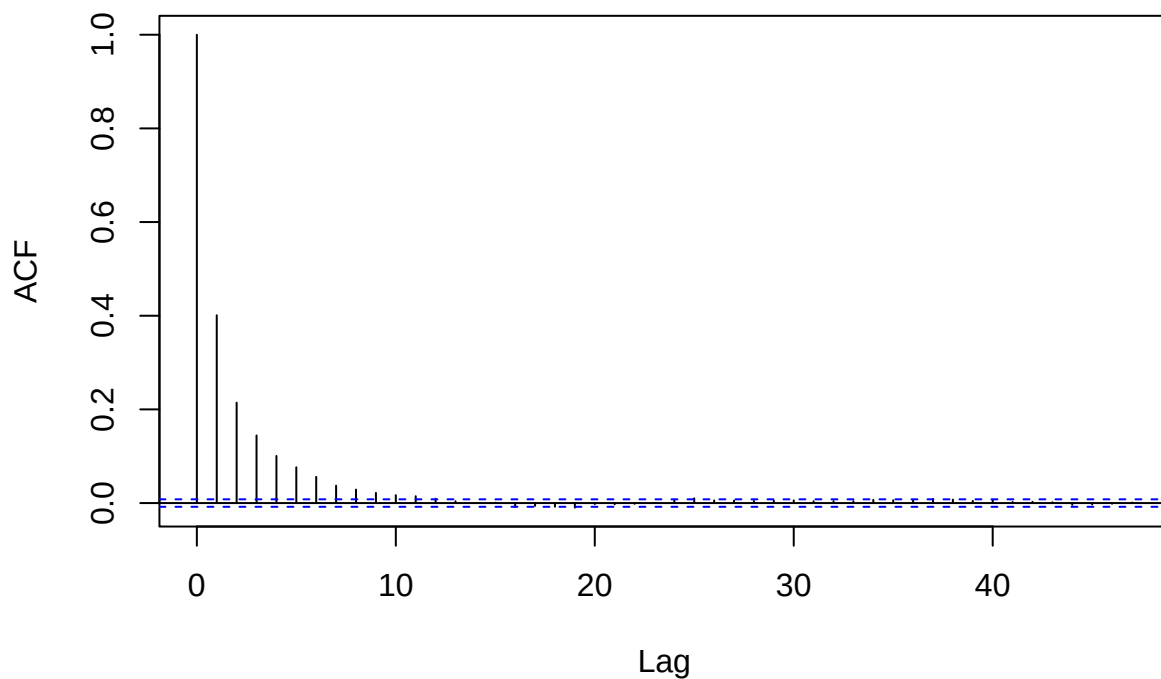
```
## ESS of sector 5 company 10: 21047.21
```

1.7 - ACF of all η_{sc} .

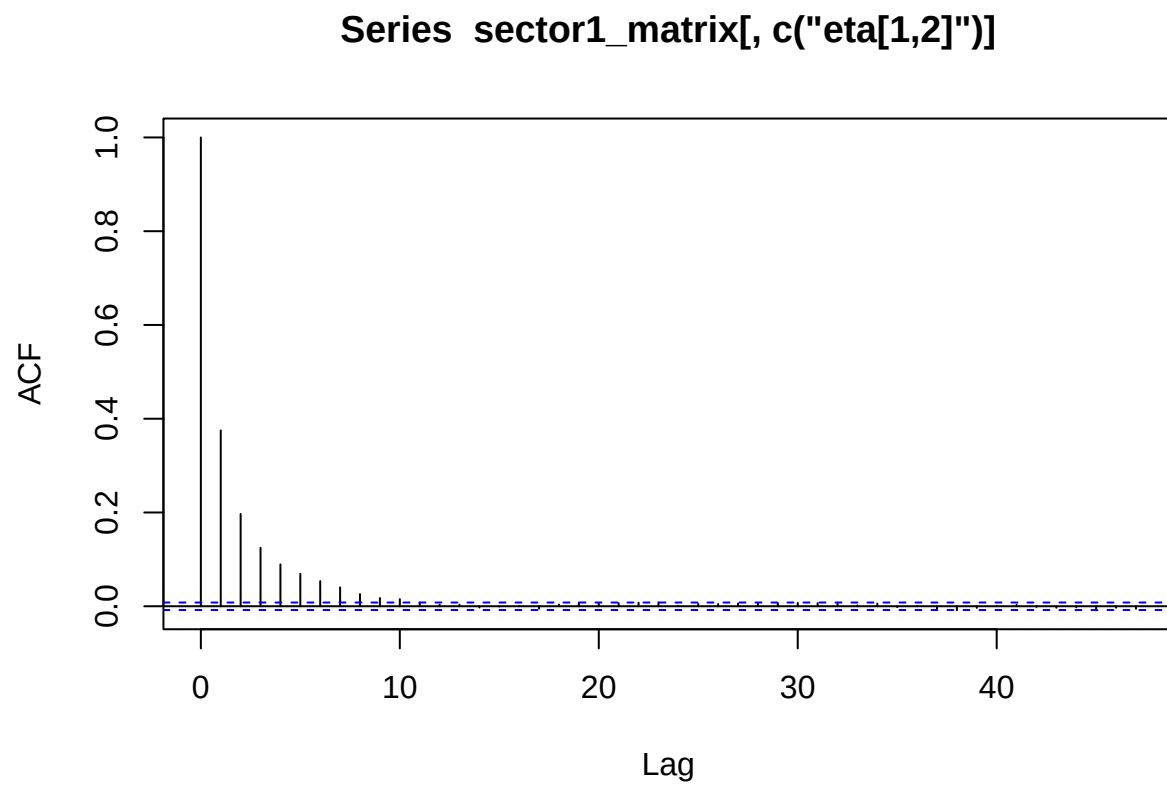
```
#sector 1
```

```
acf(sector1_matrix[, c("eta[1,1]")])
```

Series sector1_matrix[, c("eta[1,1]")]

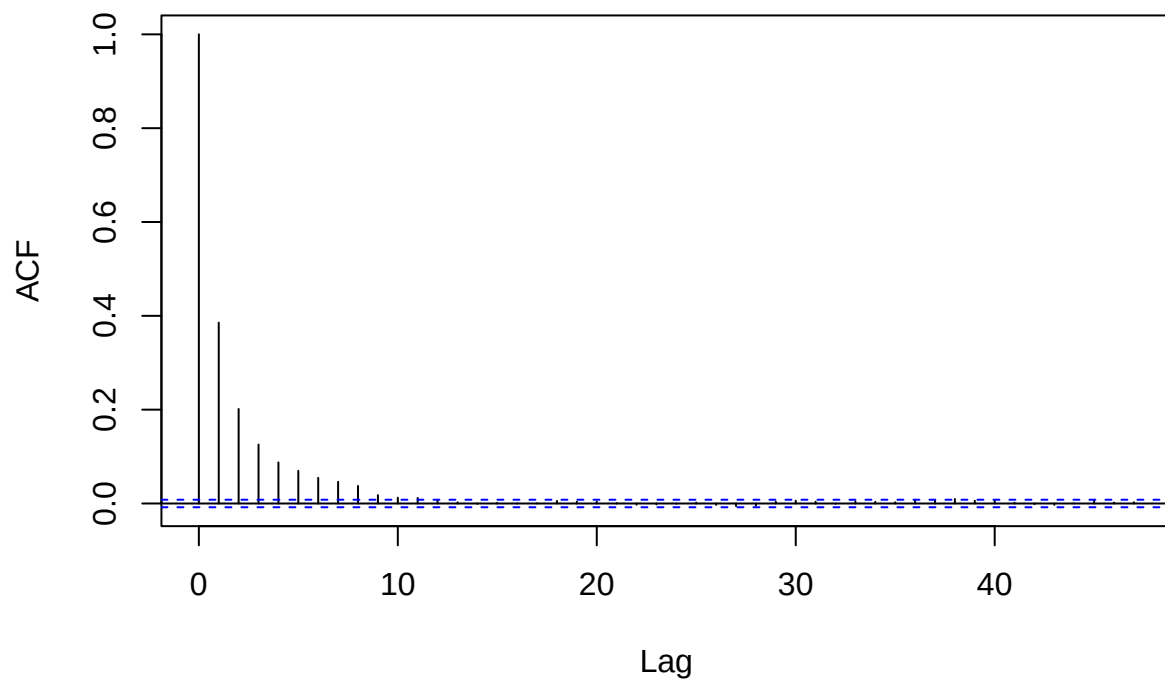


```
acf(sector1_matrix[, c("eta[1,2]")])
```



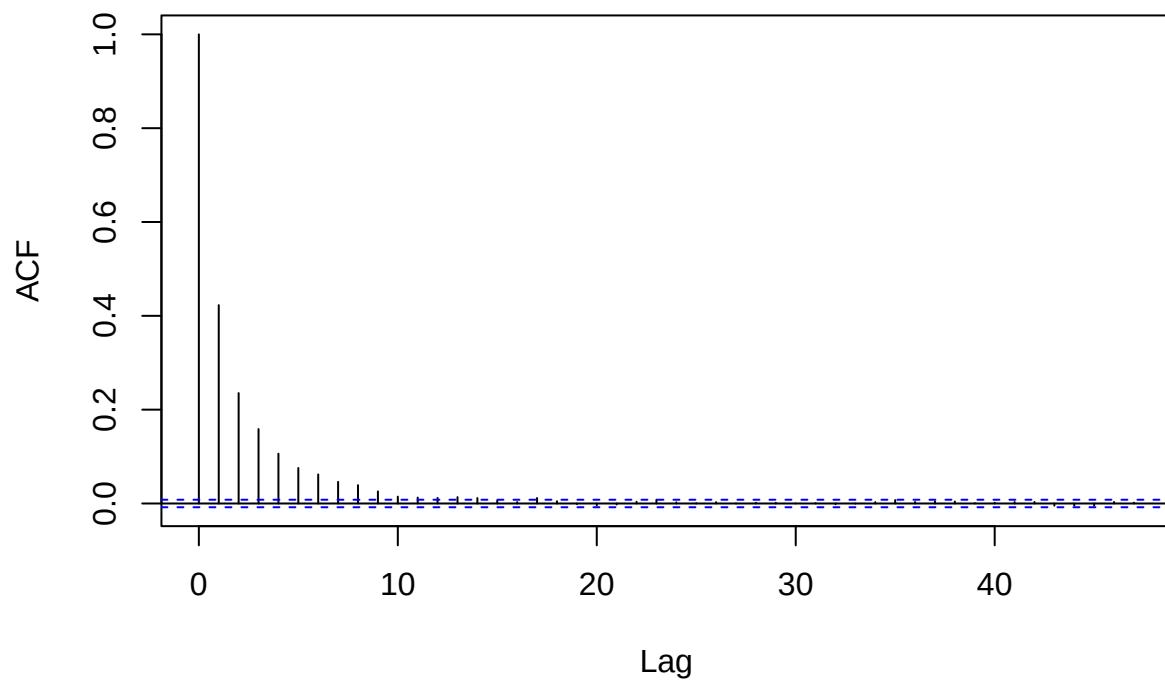
```
acf(sector1_matrix[, c("eta[1,3]")])
```

Series sector1_matrix[, c("eta[1,3]")]



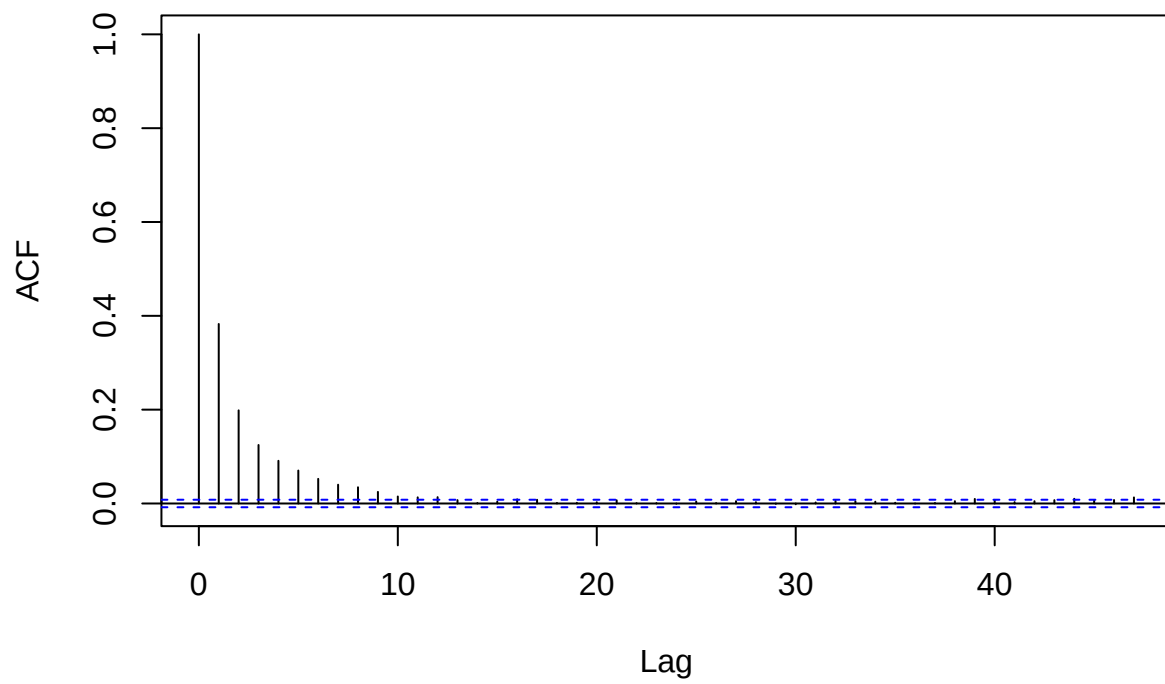
```
acf(sector1_matrix[, c("eta[1,4]")])
```


Series sector1_matrix[, c("eta[1,4]")]



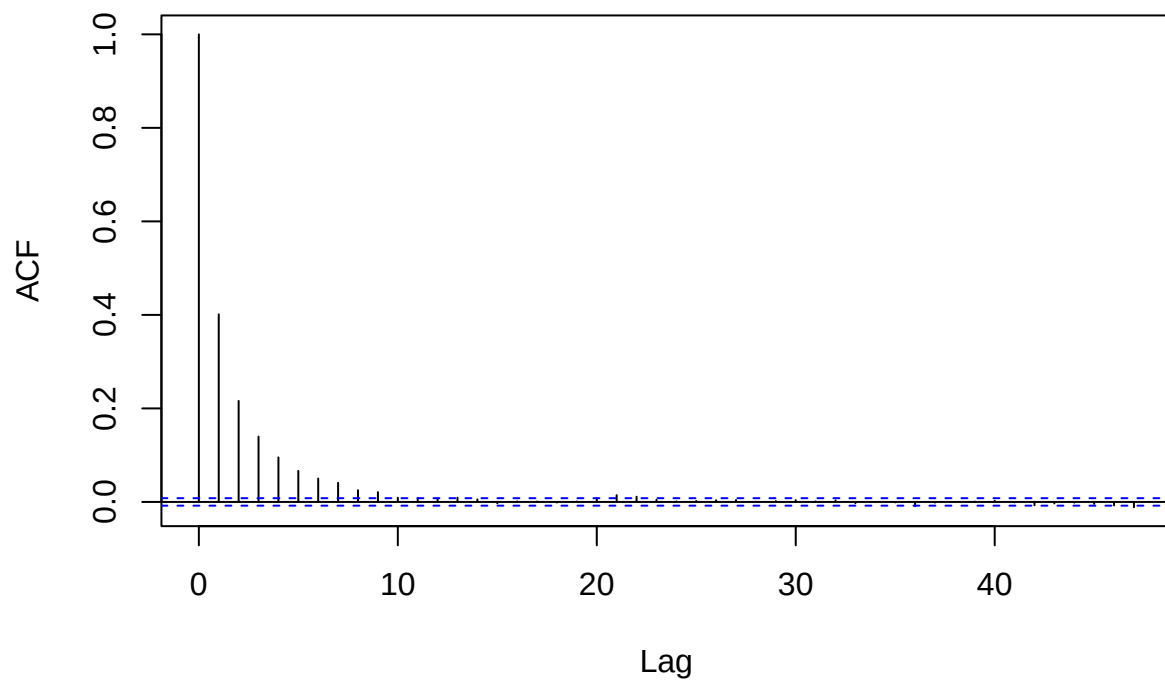
```
acf(sector1_matrix[, c("eta[1,5]")])
```

Series sector1_matrix[, c("eta[1,5]")]



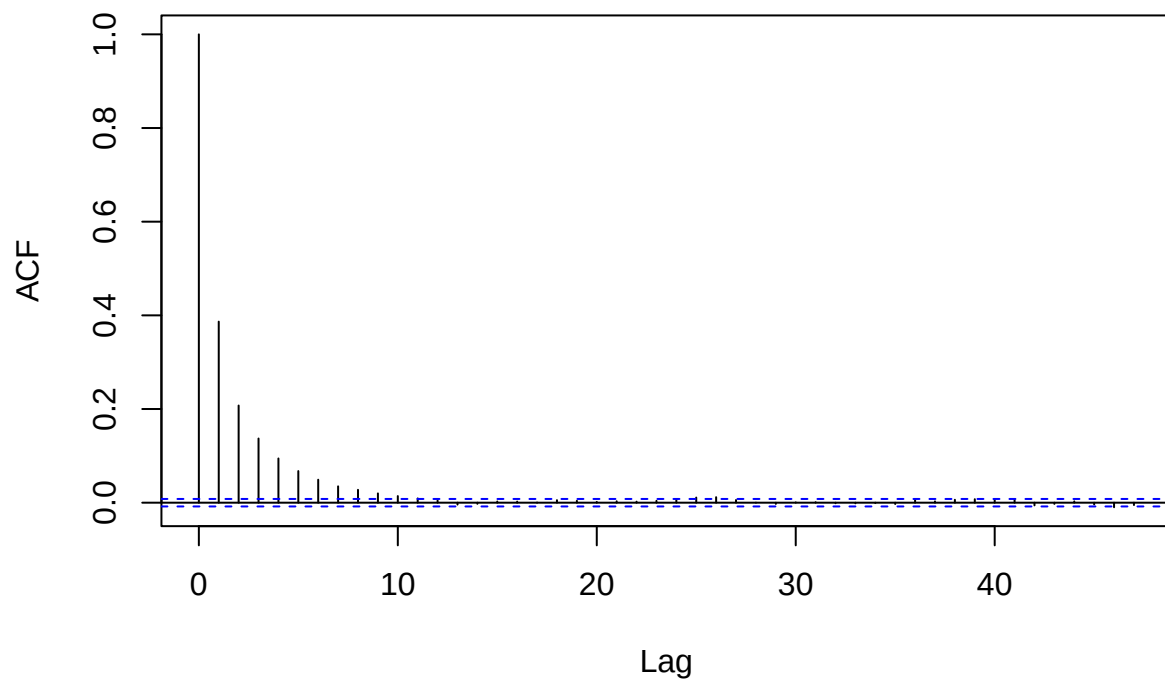
```
acf(sector1_matrix[, c("eta[1,6]")])
```

Series sector1_matrix[, c("eta[1,6]")]



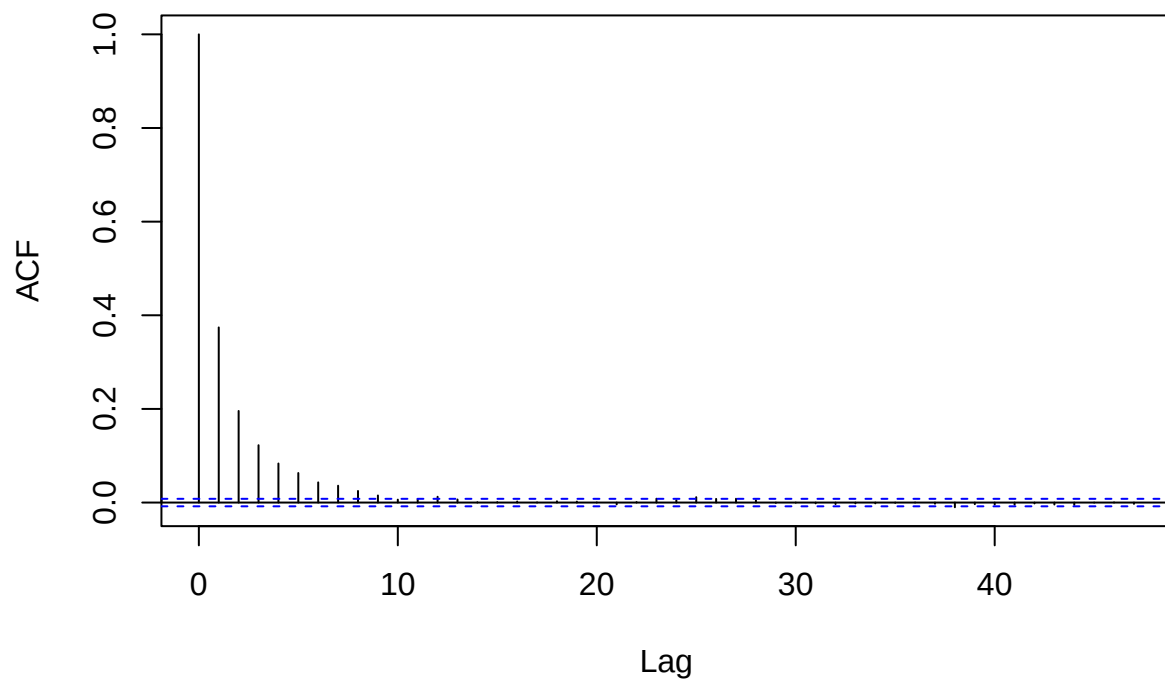
```
acf(sector1_matrix[, c("eta[1,7]")])
```

Series sector1_matrix[, c("eta[1,7]")]



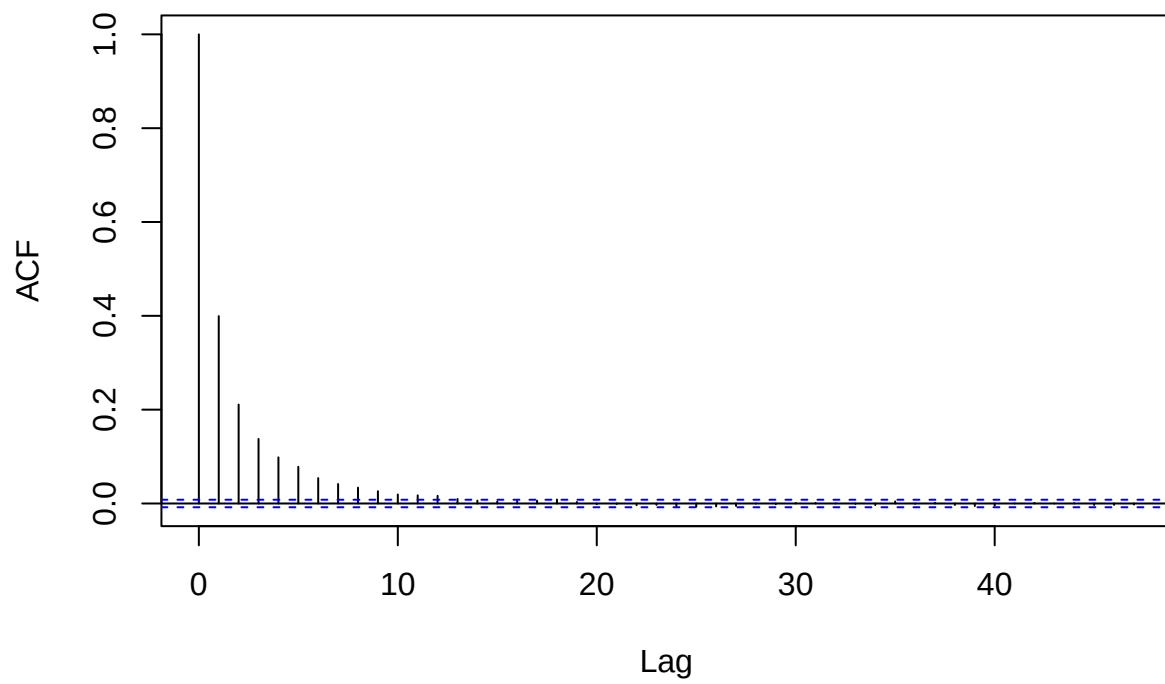
```
acf(sector1_matrix[, c("eta[1,8]")])
```

Series sector1_matrix[, c("eta[1,8]")]



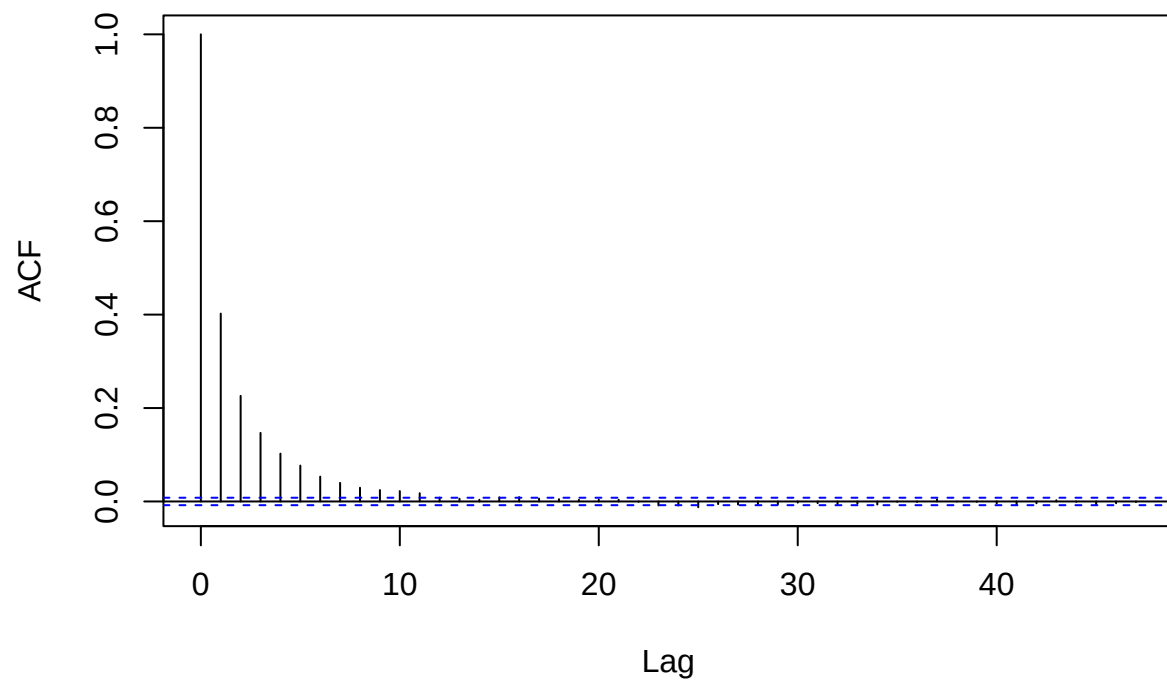
```
acf(sector1_matrix[, c("eta[1,9]")])
```

Series sector1_matrix[, c("eta[1,9]")]



```
acf(sector1_matrix[, c("eta[1,10]")])
```

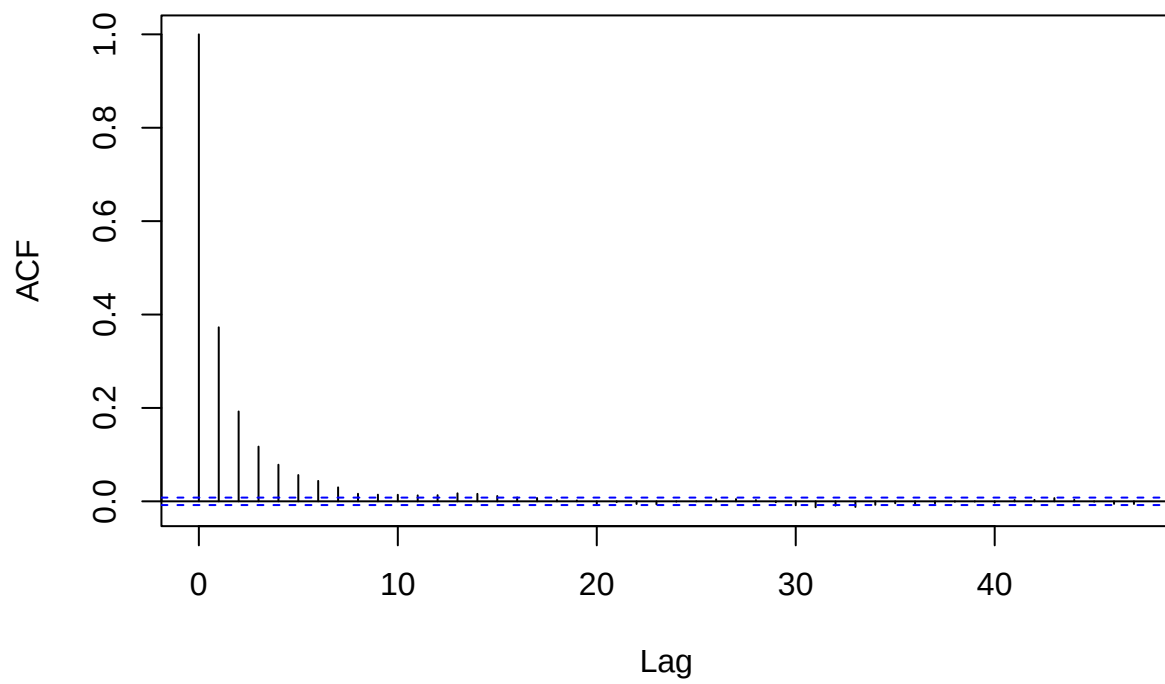
Series sector1_matrix[, c("eta[1,10]")]



```
#sector 2
```

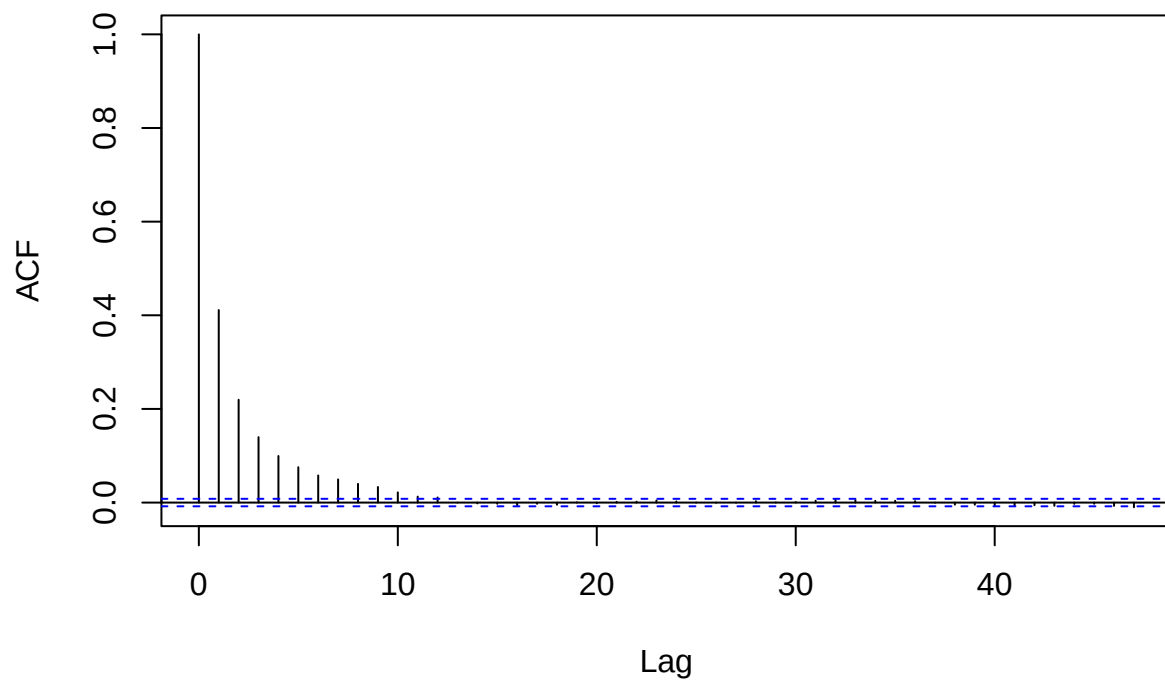
```
acf(sector2_matrix[, c("eta[2,1]")])
```

Series sector2_matrix[, c("eta[2,1]")]



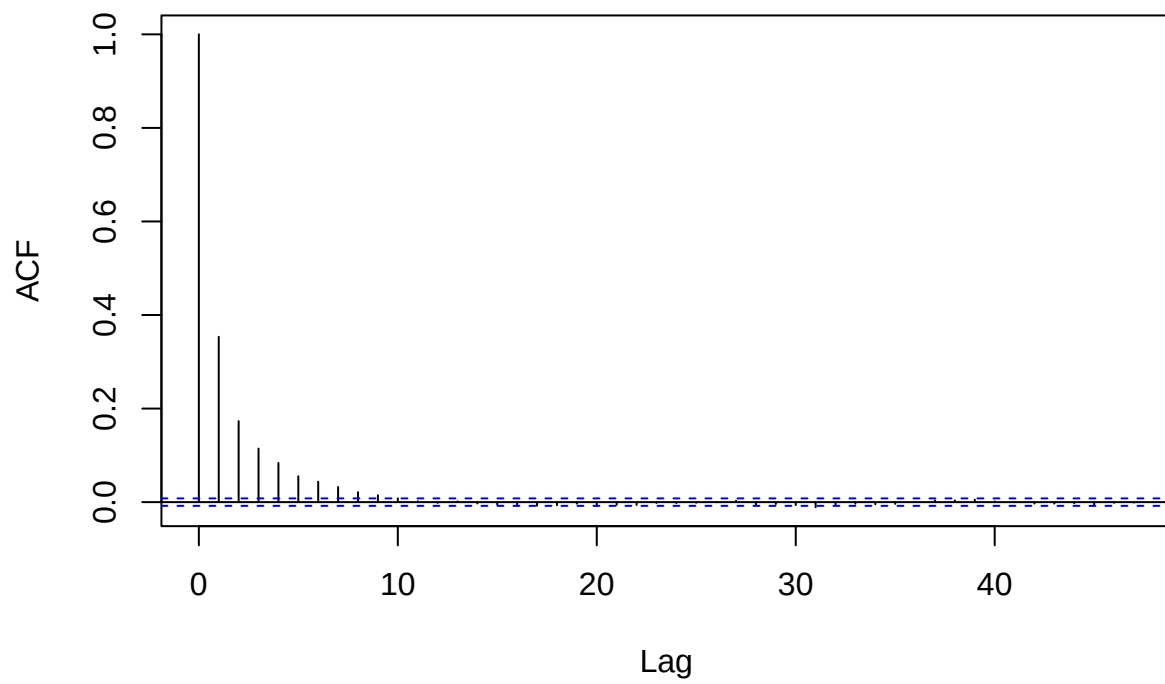
```
acf(sector2_matrix[, c("eta[2,2]")])
```


Series sector2_matrix[, c("eta[2,2]")]



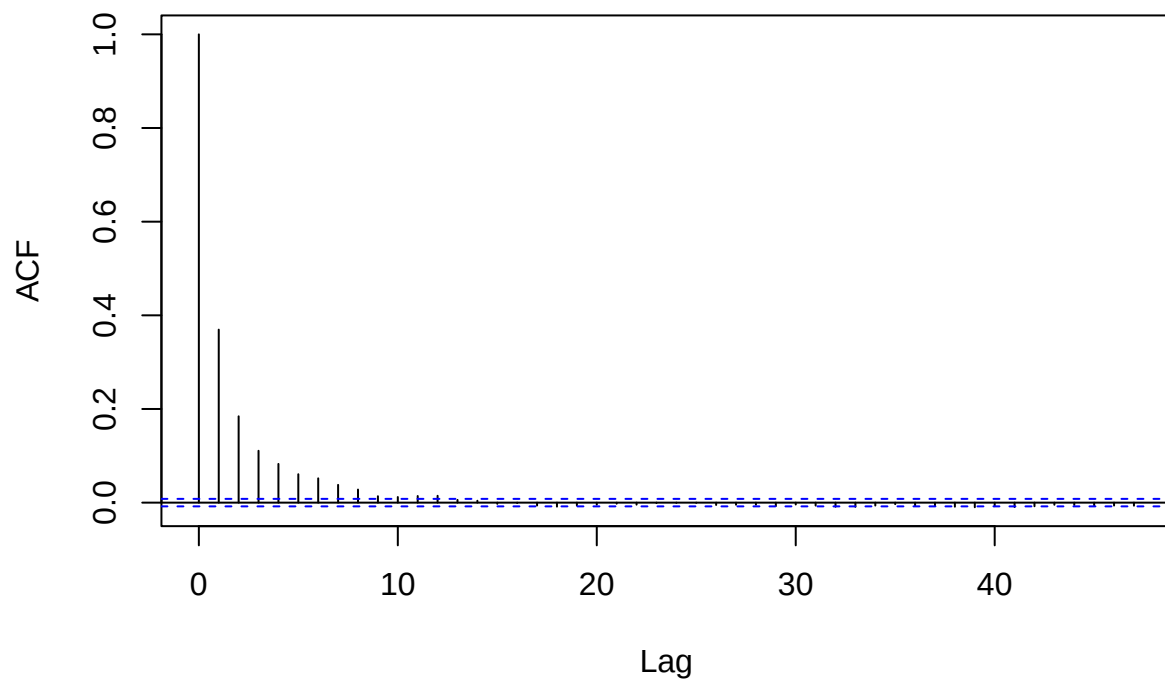
```
acf(sector2_matrix[, c("eta[2,3]")])
```

Series sector2_matrix[, c("eta[2,3]")]



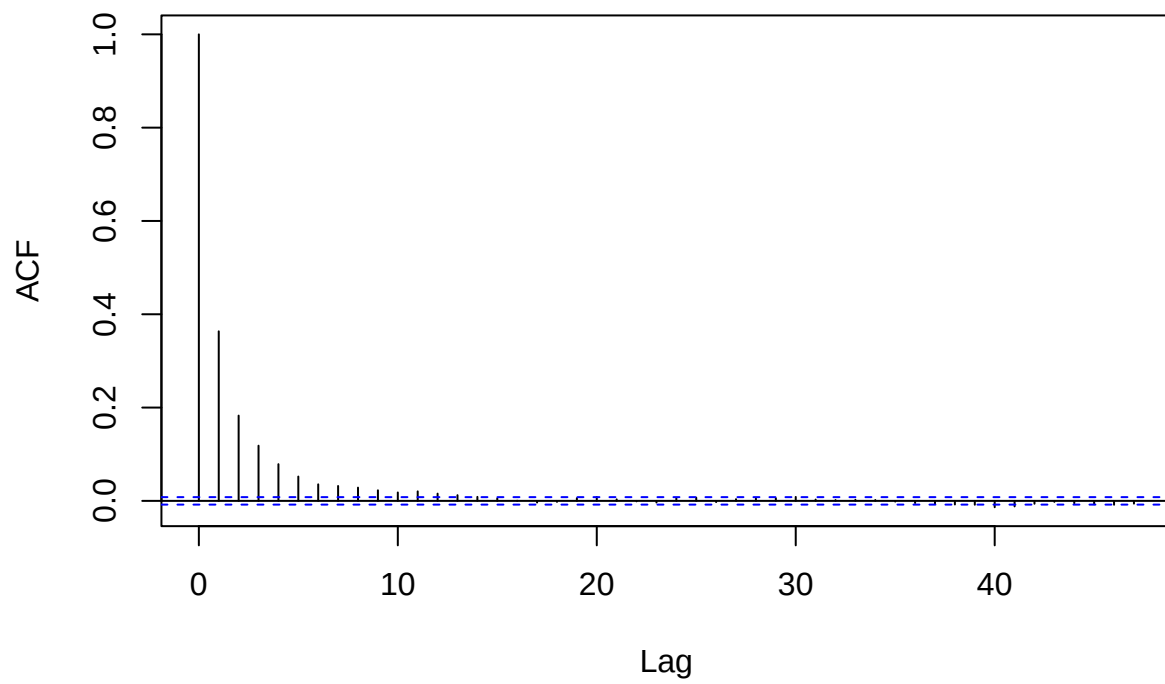
```
acf(sector2_matrix[, c("eta[2,4]")])
```

Series sector2_matrix[, c("eta[2,4]")]



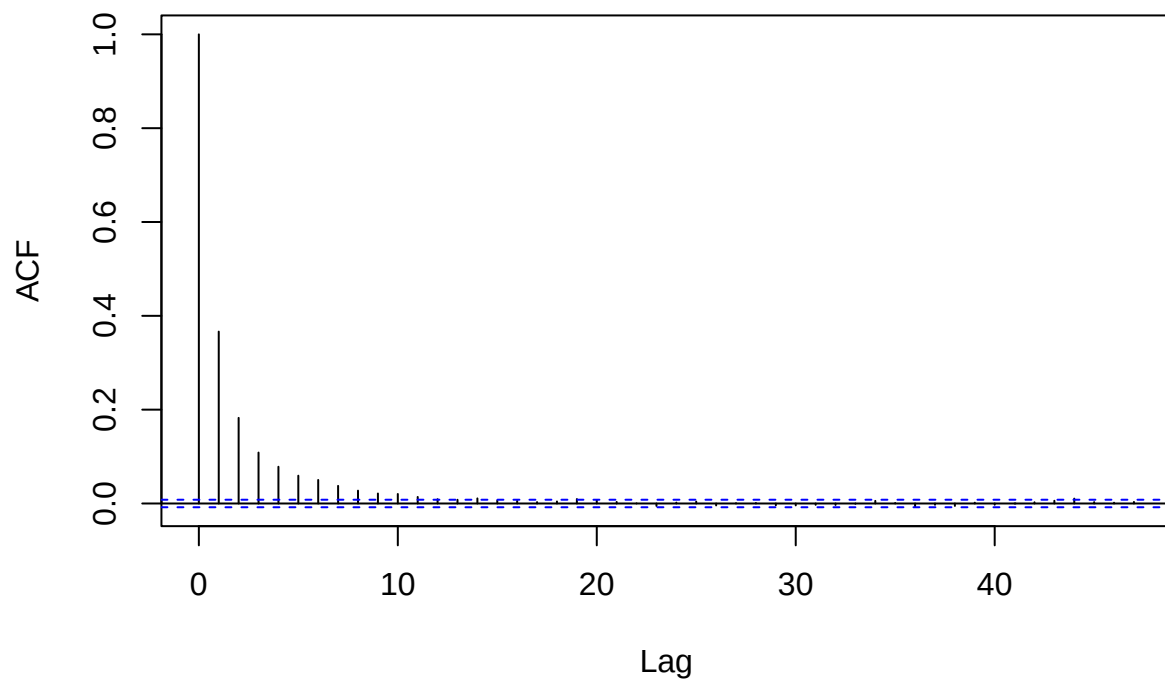
```
acf(sector2_matrix[, c("eta[2,5] ")])
```

Series sector2_matrix[, c("eta[2,5]")]



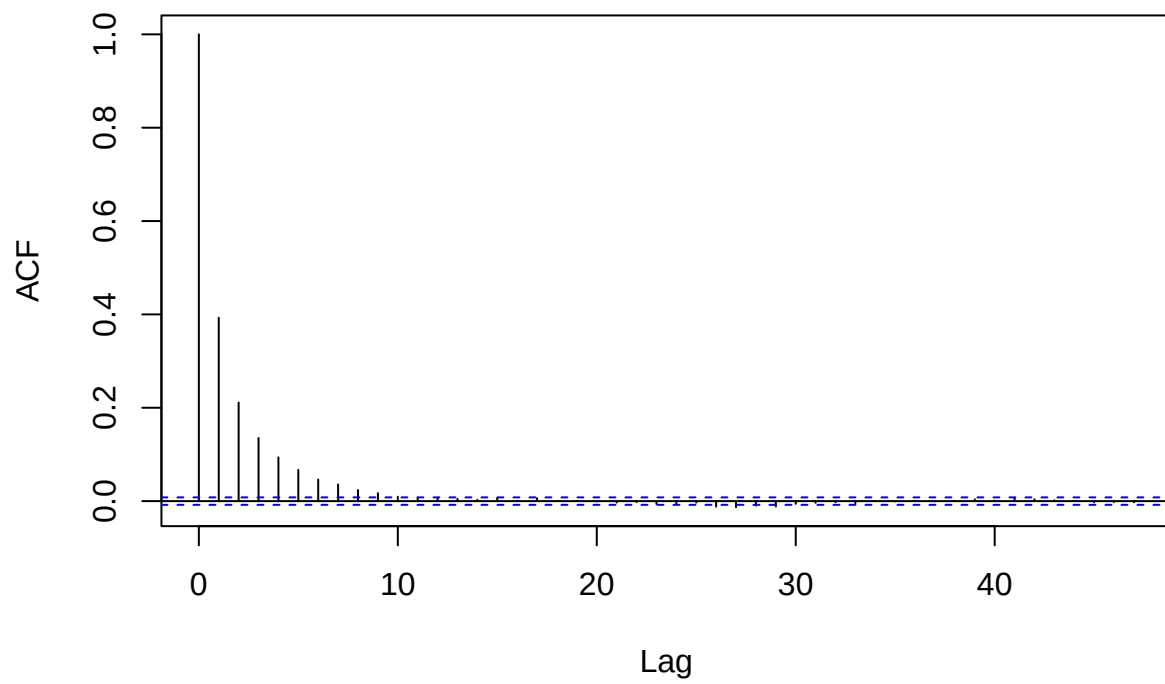
```
acf(sector2_matrix[, c("eta[2,6] ")])
```

Series sector2_matrix[, c("eta[2,6]")]



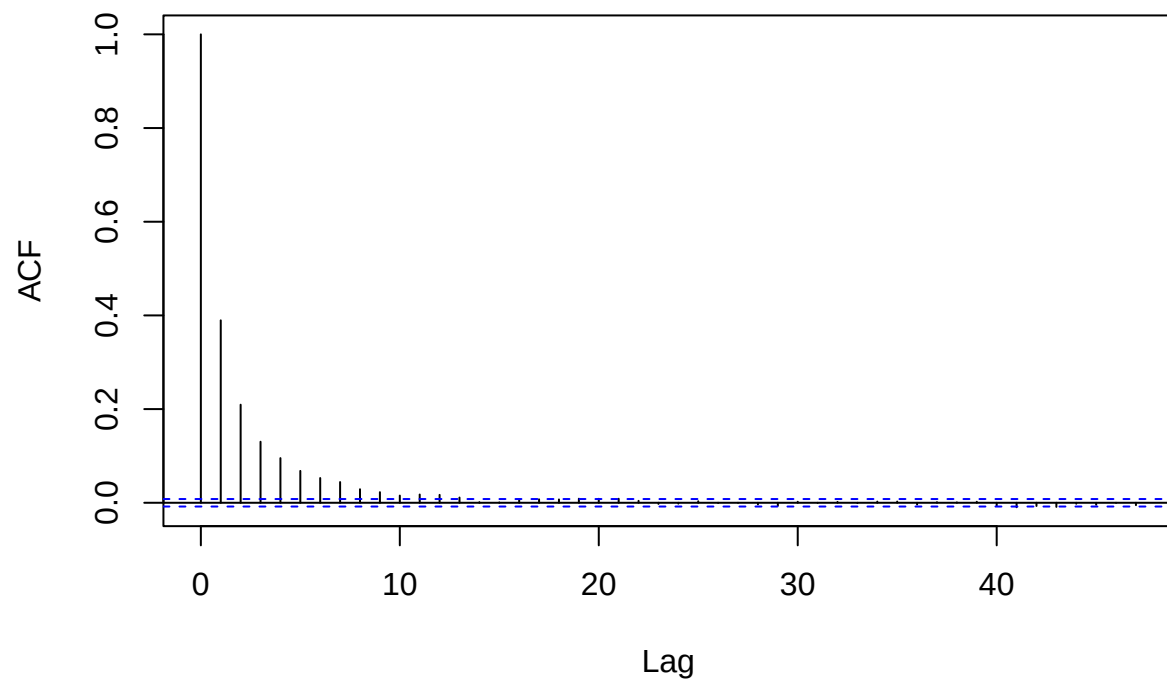
```
acf(sector2_matrix[, c("eta[2,7]")])
```

Series sector2_matrix[, c("eta[2,7]")]



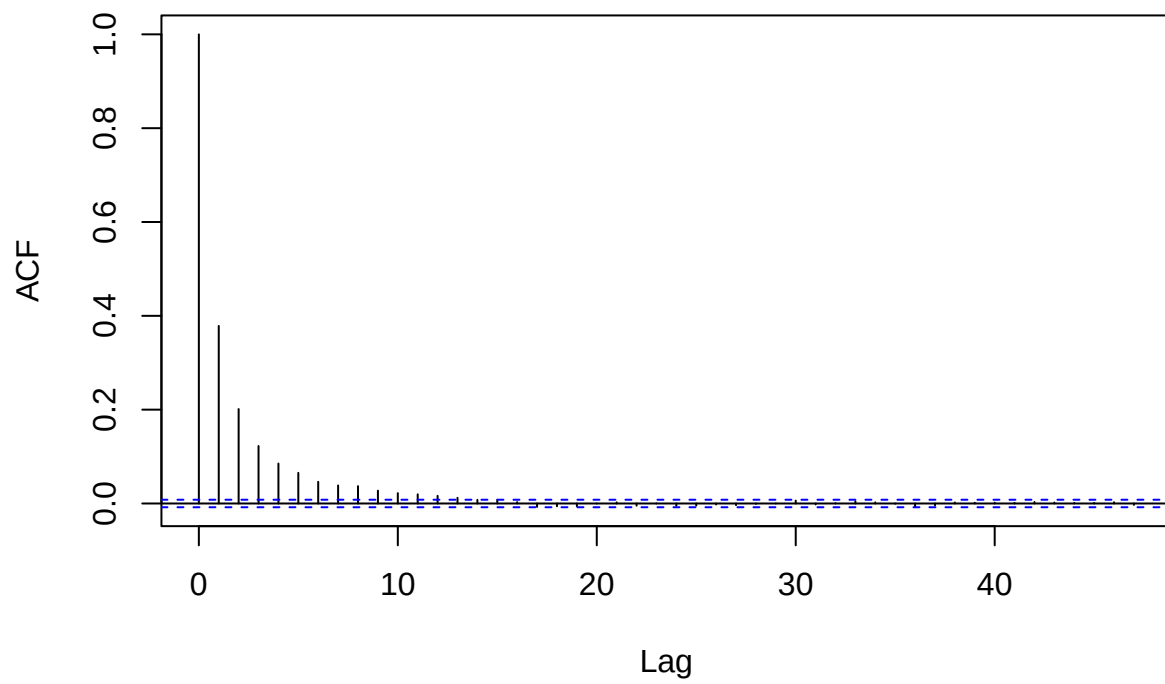
```
acf(sector2_matrix[, c("eta[2,8]")])
```

Series sector2_matrix[, c("eta[2,8]")]



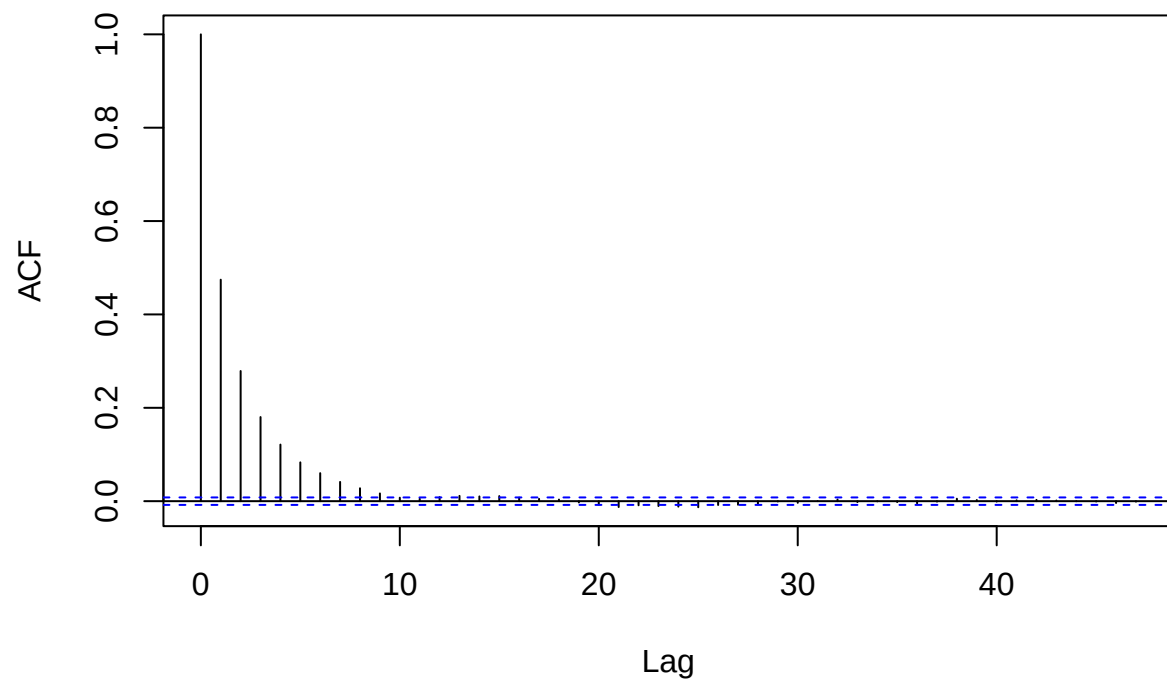
```
acf(sector2_matrix[, c("eta[2,9]")])
```

Series sector2_matrix[, c("eta[2,9]")]



```
acf(sector2_matrix[, c("eta[2,10]")])
```

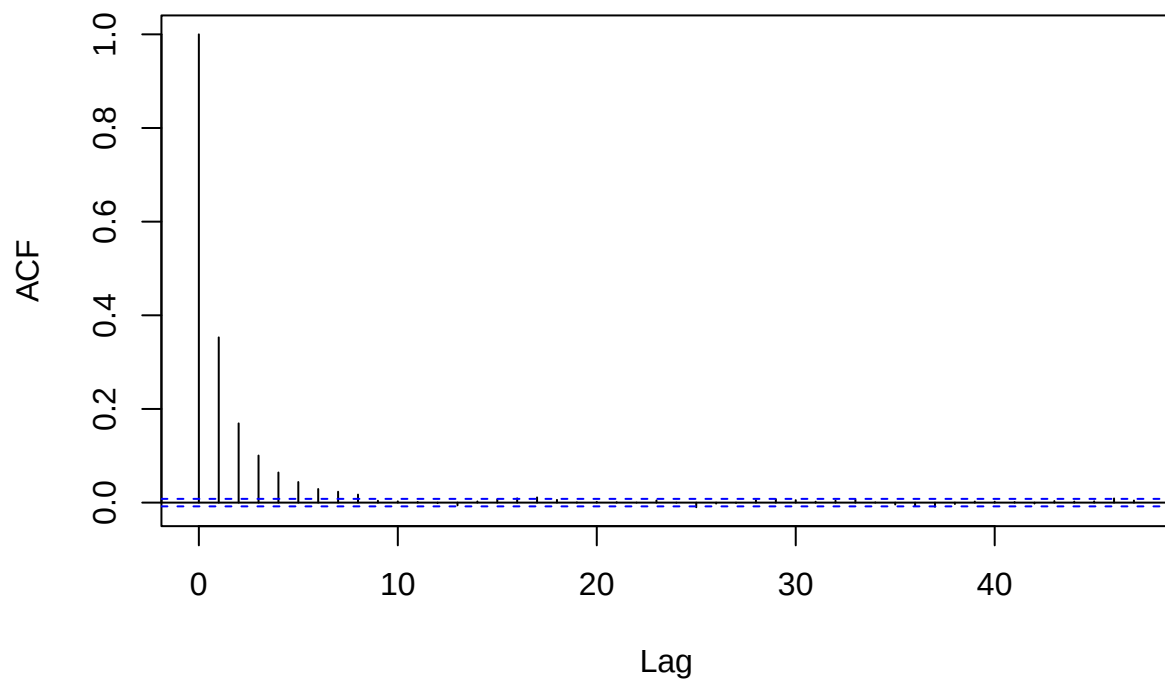

Series sector2_matrix[, c("eta[2,10]")]



```
#sector 3
```

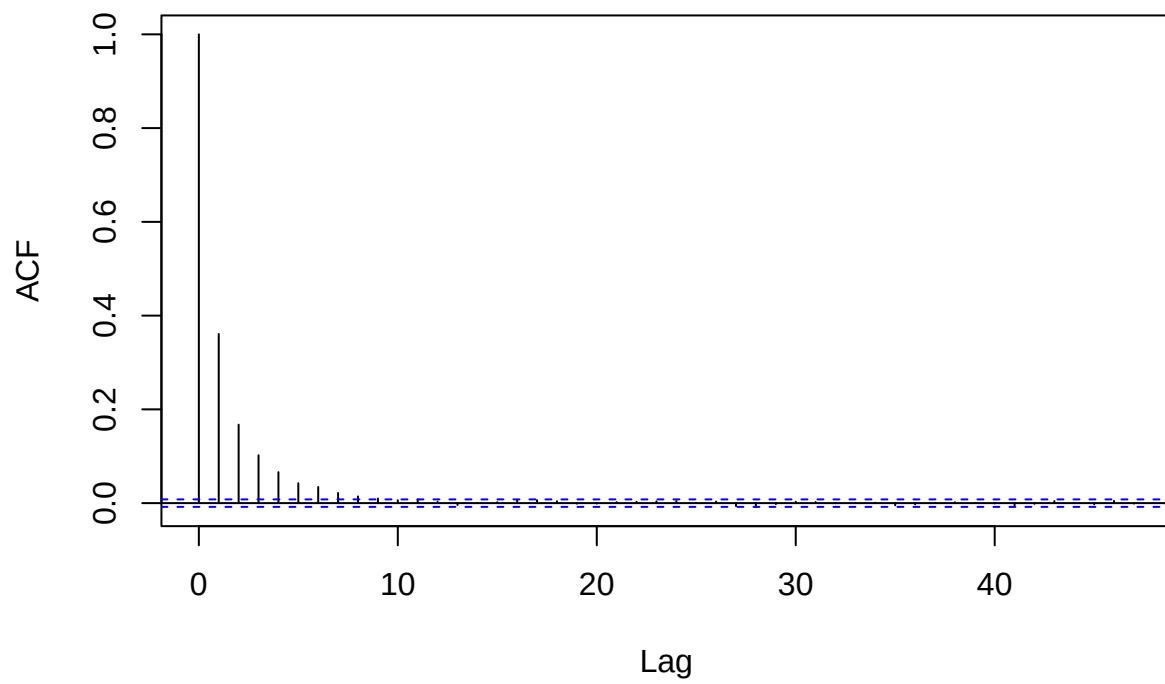
```
acf(sector3_matrix[, c("eta[3,1]")])
```

Series sector3_matrix[, c("eta[3,1]")]



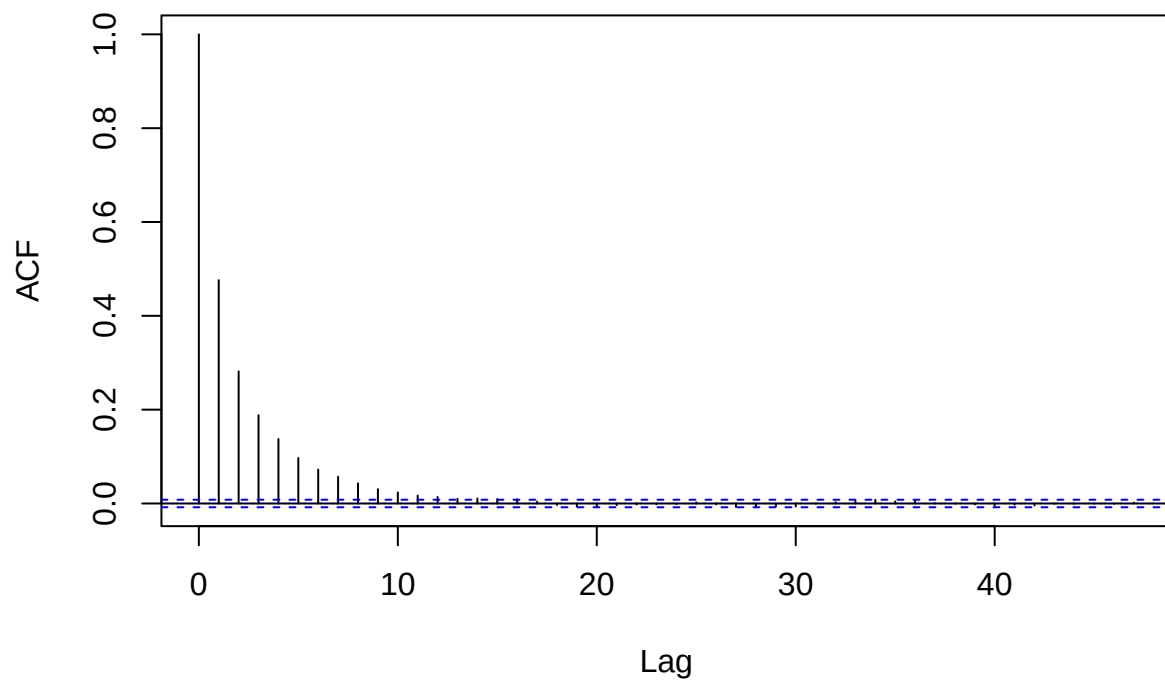
```
acf(sector3_matrix[, c("eta[3,2] ")])
```

Series sector3_matrix[, c("eta[3,2]")]



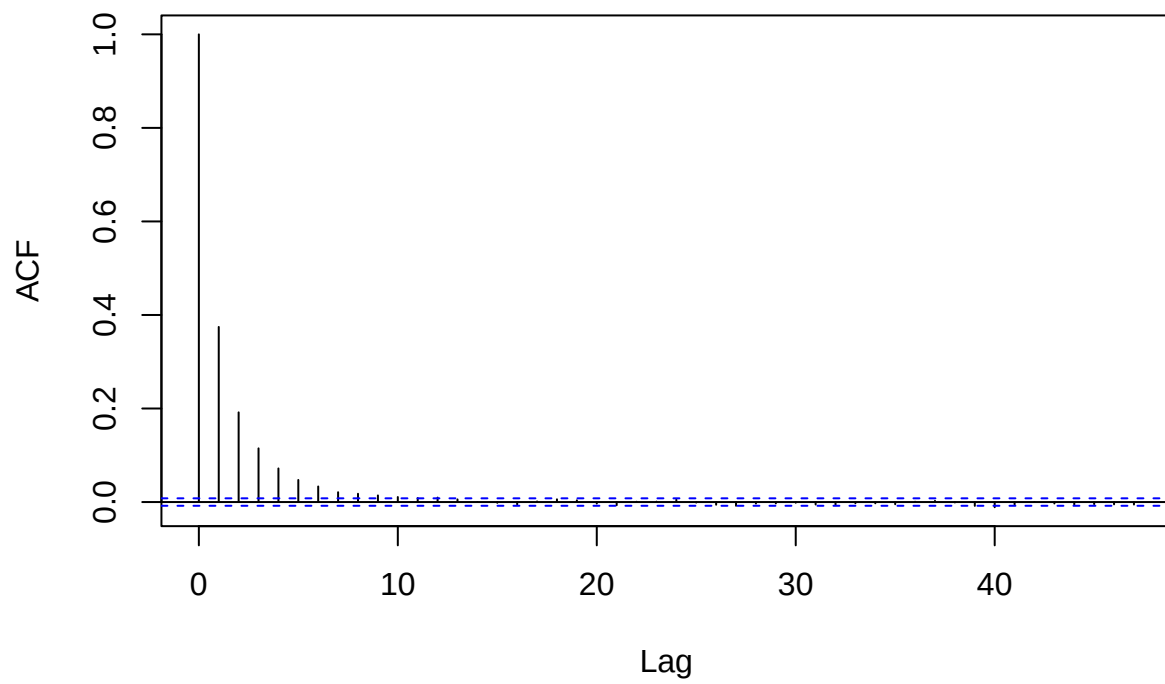
```
acf(sector3_matrix[, c("eta[3,3]")])
```

Series sector3_matrix[, c("eta[3,3]")]



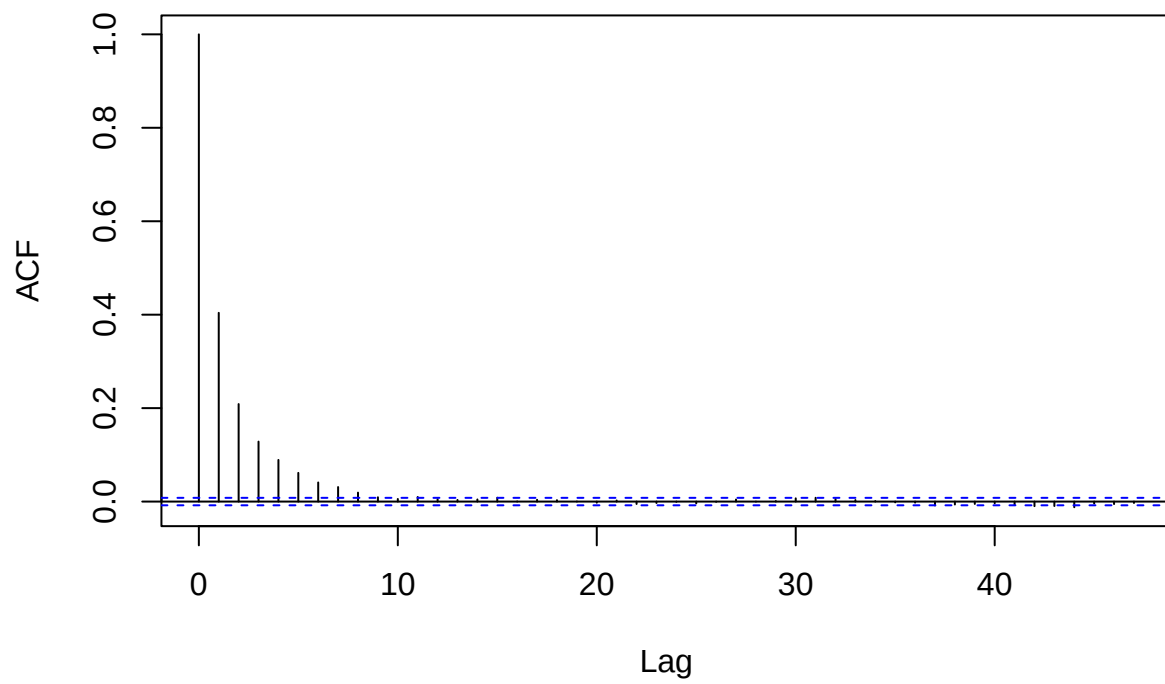
```
acf(sector3_matrix[, c("eta[3,4]")])
```

Series sector3_matrix[, c("eta[3,4]")]



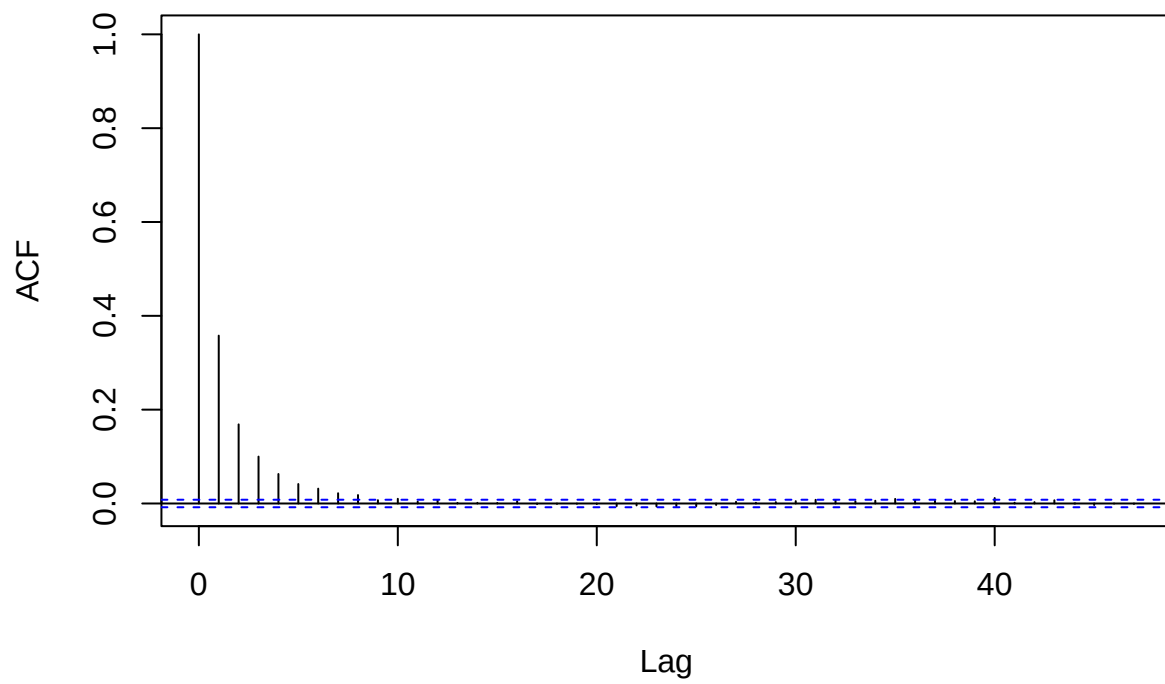
```
acf(sector3_matrix[, c("eta[3,5] ")])
```

Series sector3_matrix[, c("eta[3,5]")]



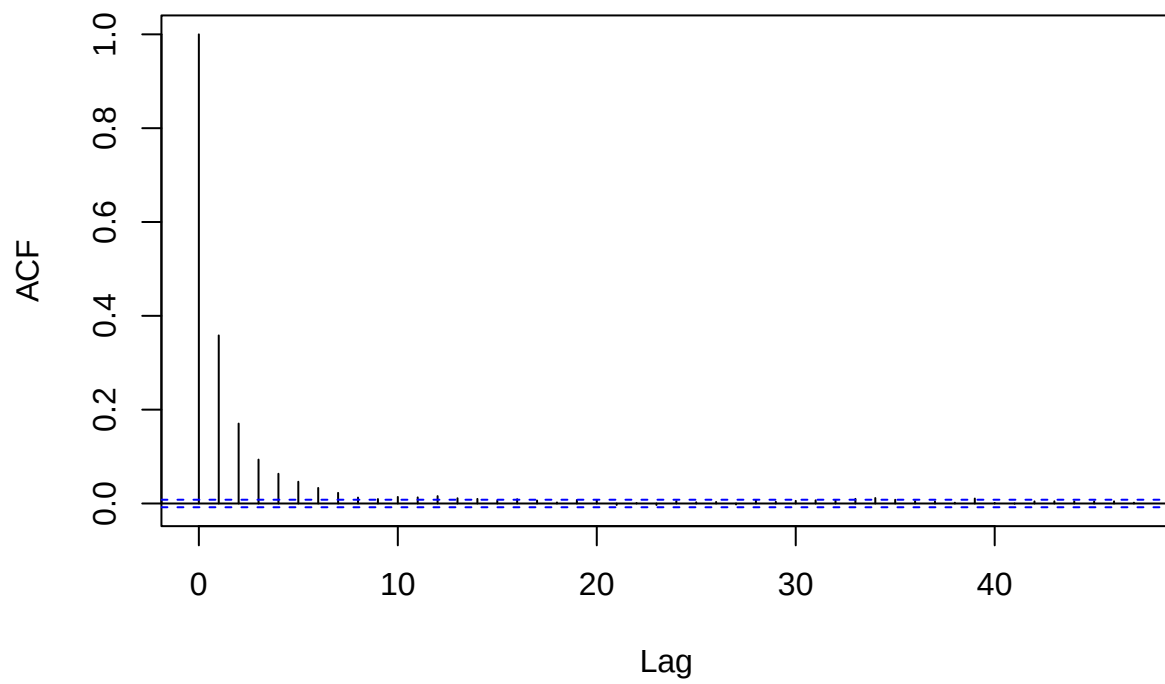
```
acf(sector3_matrix[, c("eta[3,6]")])
```

Series sector3_matrix[, c("eta[3,6]")]



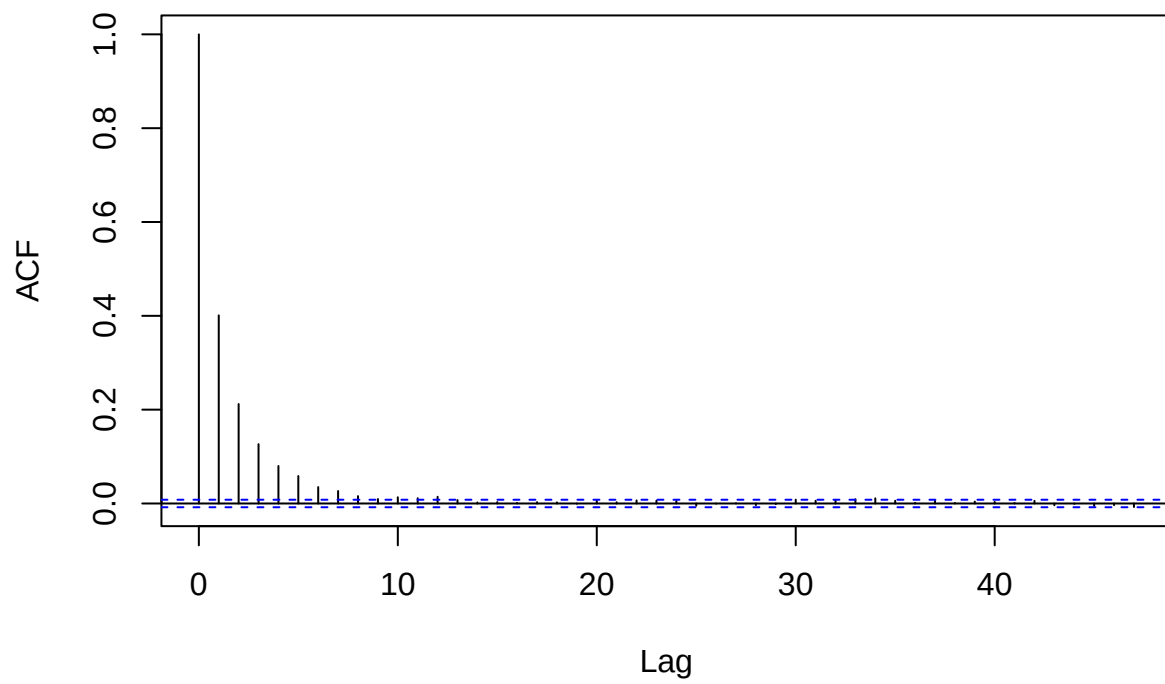
```
acf(sector3_matrix[, c("eta[3,7]")])
```

Series sector3_matrix[, c("eta[3,7]")]



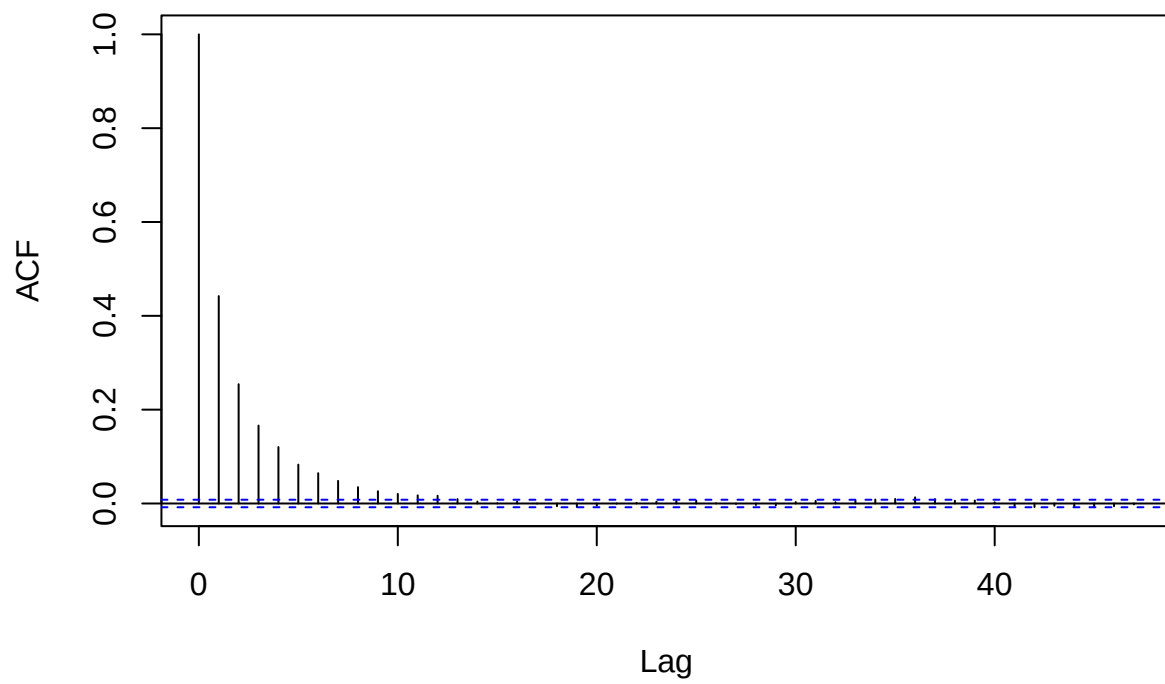
```
acf(sector3_matrix[, c("eta[3,8]")])
```


Series sector3_matrix[, c("eta[3,8]")]



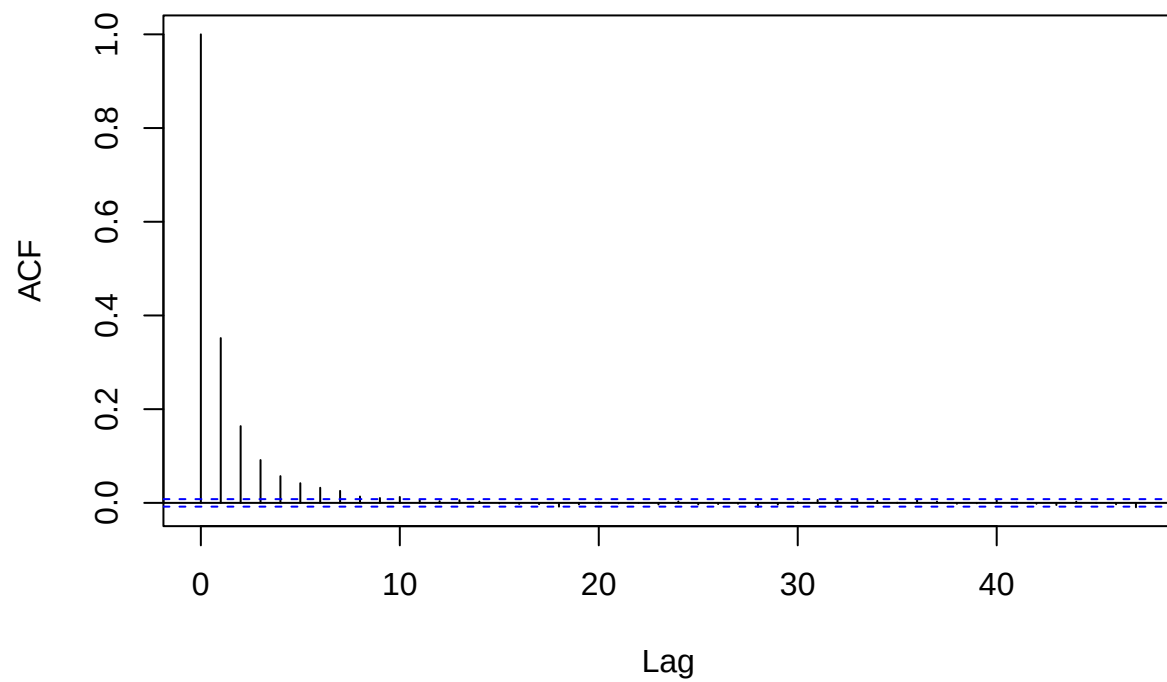
```
acf(sector3_matrix[, c("eta[3,9]")])
```

Series sector3_matrix[, c("eta[3,9]")]



```
acf(sector3_matrix[, c("eta[3,10]")])
```

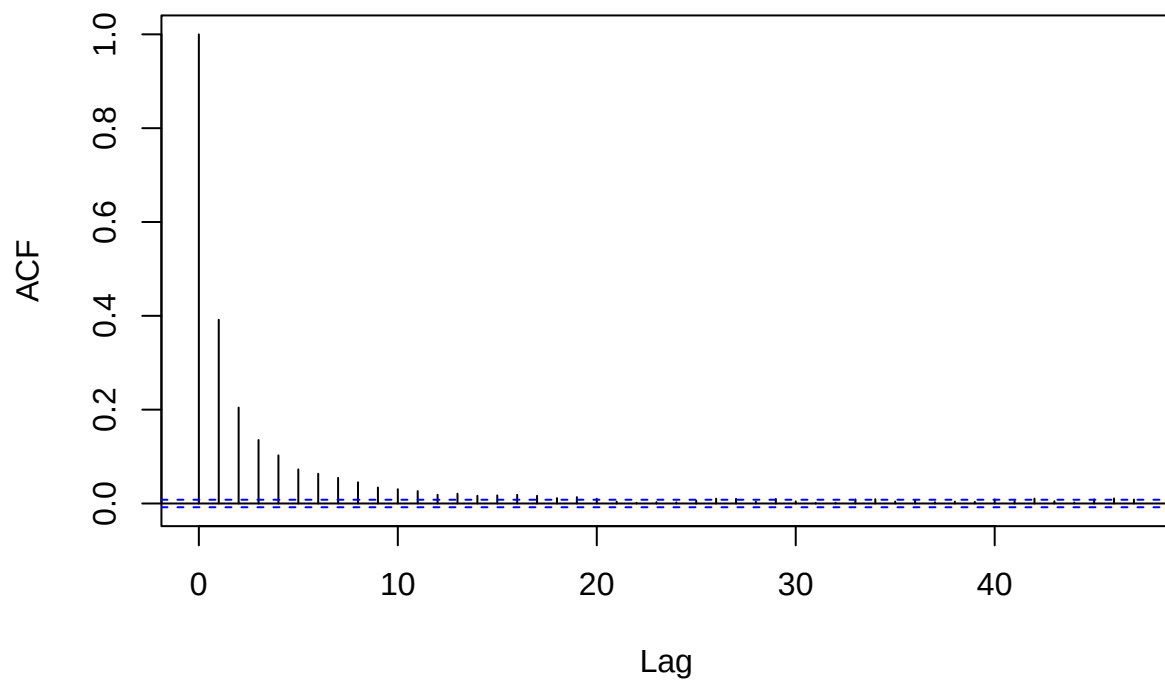
Series sector3_matrix[, c("eta[3,10]")]



```
#sector 4
```

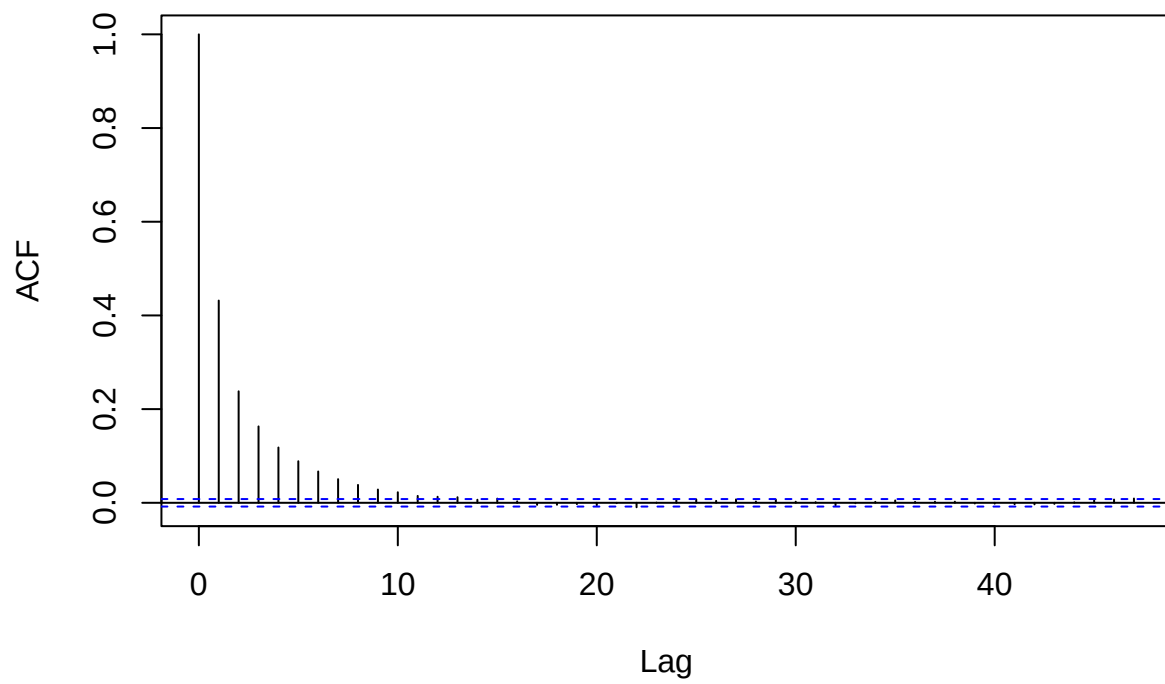
```
acf(sector4_matrix[, c("eta[4,1]")])
```

Series sector4_matrix[, c("eta[4,1]")]



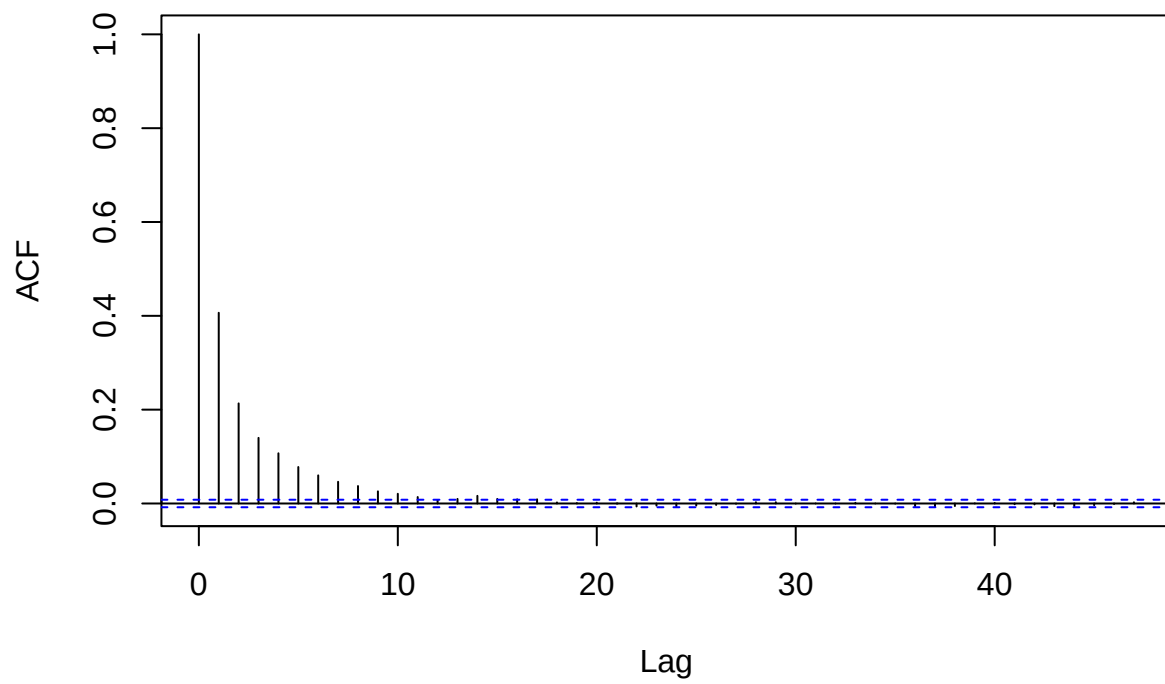
```
acf(sector4_matrix[, c("eta[4,2] ")])
```

Series sector4_matrix[, c("eta[4,2]")]



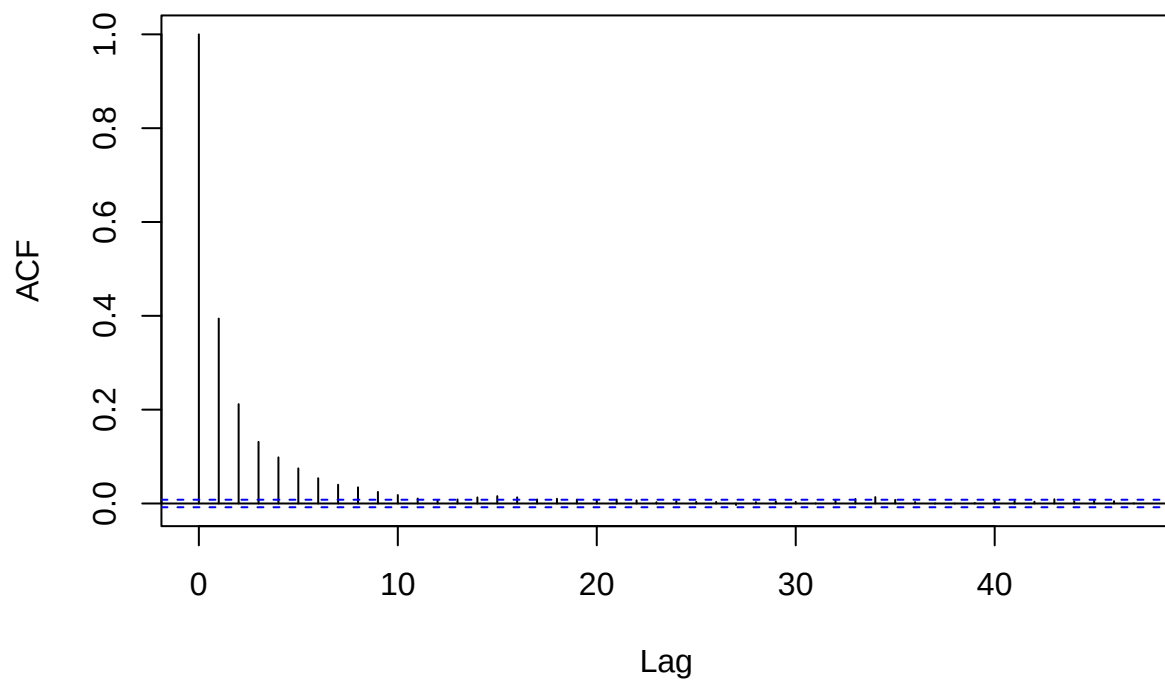
```
acf(sector4_matrix[, c("eta[4,3]")])
```

Series sector4_matrix[, c("eta[4,3]")]



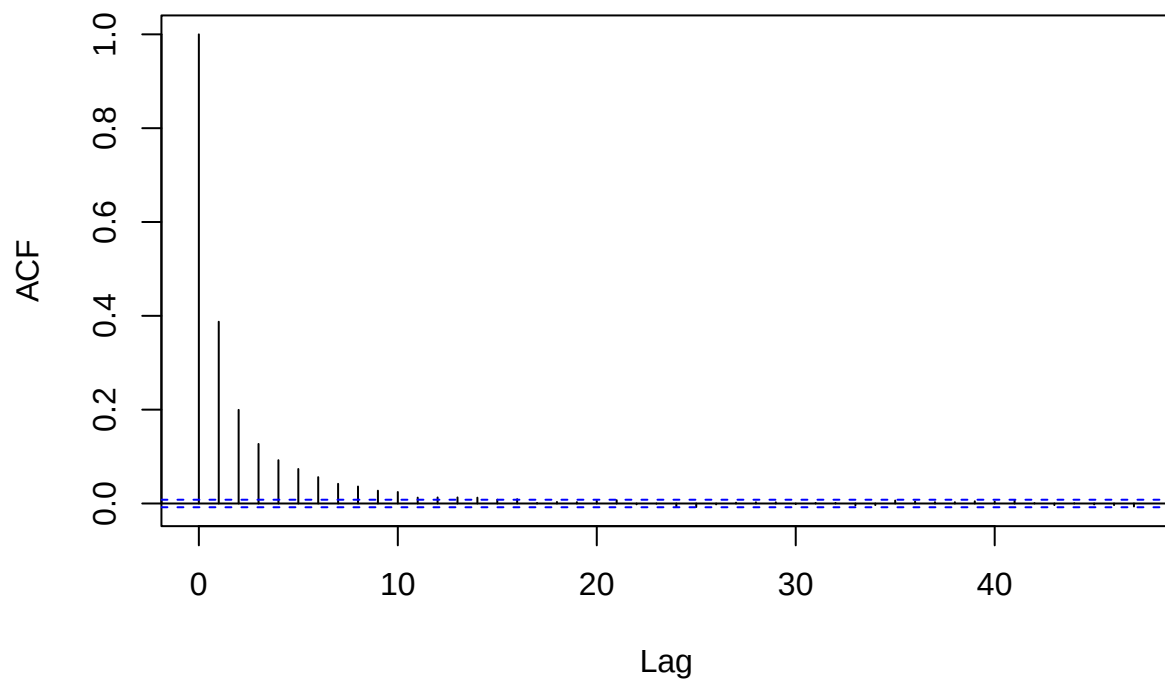
```
acf(sector4_matrix[, c("eta[4,4]")])
```

Series sector4_matrix[, c("eta[4,4]")]



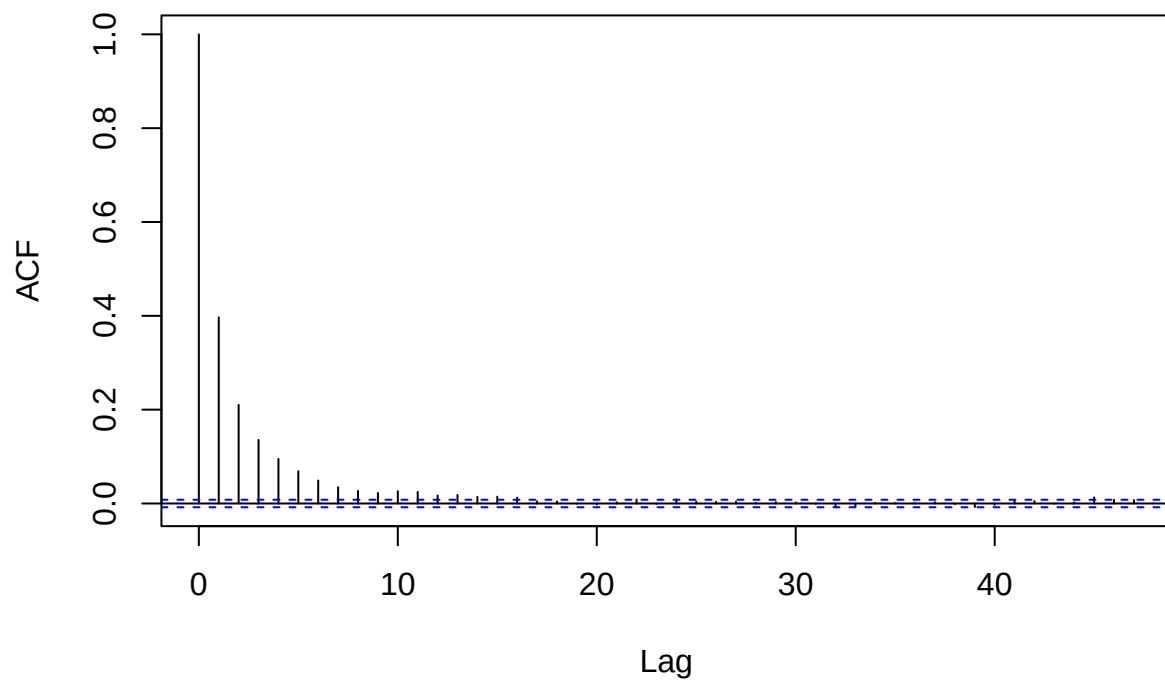
```
acf(sector4_matrix[, c("eta[4,5]")])
```

Series sector4_matrix[, c("eta[4,5]")]



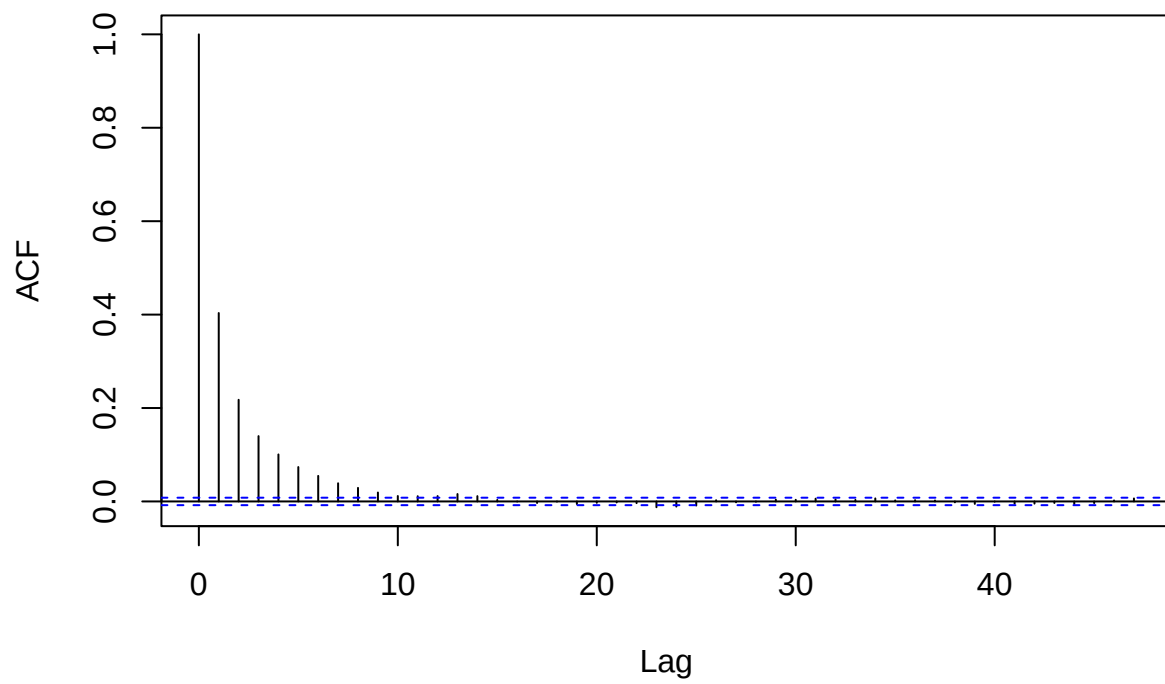
```
acf(sector4_matrix[, c("eta[4,6]")])
```


Series sector4_matrix[, c("eta[4,6]")]



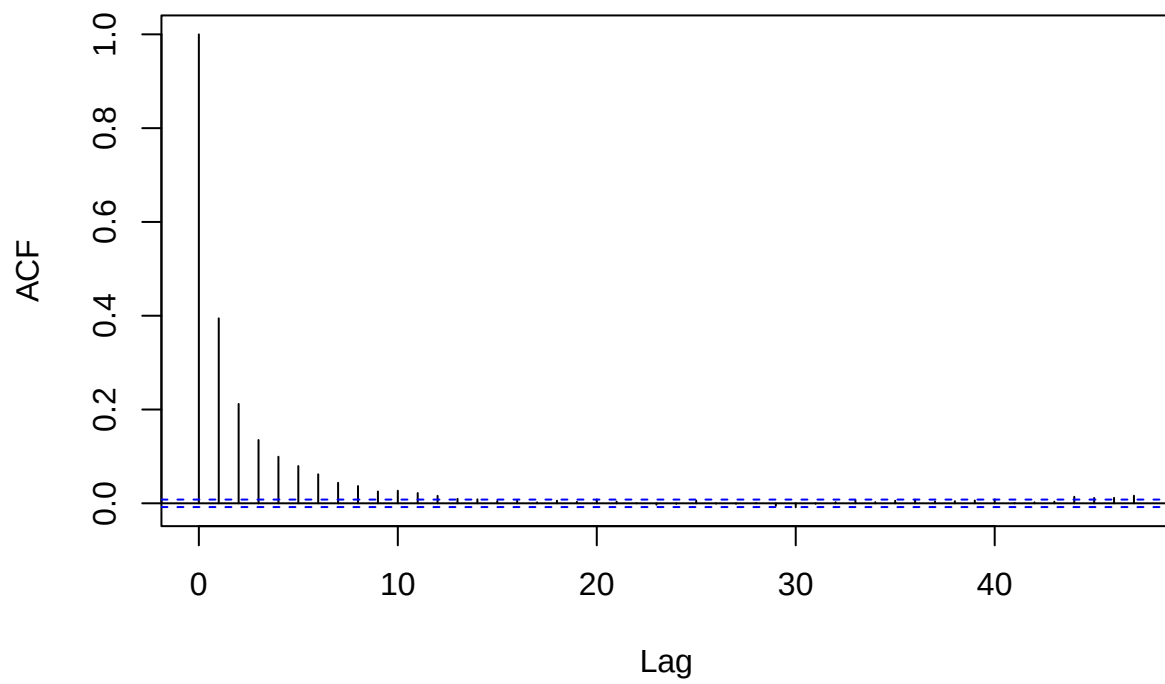
```
acf(sector4_matrix[, c("eta[4,7]")])
```

Series sector4_matrix[, c("eta[4,7]")]



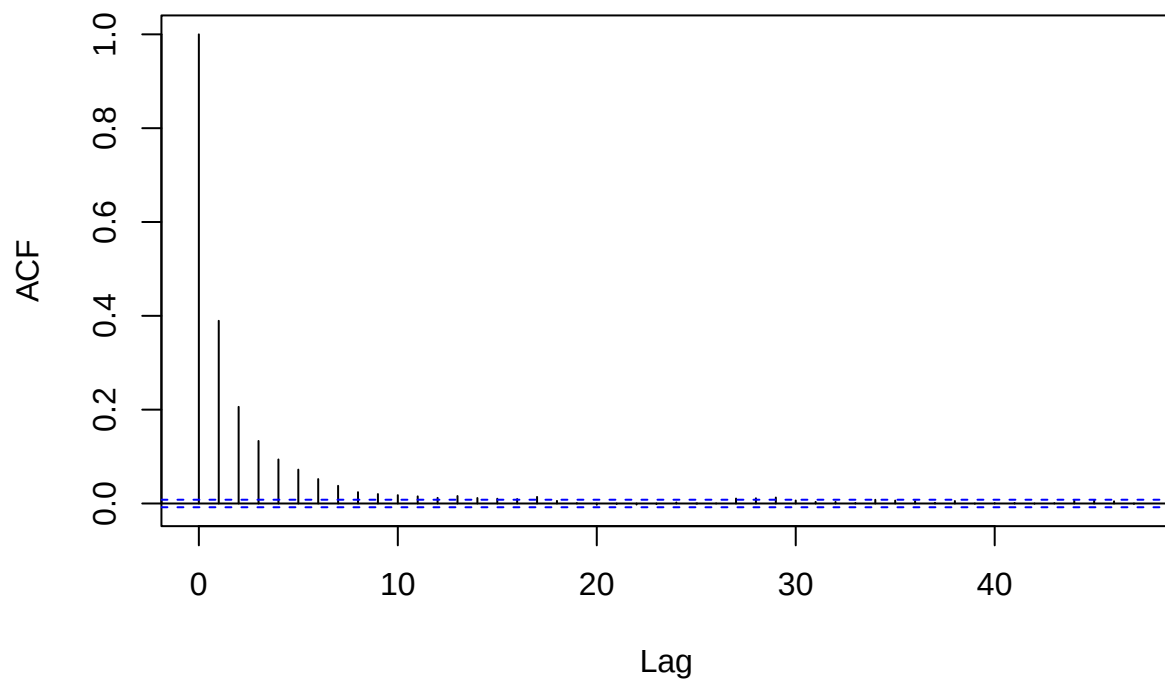
```
acf(sector4_matrix[, c("eta[4,8]")])
```

Series sector4_matrix[, c("eta[4,8]")]



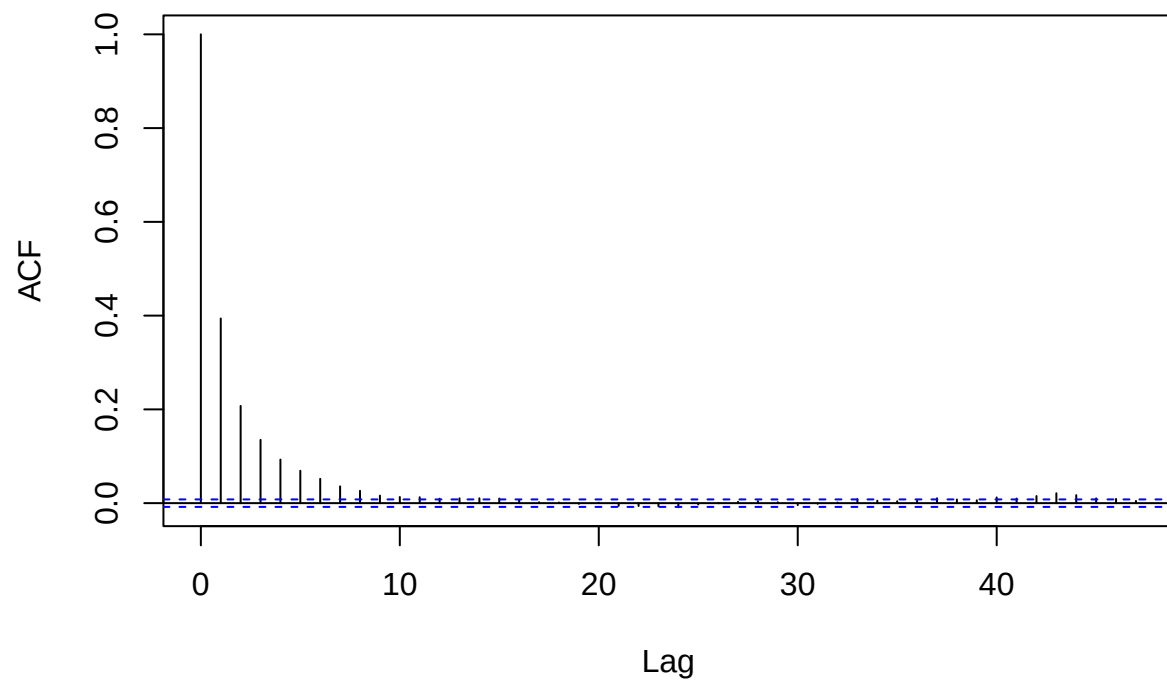
```
acf(sector4_matrix[, c("eta[4,9]")])
```

Series sector4_matrix[, c("eta[4,9]")]



```
acf(sector4_matrix[, c("eta[4,10]")])
```

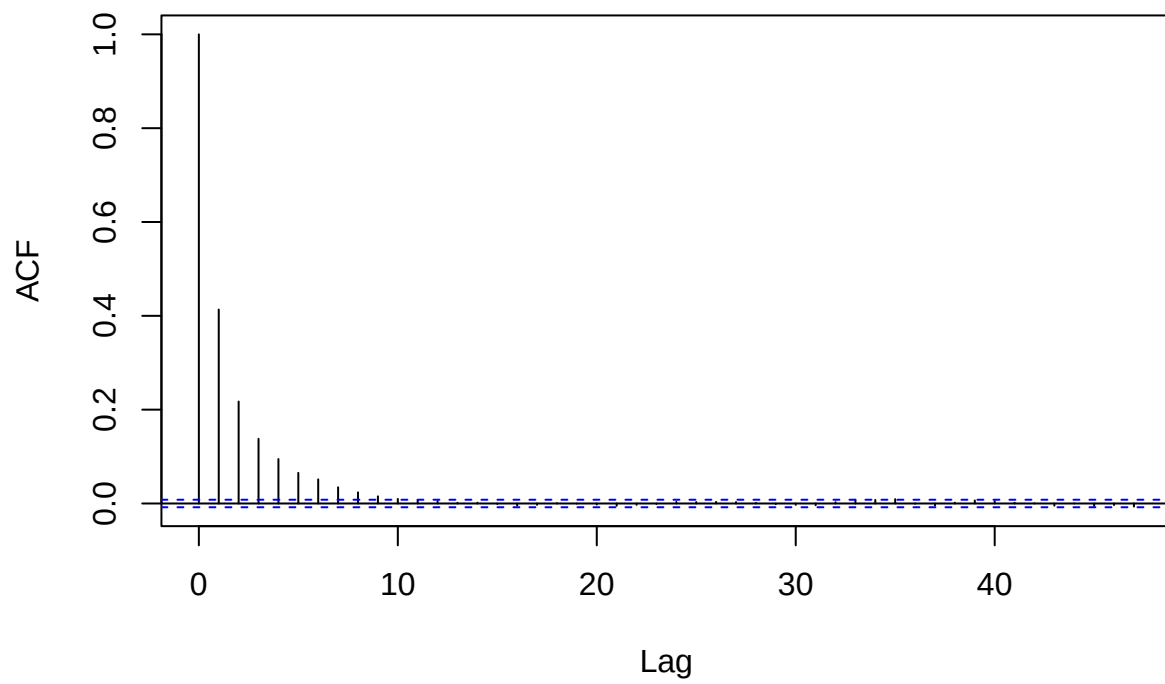
Series sector4_matrix[, c("eta[4,10]")]



```
#sector 5
```

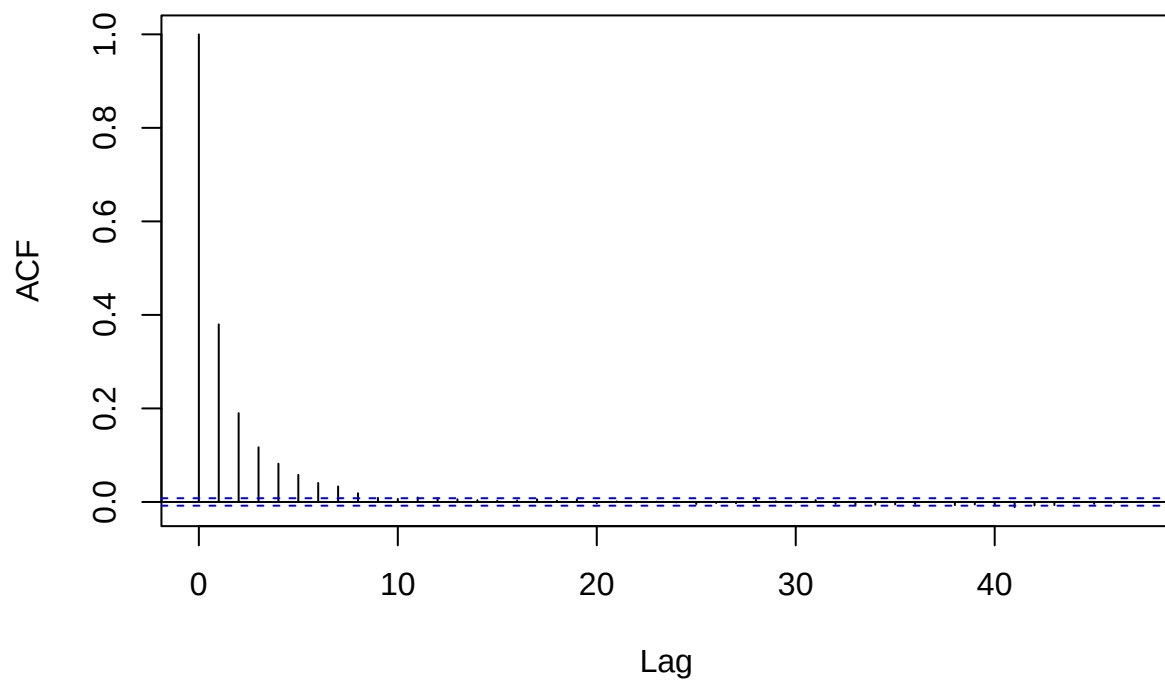
```
acf(sector5_matrix[, c("eta[5,1]")])
```

Series sector5_matrix[, c("eta[5,1]")]



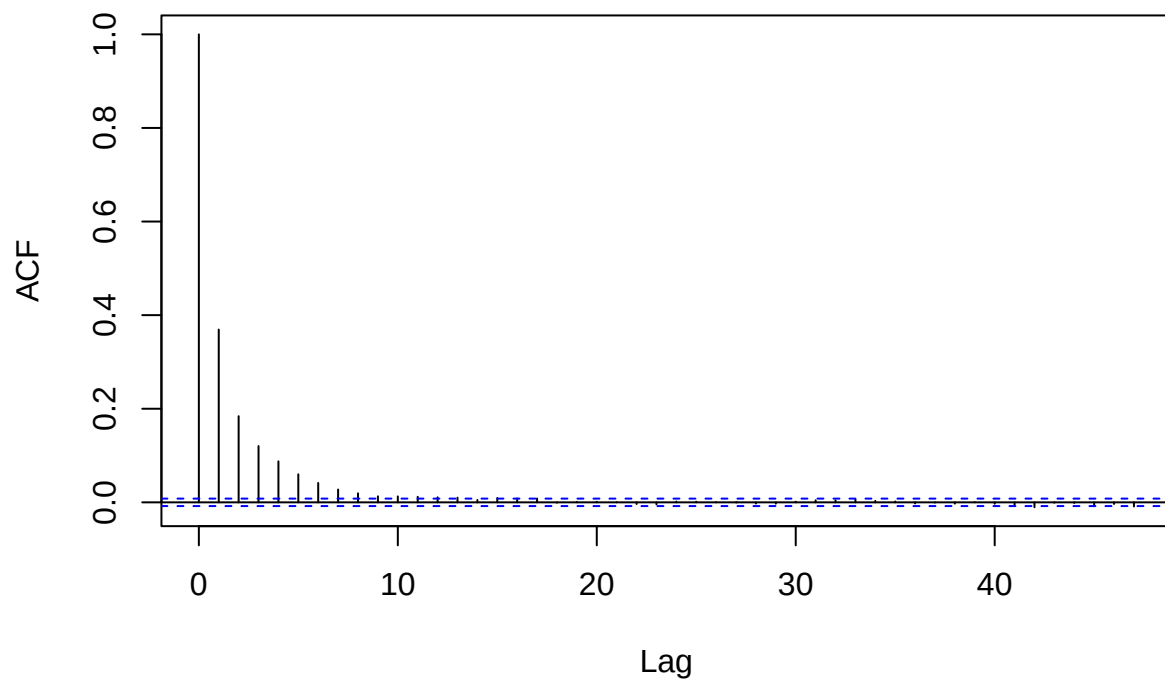
```
acf(sector5_matrix[, c("eta[5,2] ")])
```

Series sector5_matrix[, c("eta[5,2]")]



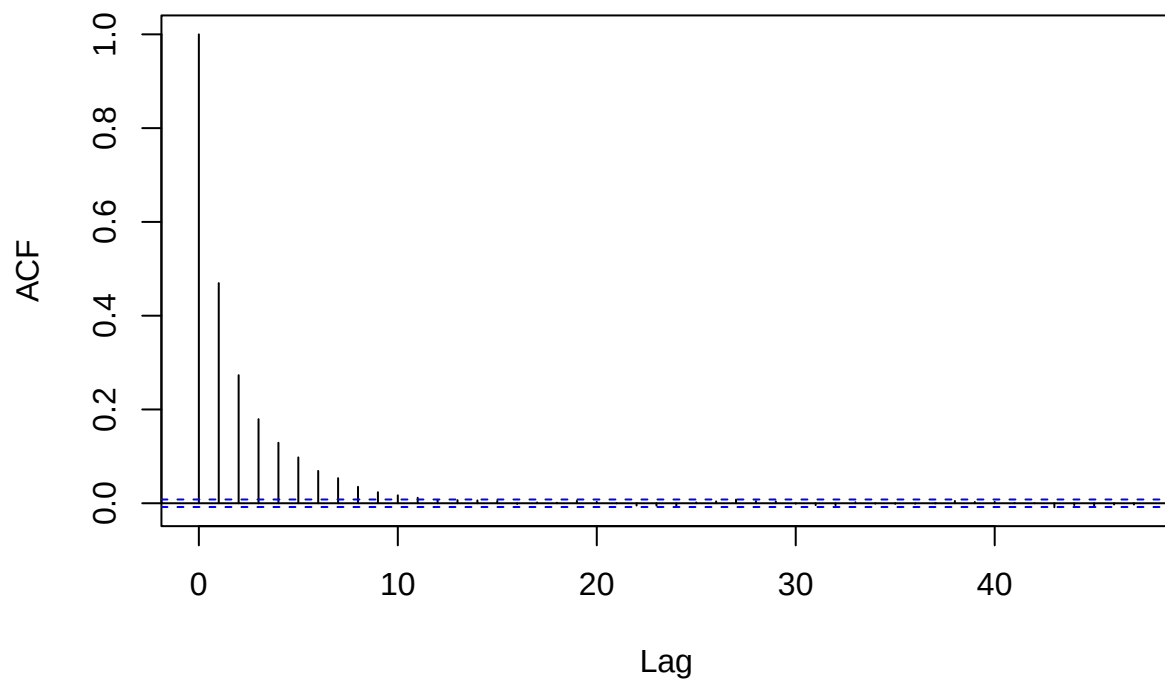
```
acf(sector5_matrix[, c("eta[5,3] ")])
```

Series sector5_matrix[, c("eta[5,3]")]



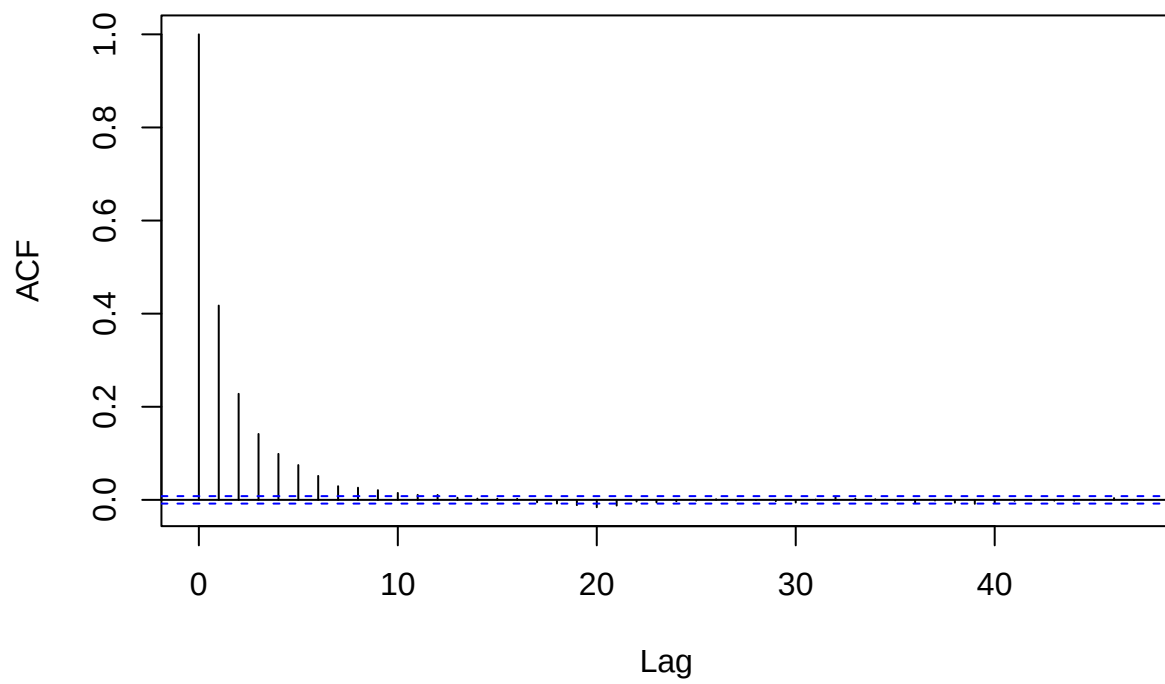
```
acf(sector5_matrix[, c("eta[5,4] ")])
```


Series sector5_matrix[, c("eta[5,4]")]



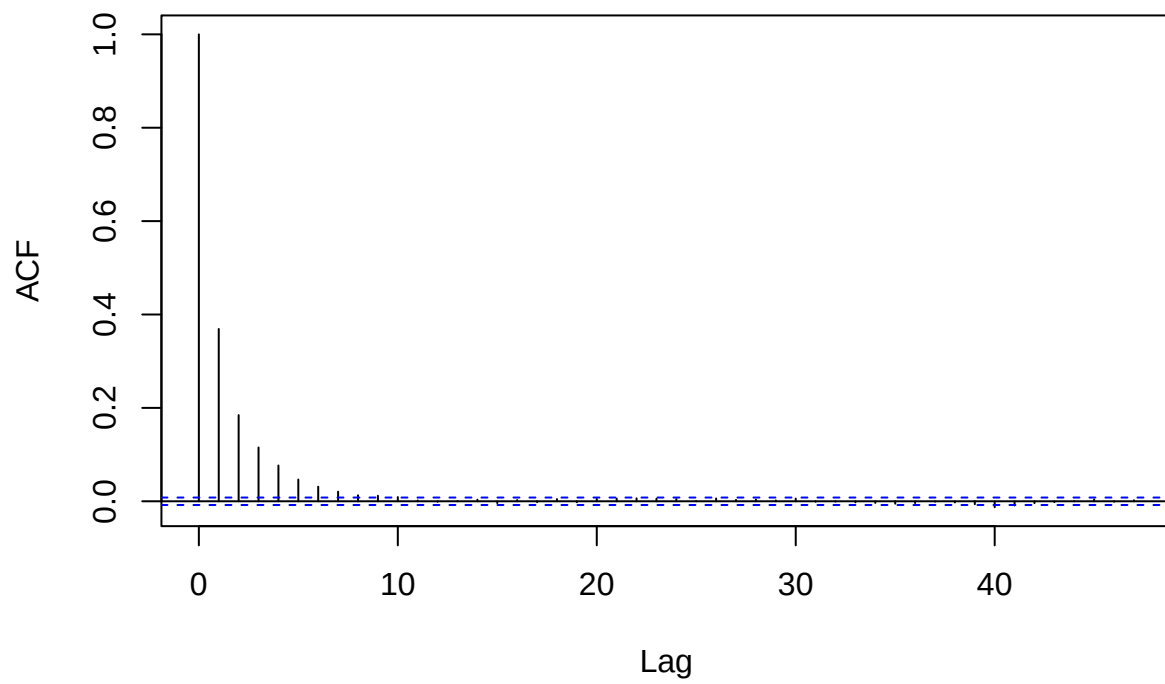
```
acf(sector5_matrix[, c("eta[5,5] ")])
```

Series sector5_matrix[, c("eta[5,5]")]



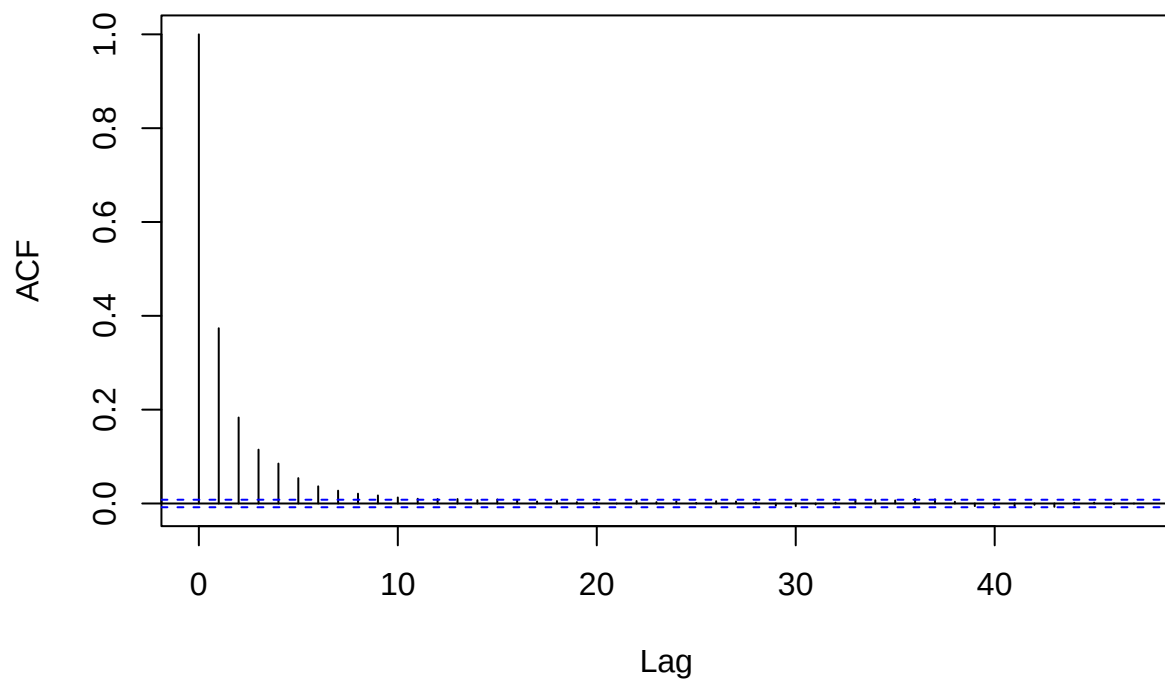
```
acf(sector5_matrix[, c("eta[5,6] ")])
```

Series sector5_matrix[, c("eta[5,6]")]



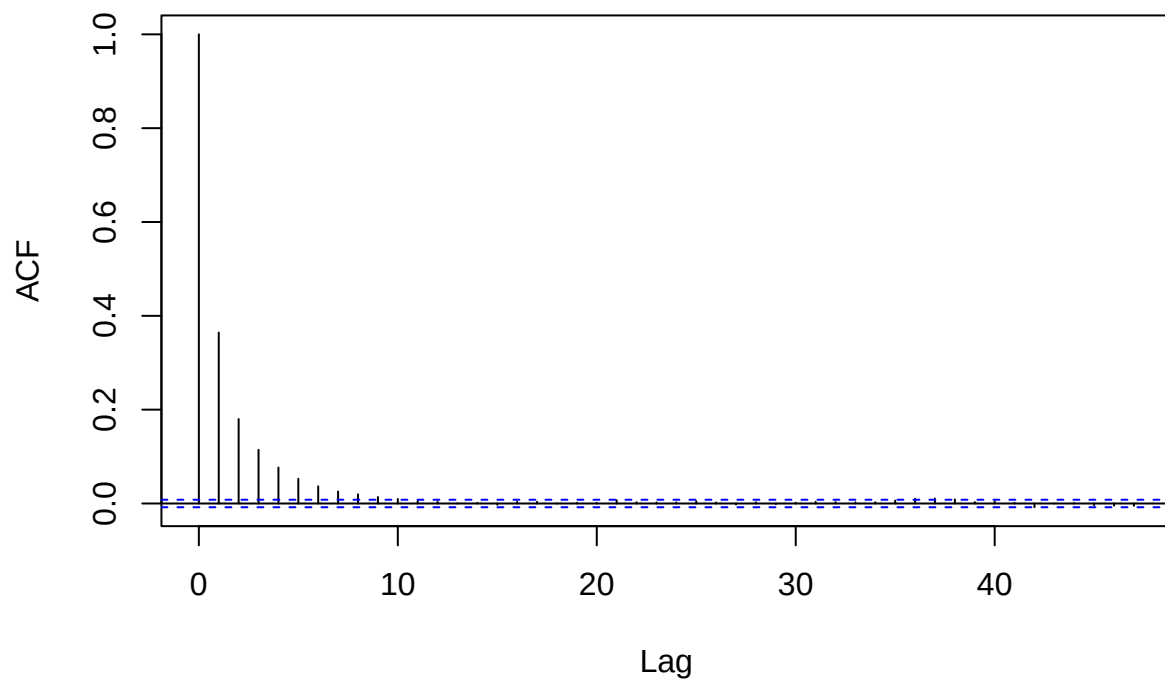
```
acf(sector5_matrix[, c("eta[5,7] ")])
```

Series sector5_matrix[, c("eta[5,7]")]



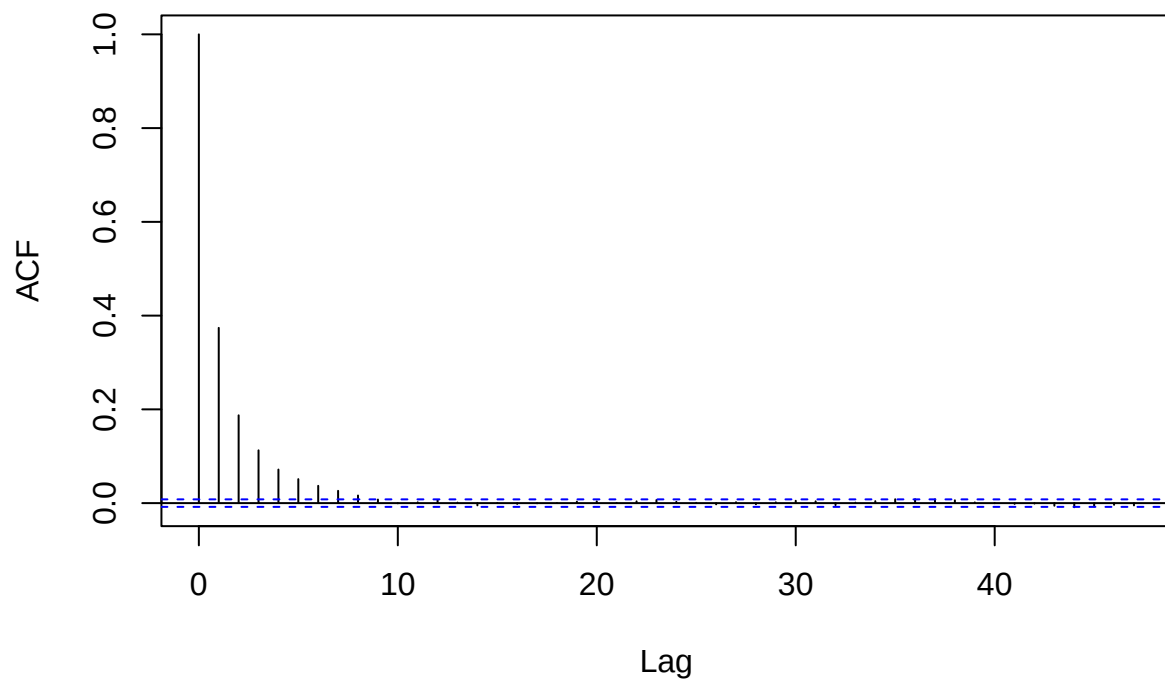
```
acf(sector5_matrix[, c("eta[5,8] ")])
```

Series sector5_matrix[, c("eta[5,8]")]



```
acf(sector5_matrix[, c("eta[5,9] ")])
```

Series sector5_matrix[, c("eta[5,9]")]



```
acf(sector5_matrix[, c("eta[5,10]")])
```

Series sector5_matrix[, c("eta[5,10]")]

